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Van Baelen et al.

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(54) **TRAY ARRANGEMENTS FOR CASSETTES**

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patent is extended or adjusted under 35
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This patent is subject to a terminal dis-
claimer.

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(65) **Prior Publication Data**

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Related U.S. Application Data

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Feb. 18, 2022, now Pat. No. 11,971,598.

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18, 2021.

(51) **Int. Cl.**
G02B 6/00 (2006.01)
G02B 6/44 (2006.01)

(52) **U.S. Cl.**

CPC **G02B 6/4455** (2013.01); **G02B 6/44526**
(2023.05); **G02B 6/44528** (2023.05); **G02B**
6/4453 (2013.01)

(58) **Field of Classification Search**

CPC G02B 6/4453; G02B 6/4452; G02B
6/44526; G02B 6/44528; G02B 6/00
See application file for complete search history.

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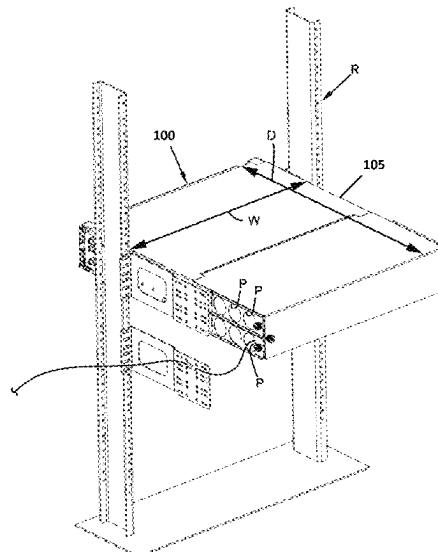
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(57) **ABSTRACT**

A communications panel includes a chassis receiving one or
more tray arrangements that each support one or more
cassettes. Each cassette carries a plurality of ports at which
connections are made between front and rear plug connec-
tors. Each tray arrangement includes guides along which the
cassettes slidably mount. The guides and cassettes are con-
figured to enable cassettes of various size to mount to the
same tray without reconfiguring the guides.

20 Claims, 45 Drawing Sheets



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FIG. 1

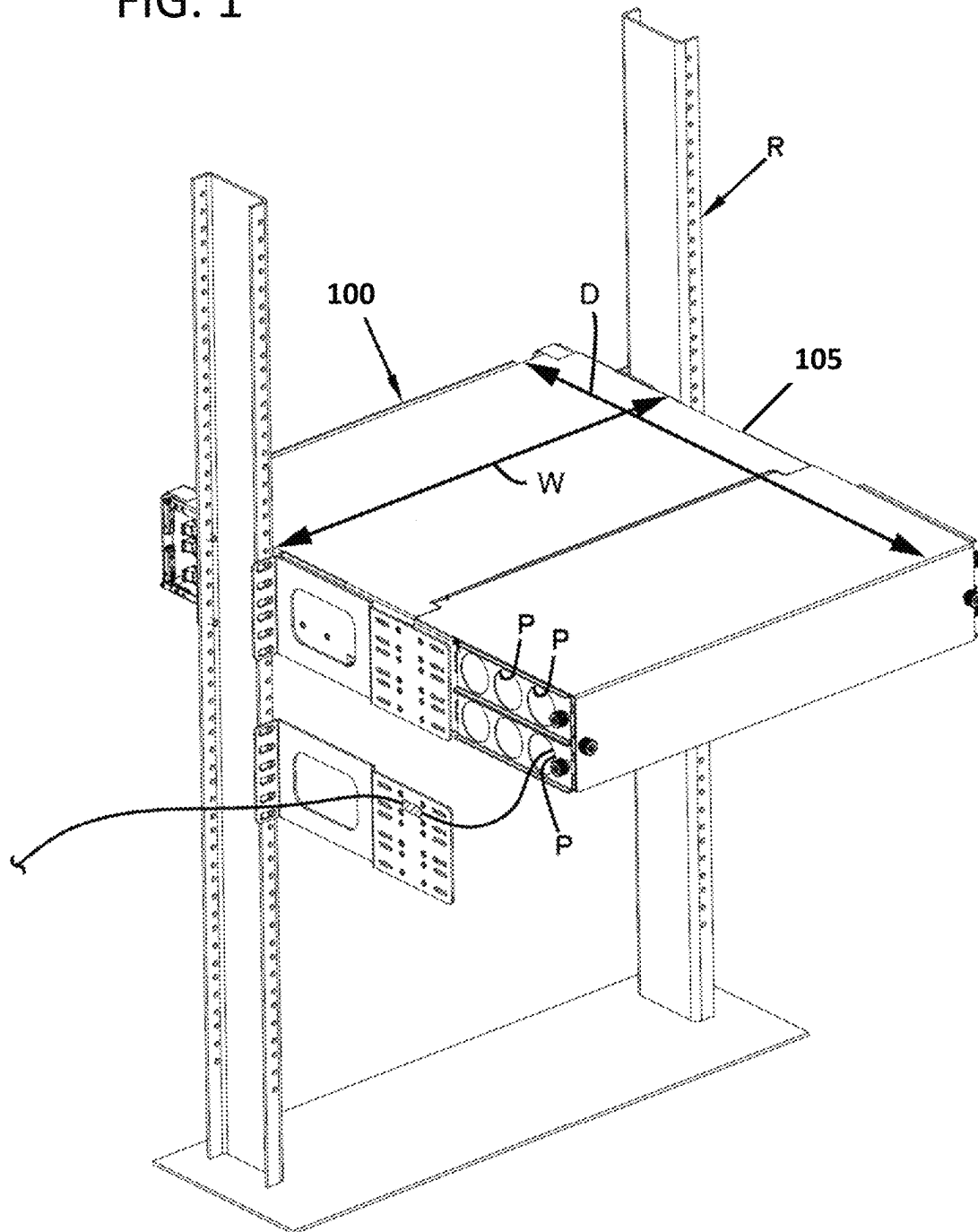
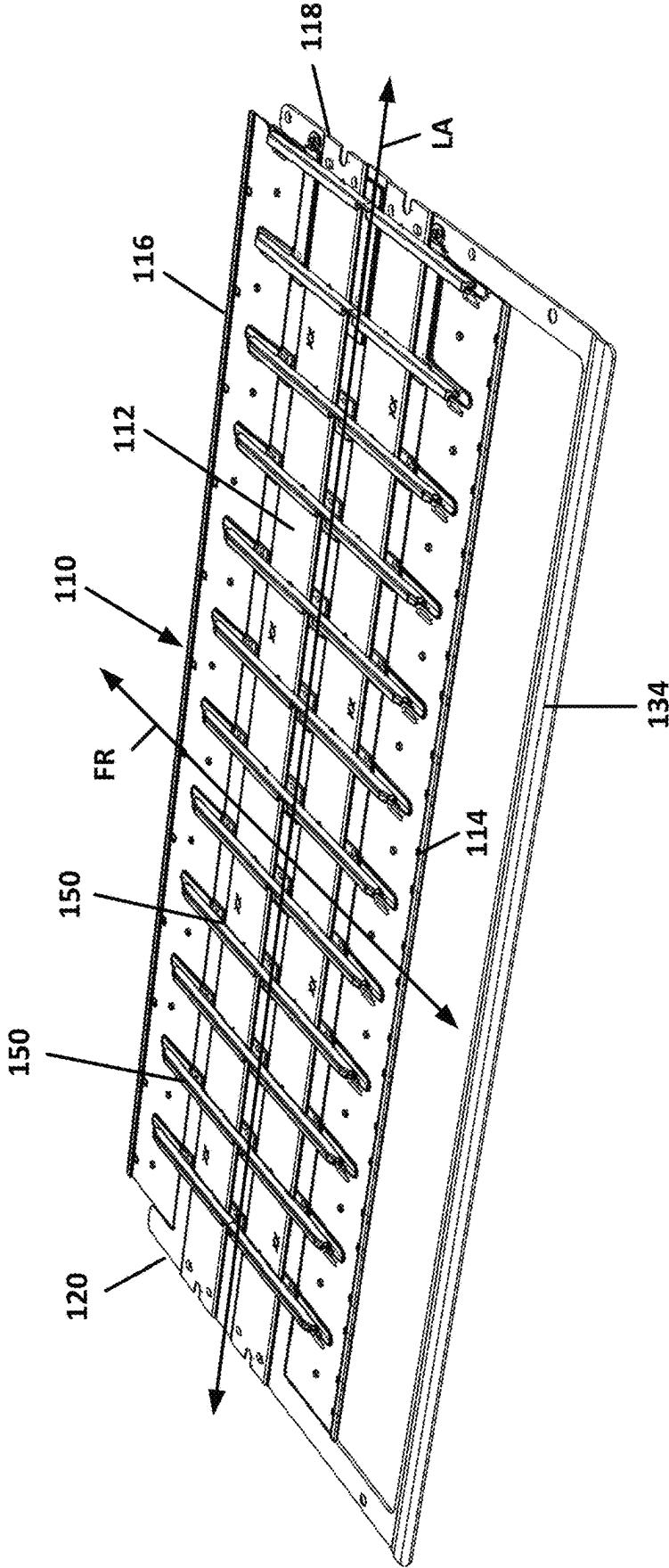


FIG. 2



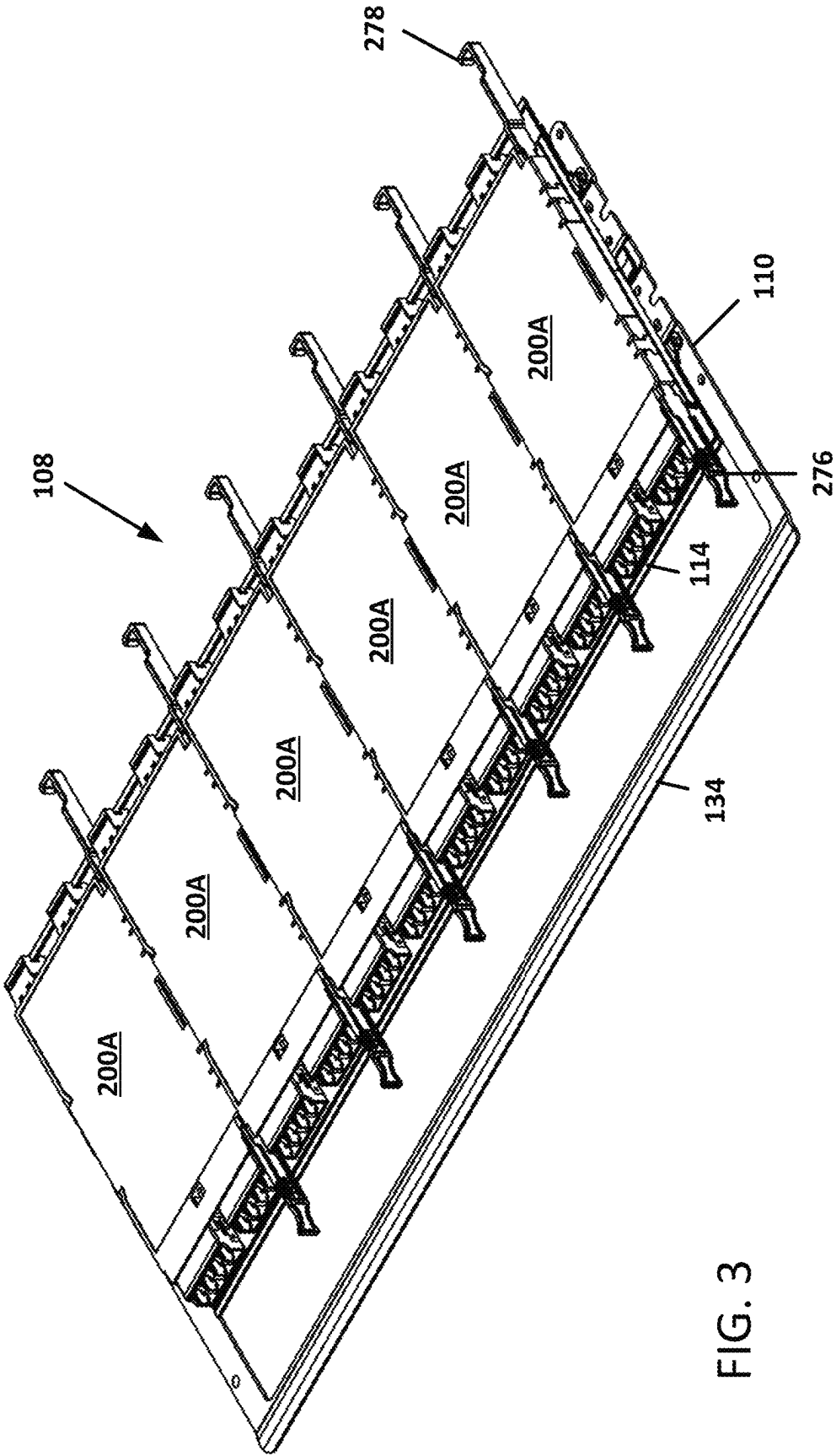


FIG. 3

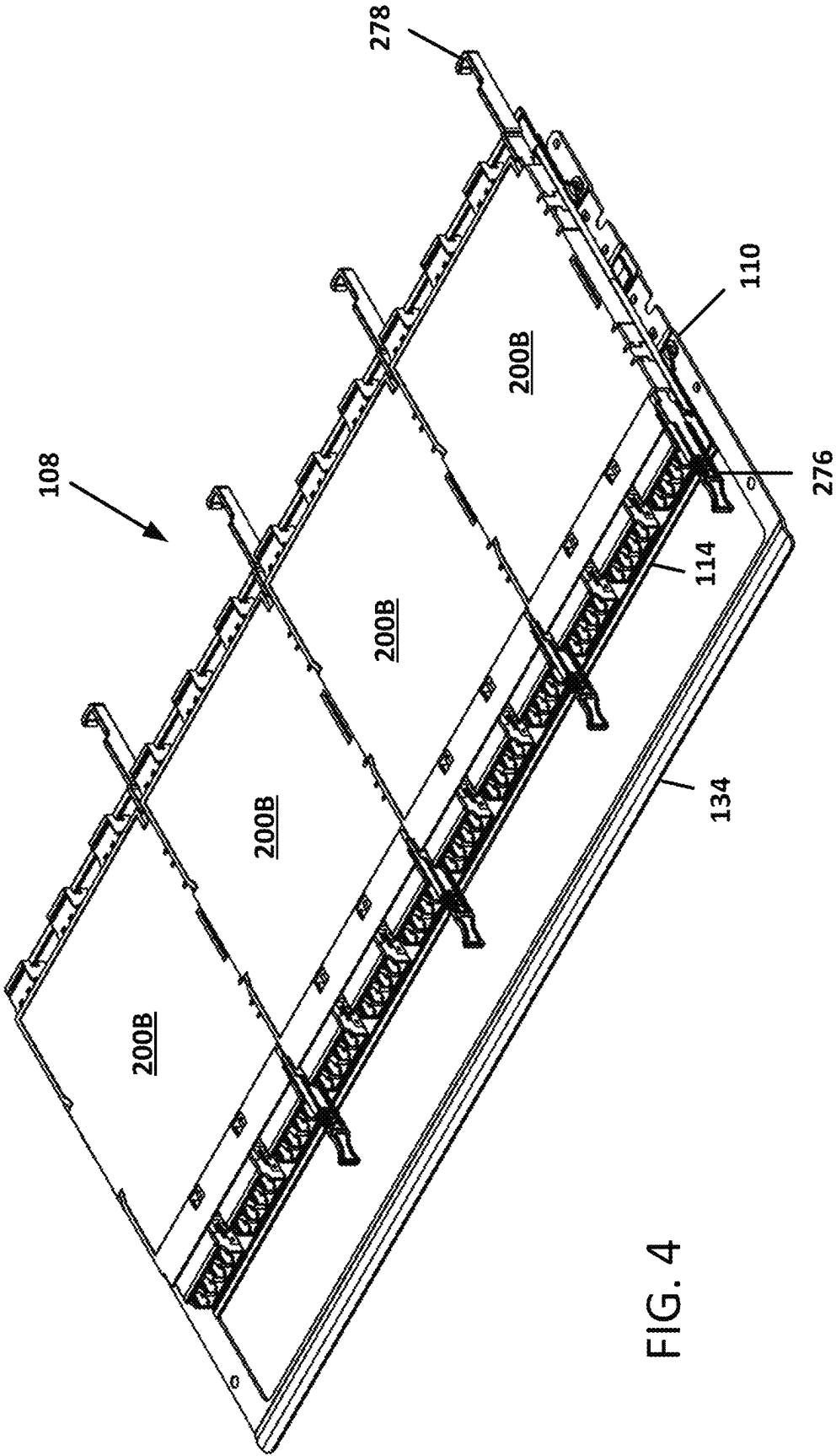


FIG. 4

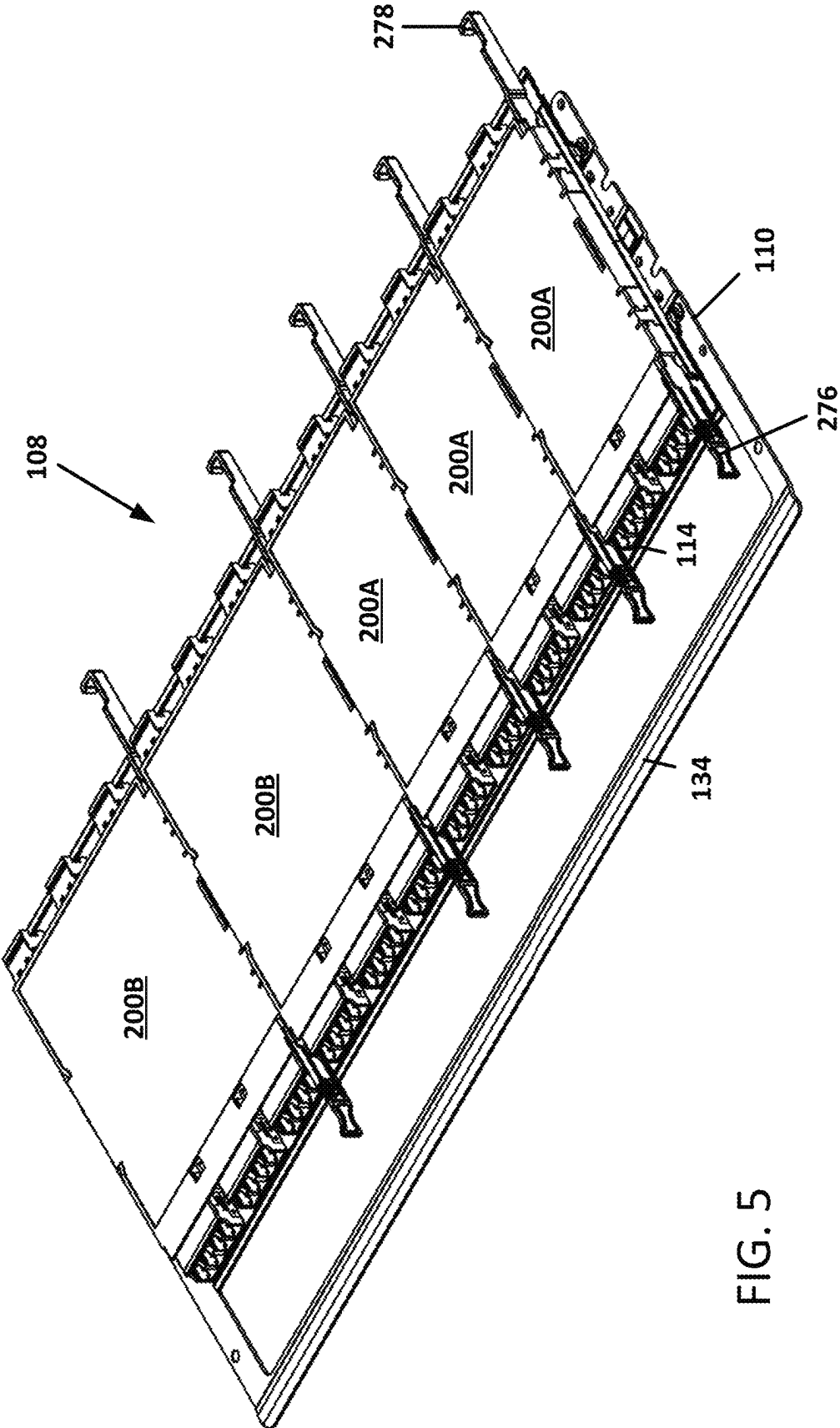


FIG. 5

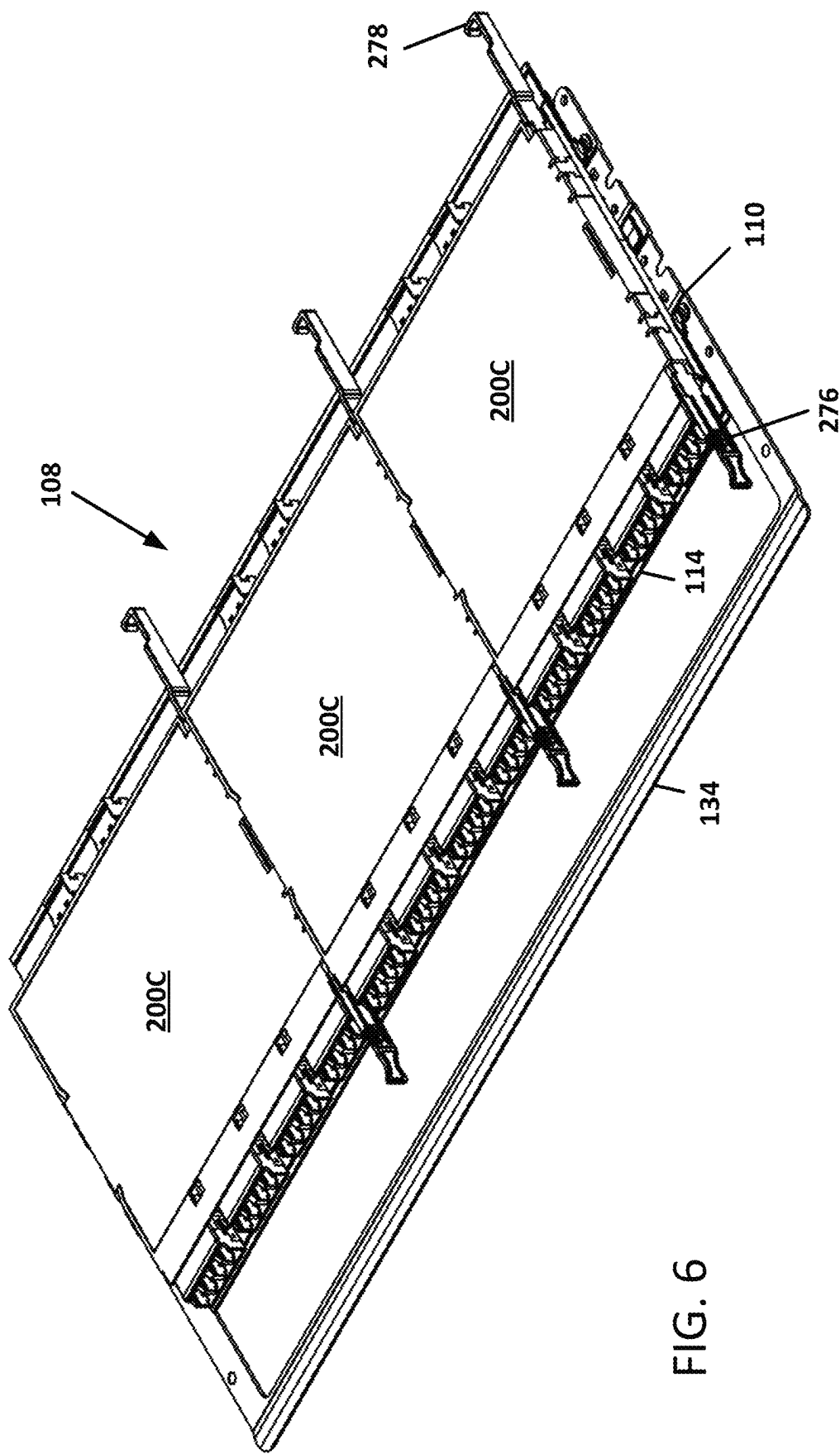


FIG. 6

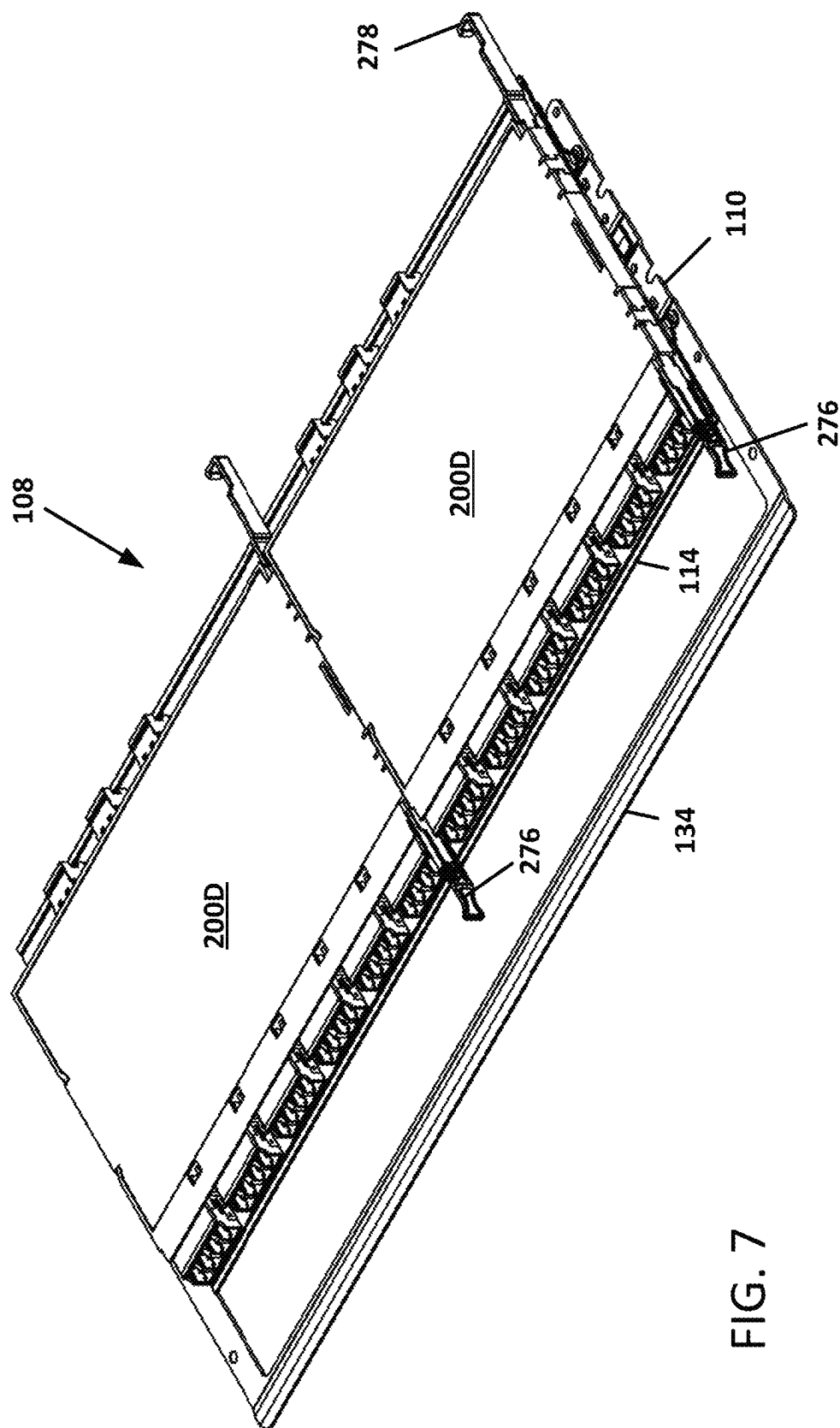
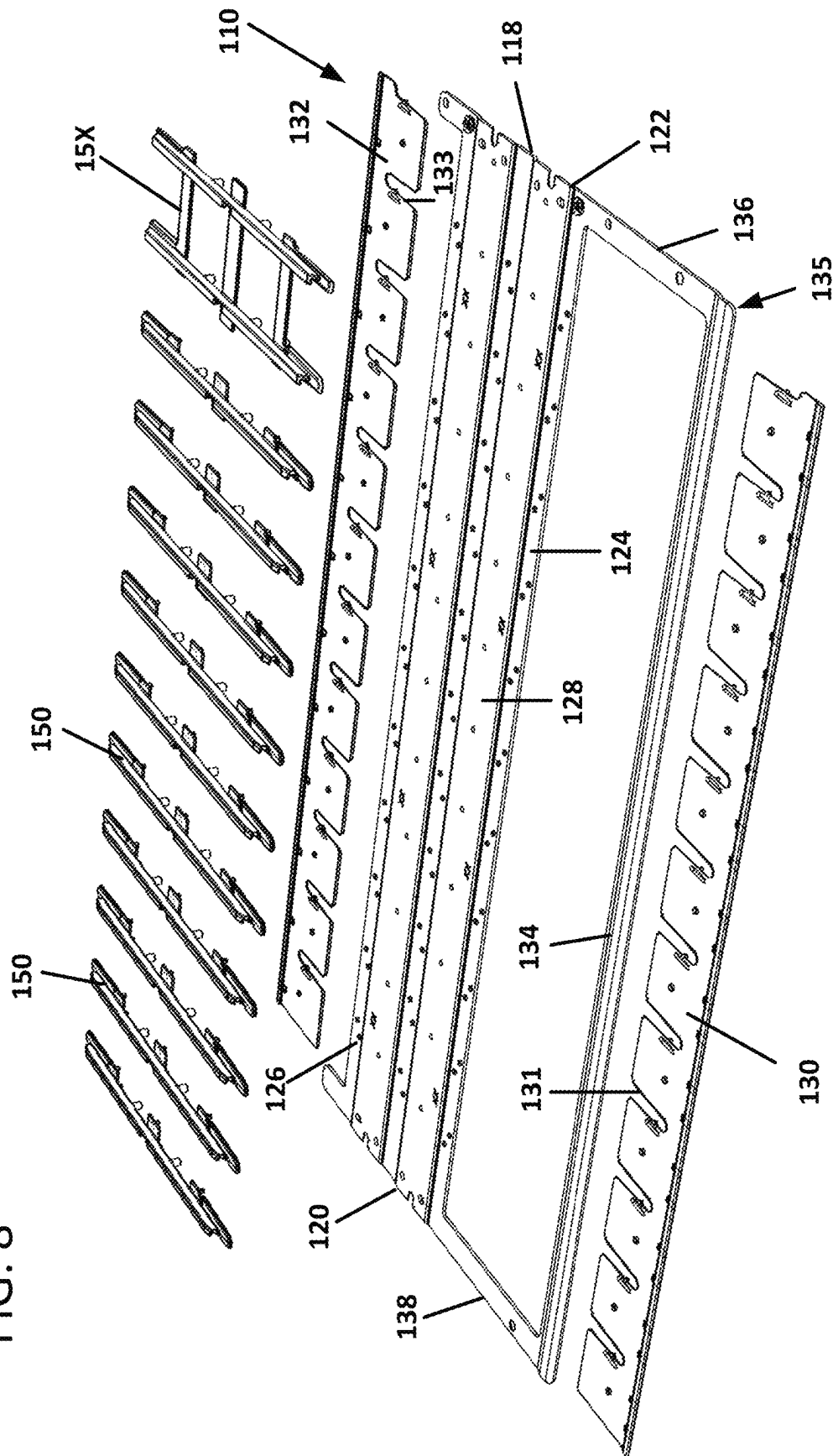
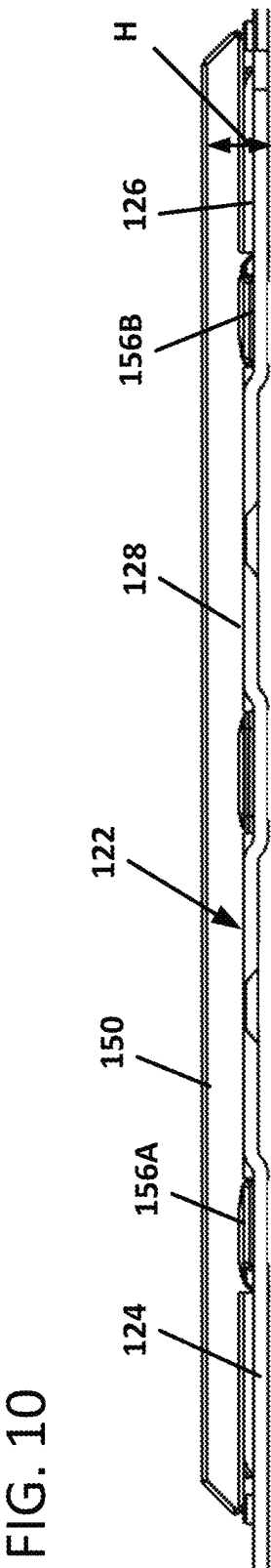
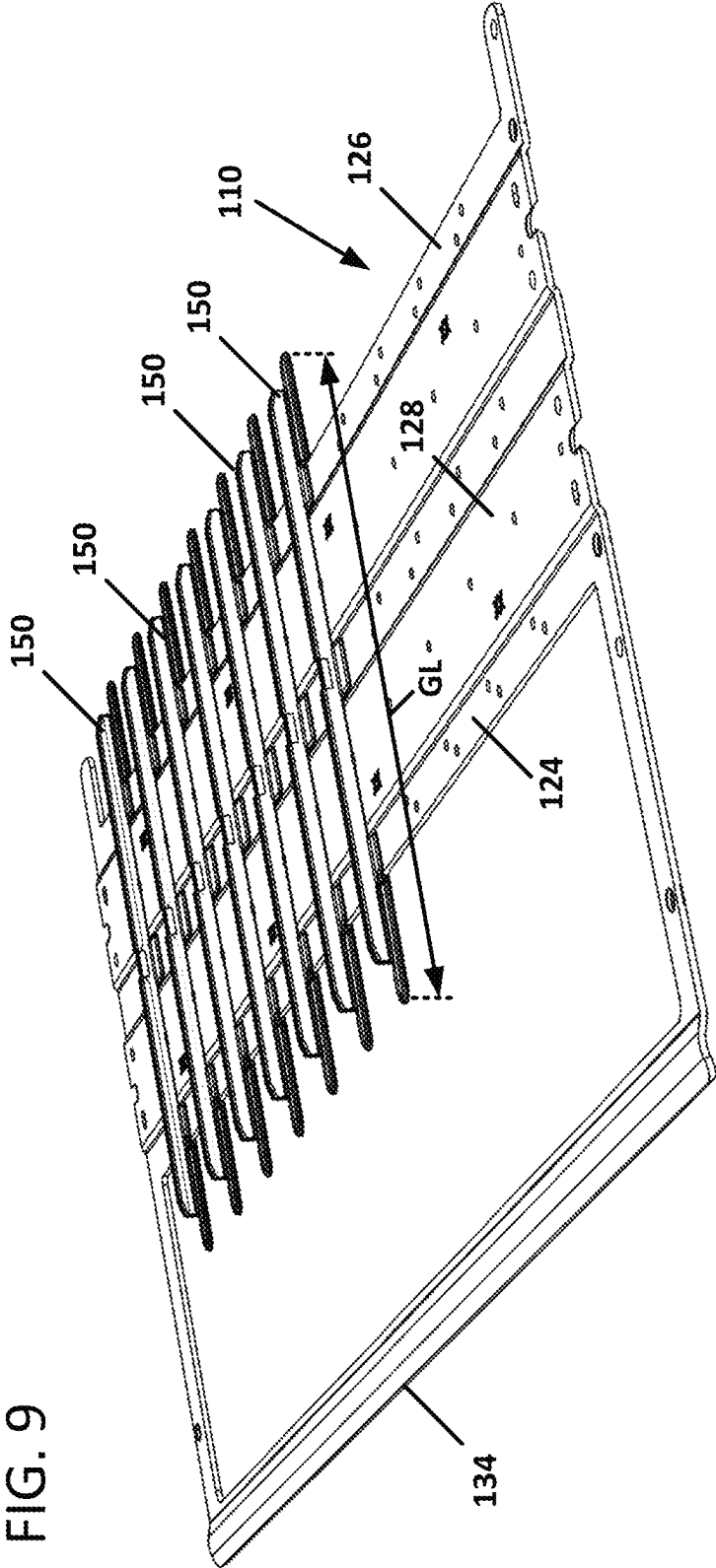
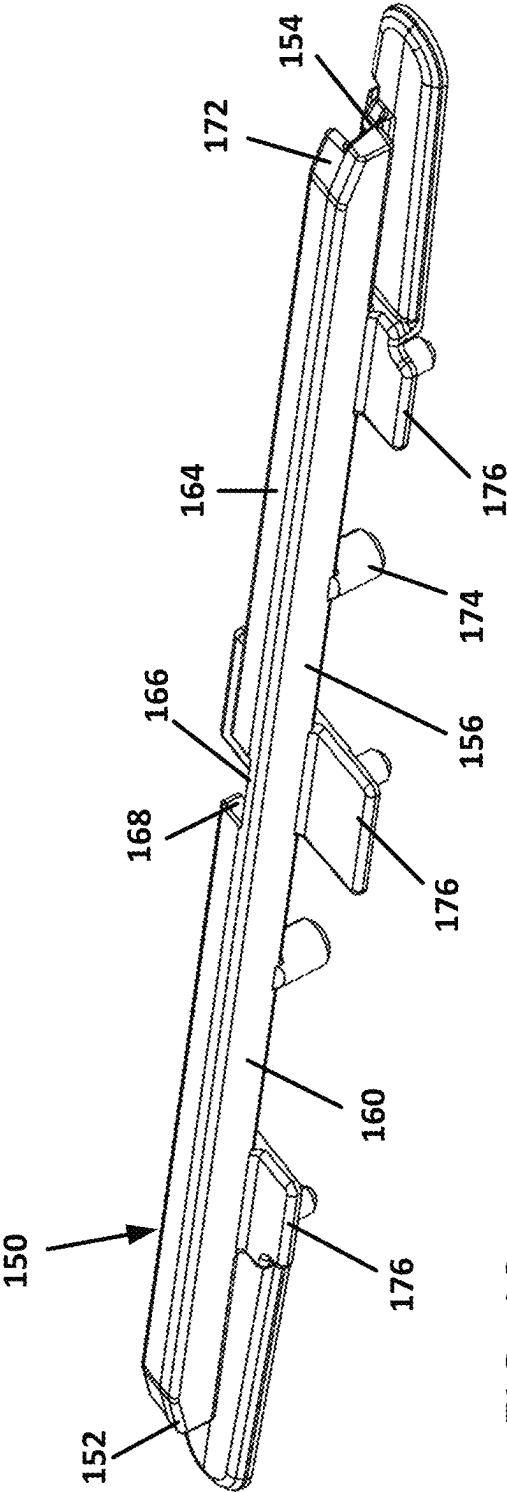
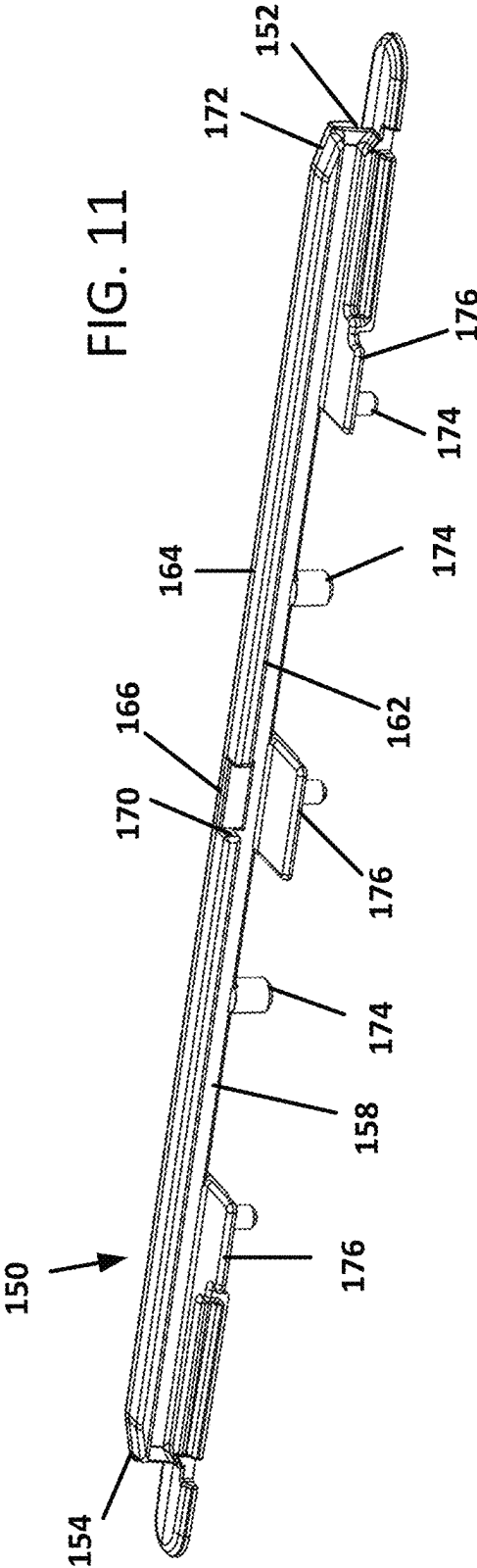


FIG. 7

8
G.
F.







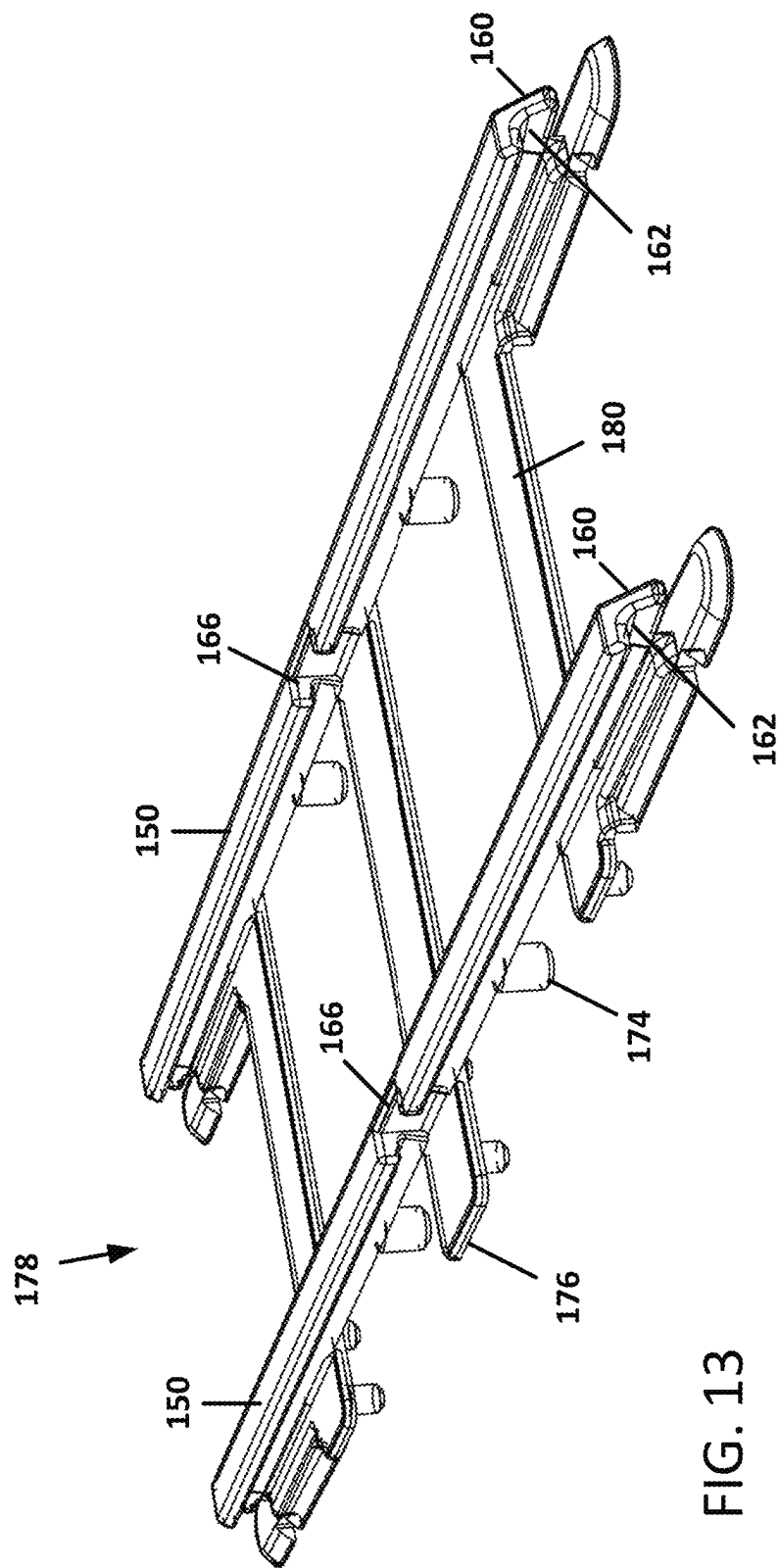
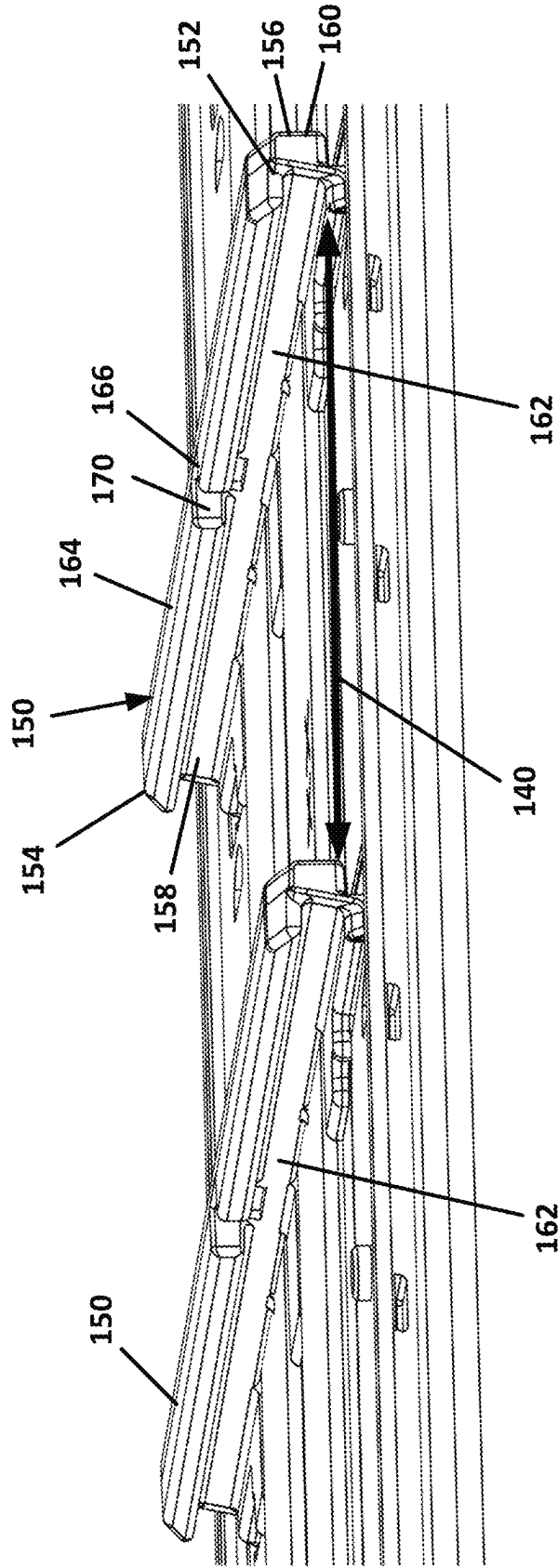


FIG. 14



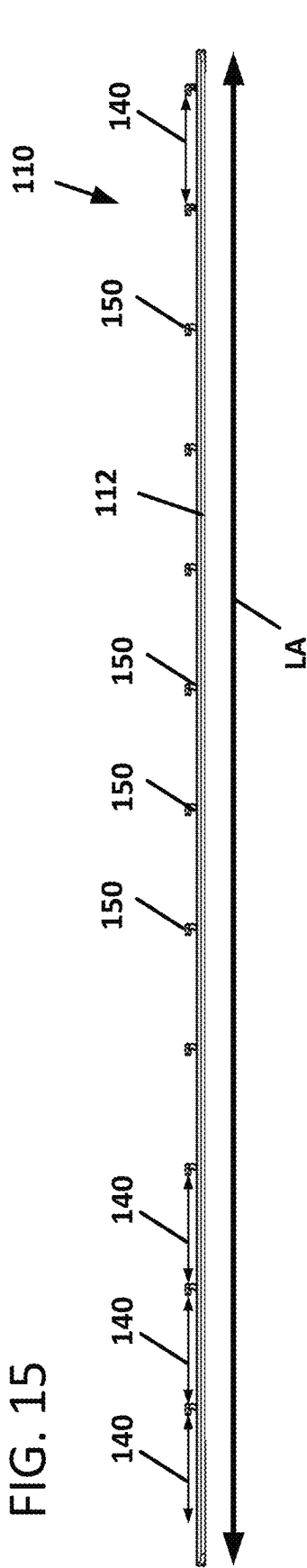


FIG. 16

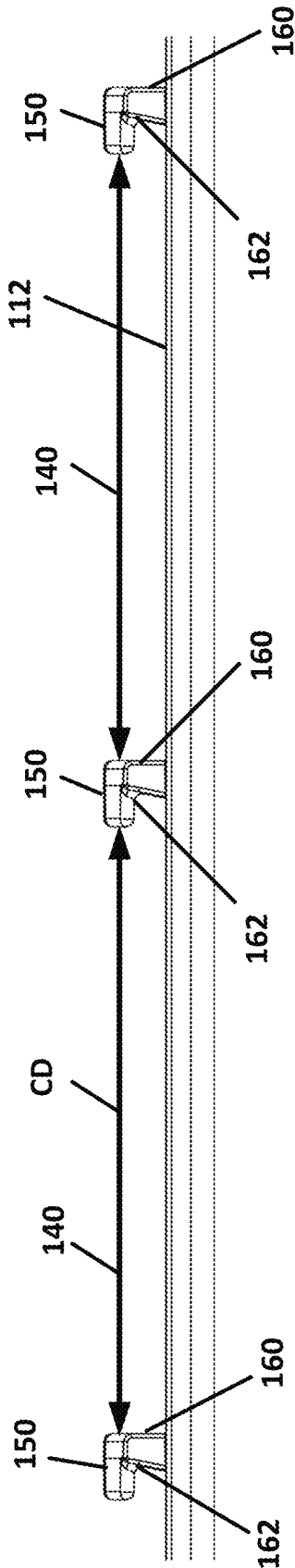


FIG. 17

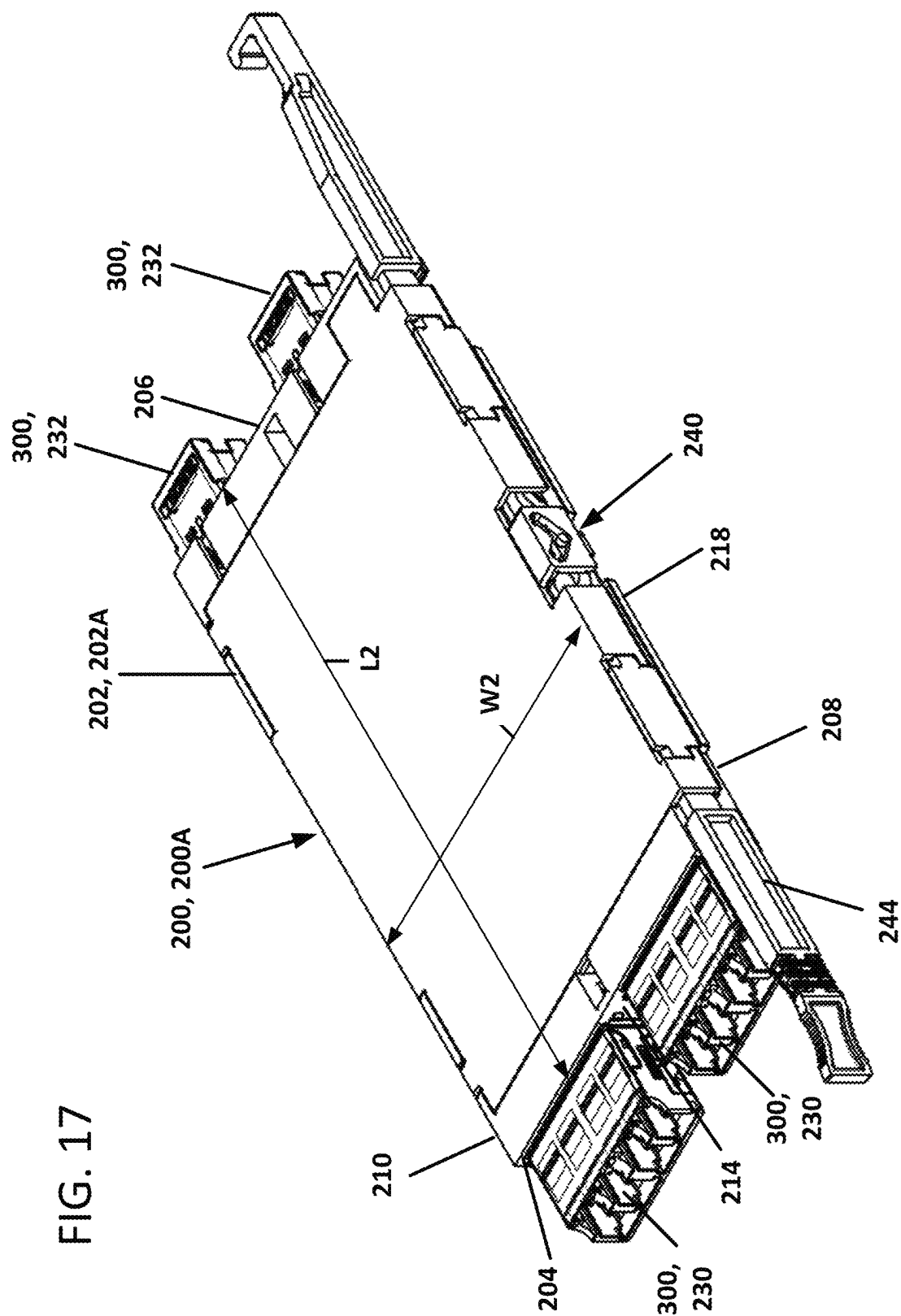
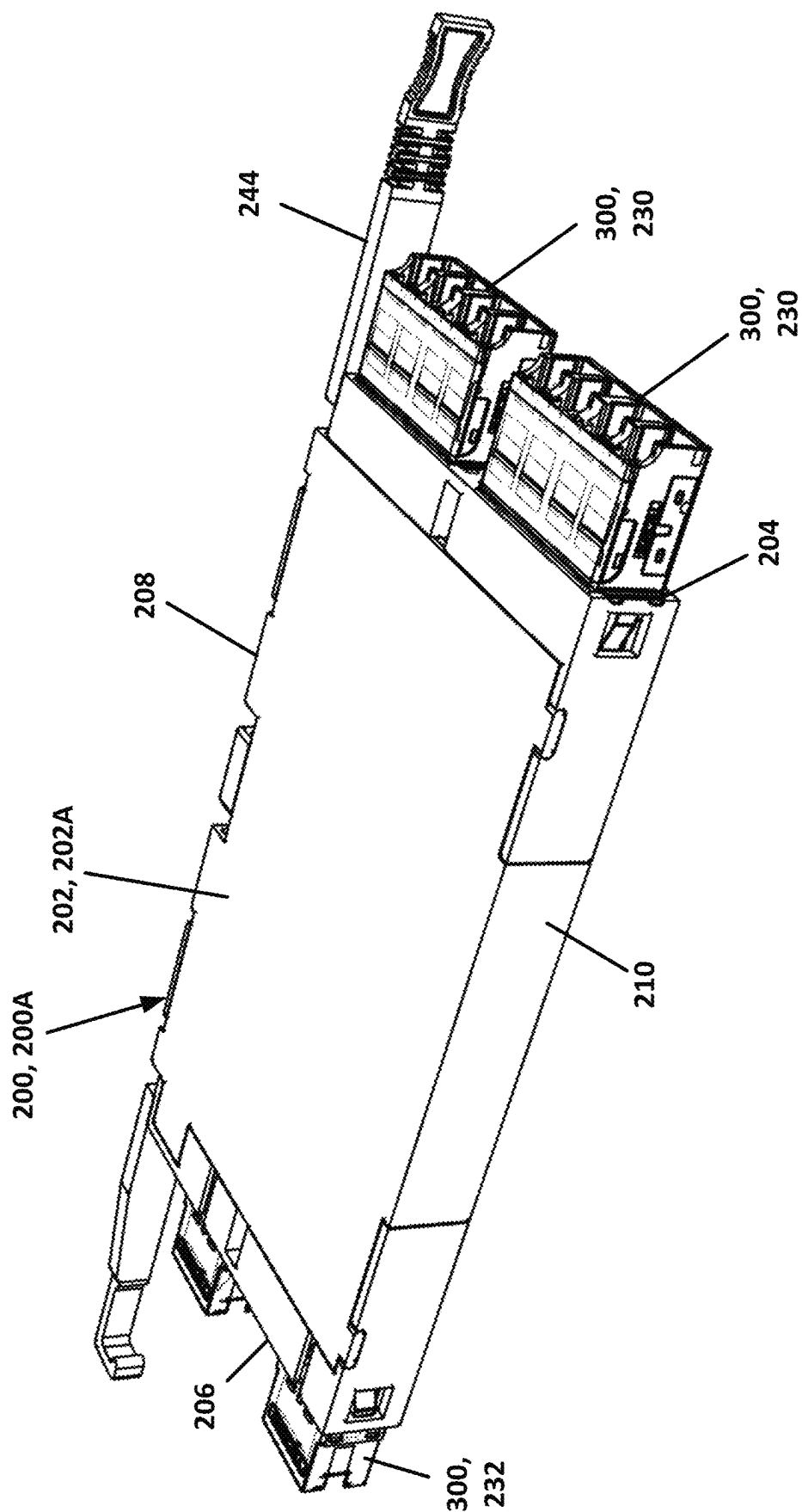
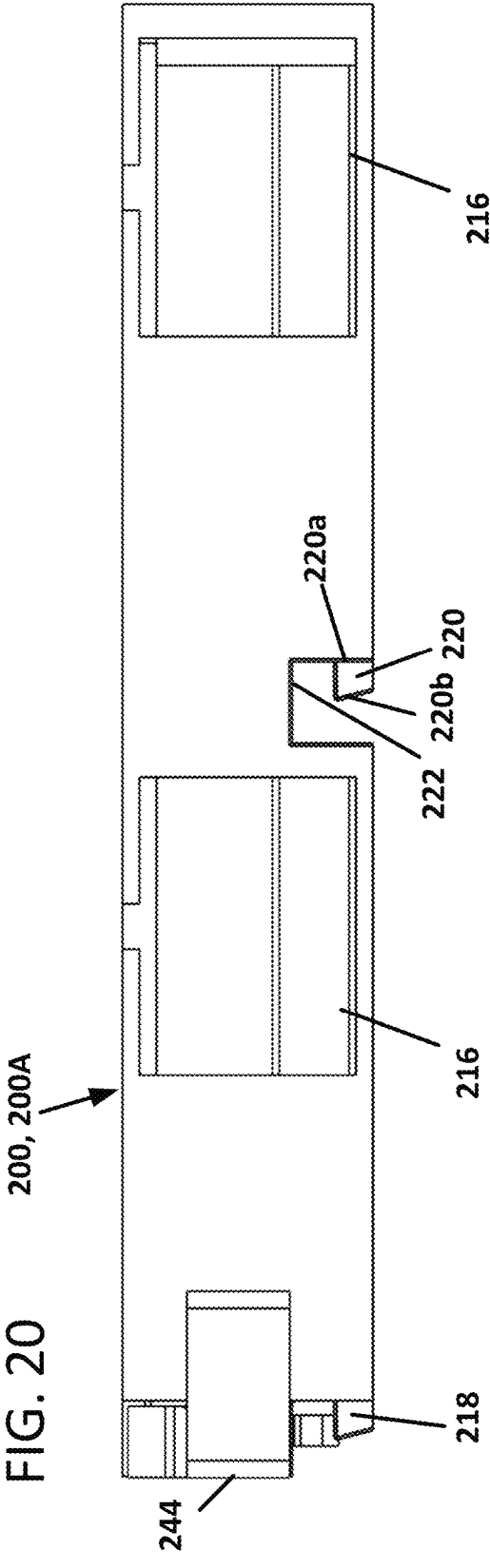
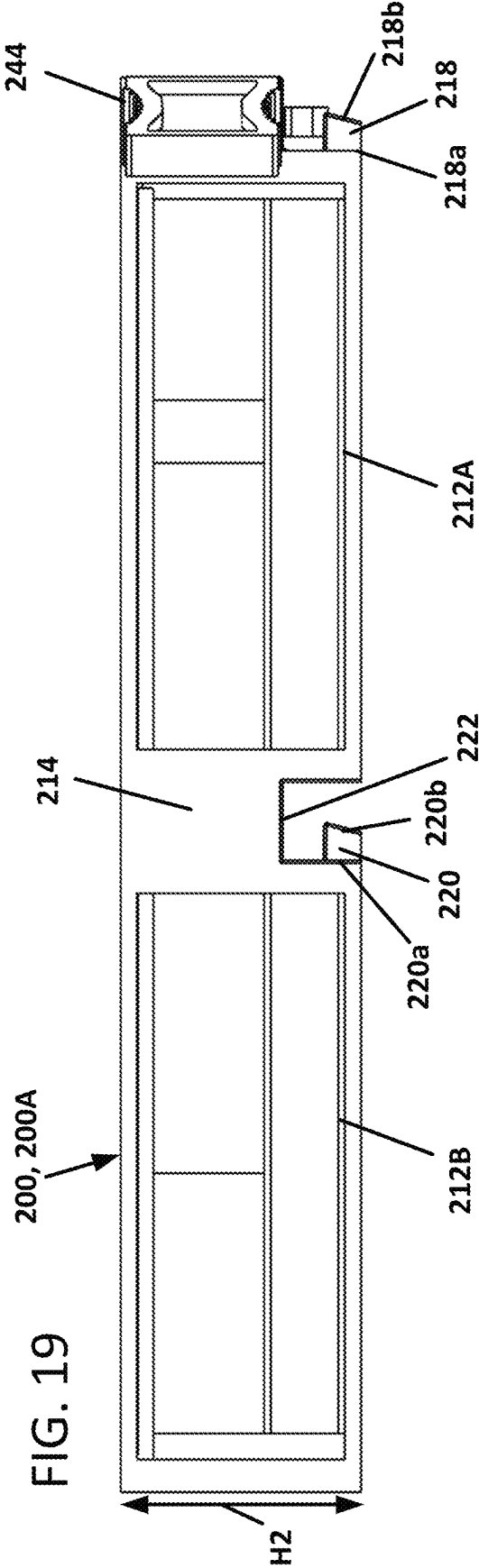
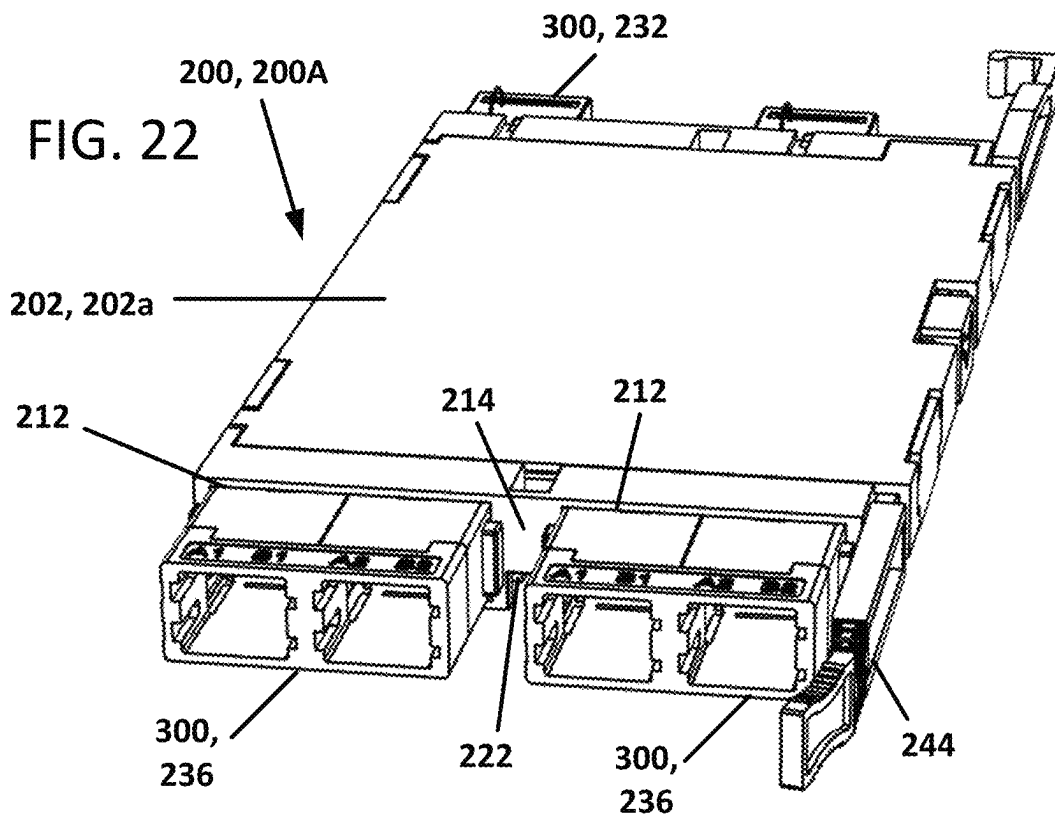
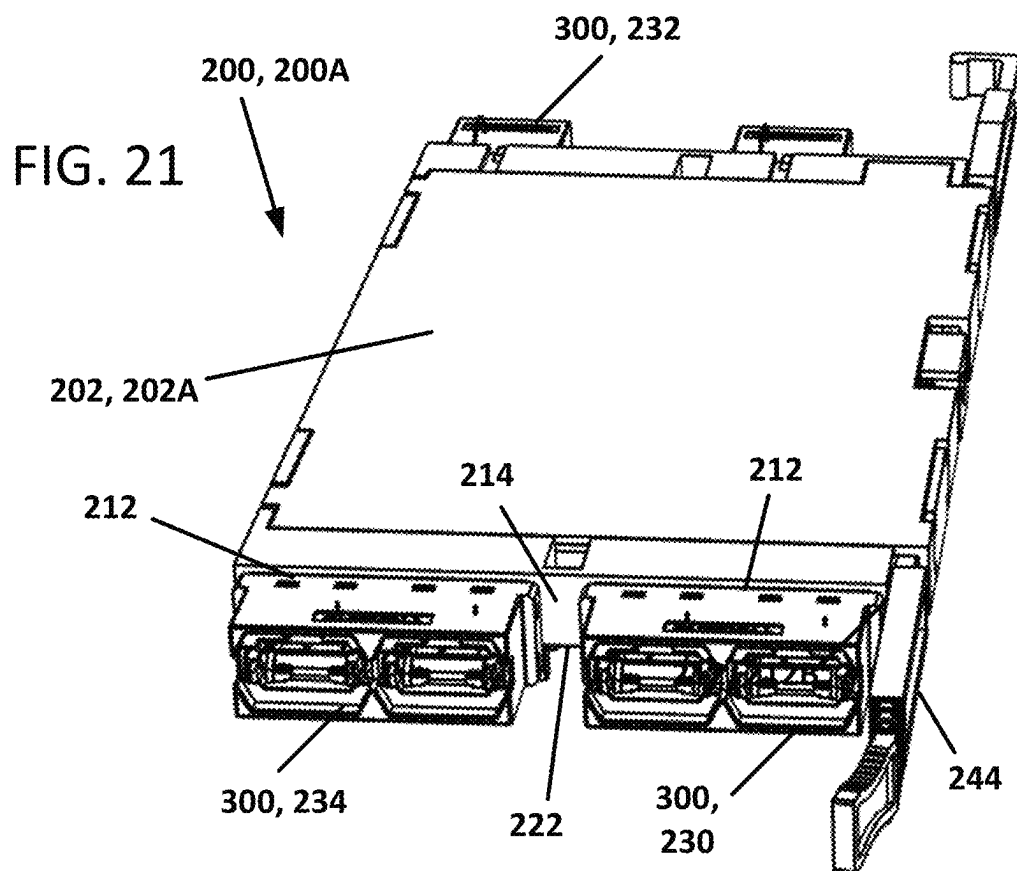
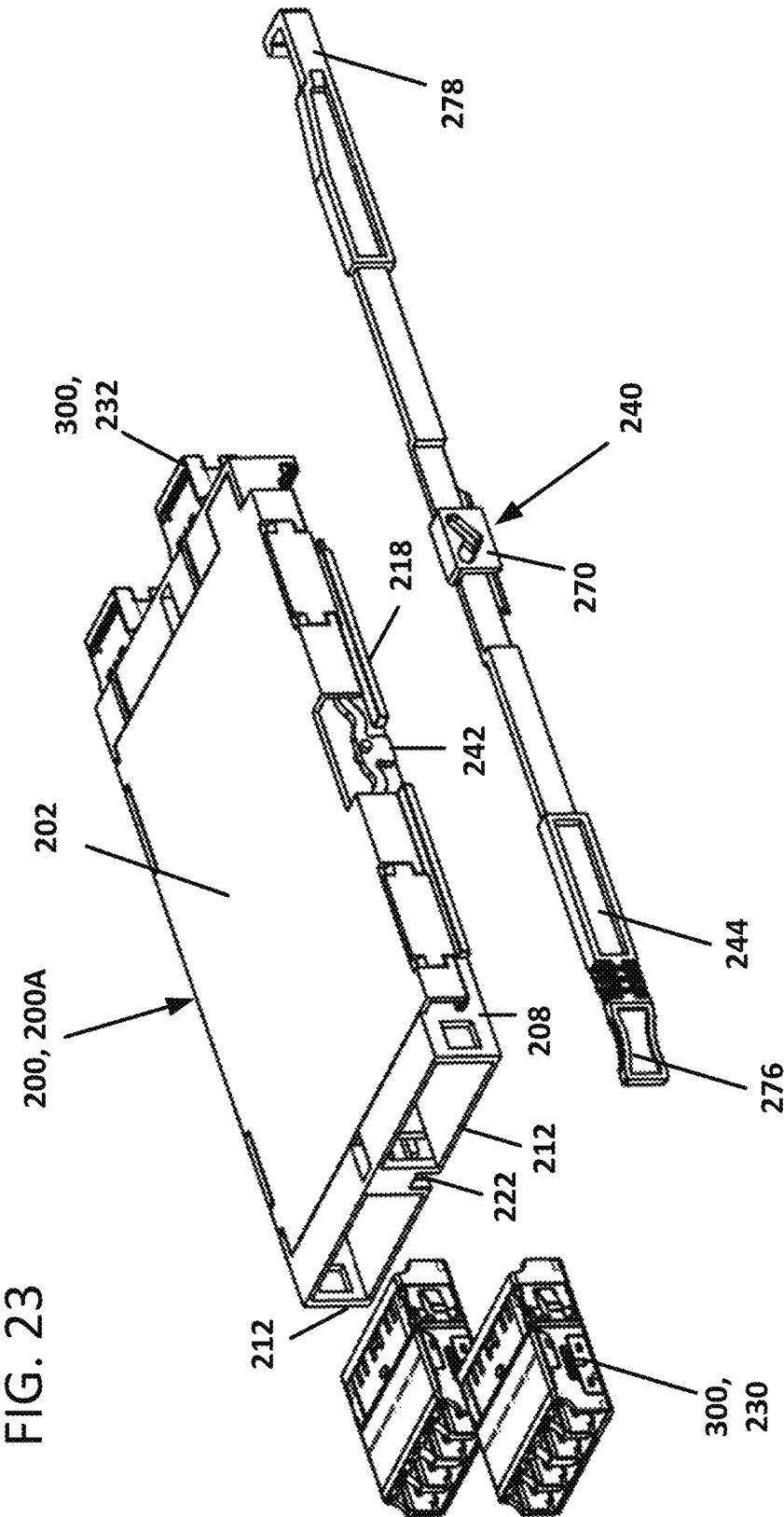


FIG. 18









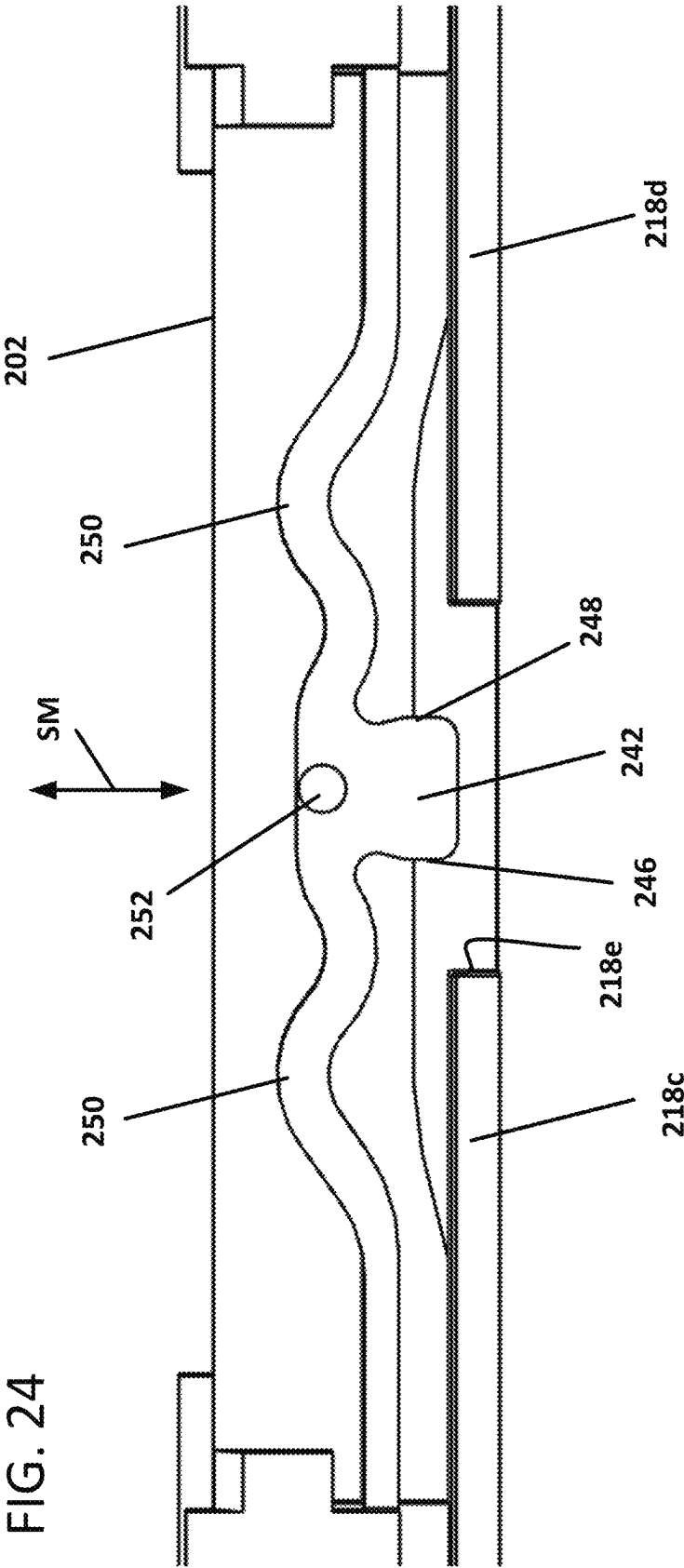
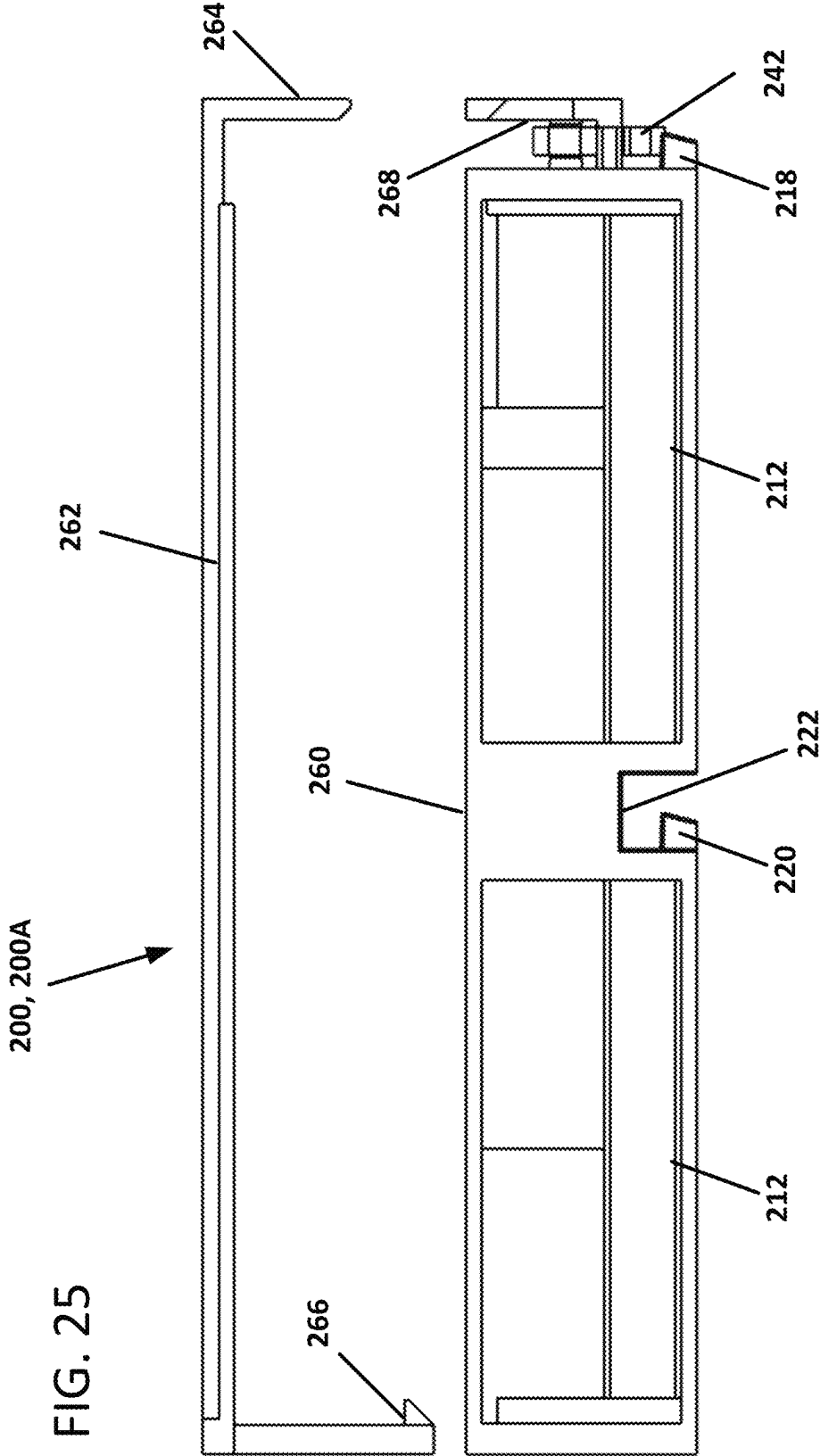


FIG. 24



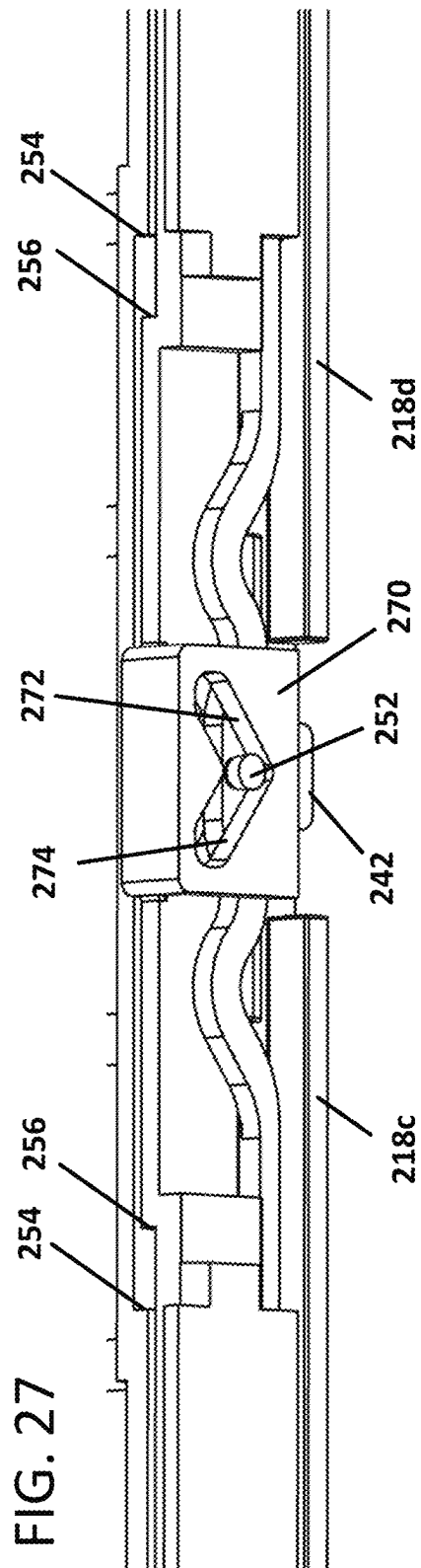
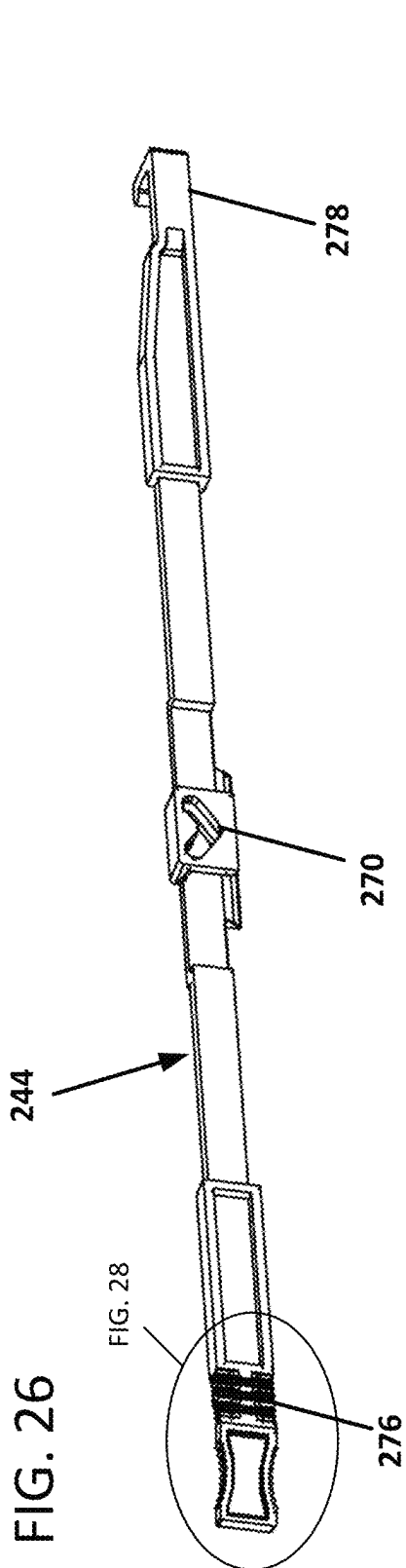


FIG. 28

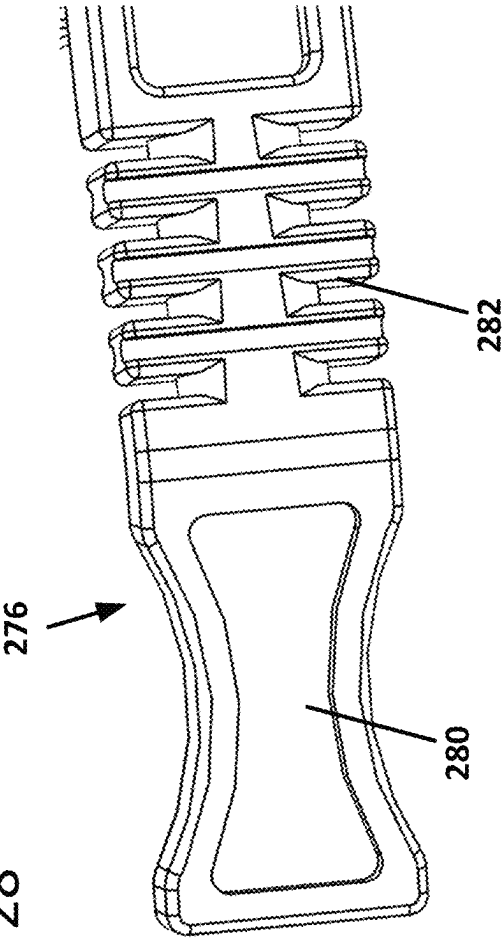


FIG. 29

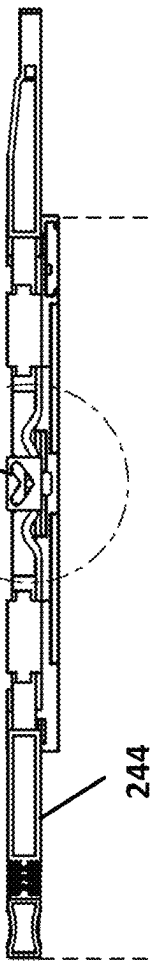


FIG. 30

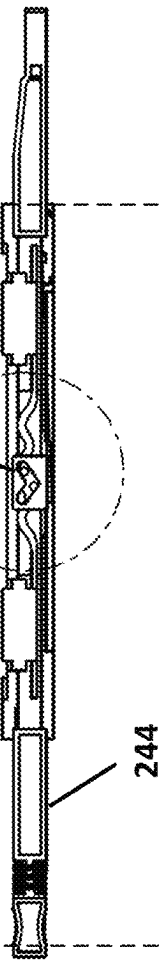


FIG. 31

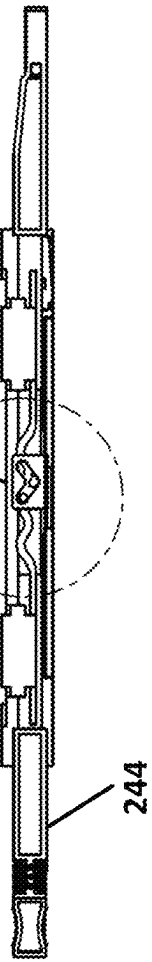


FIG. 29A

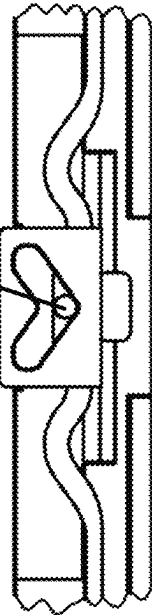


FIG. 30A

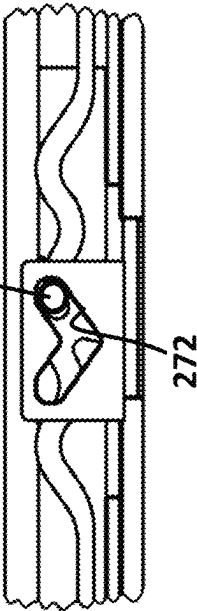
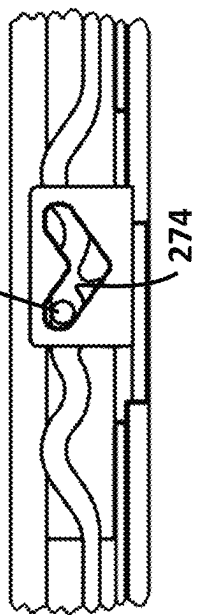


FIG. 31A



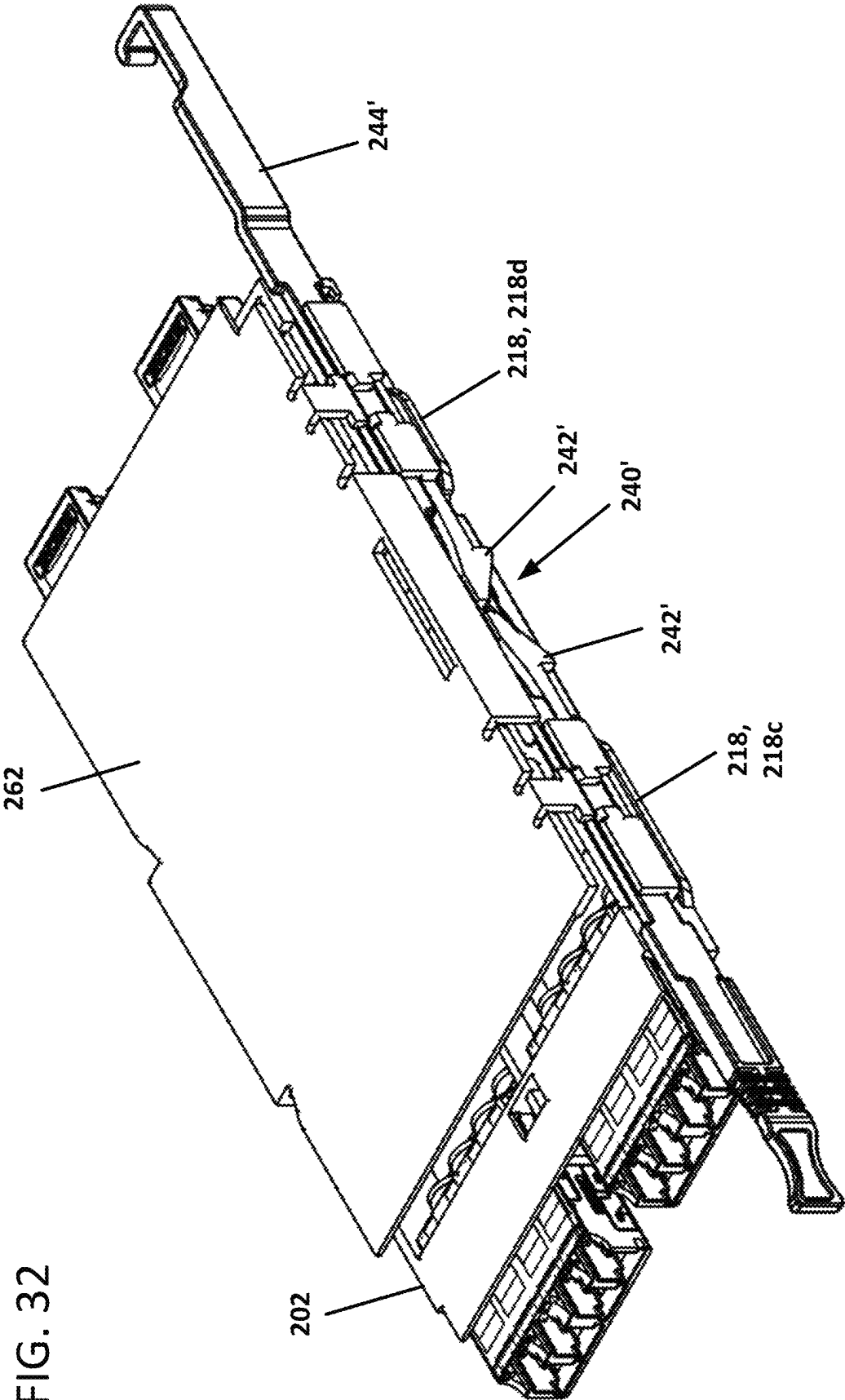


FIG. 33

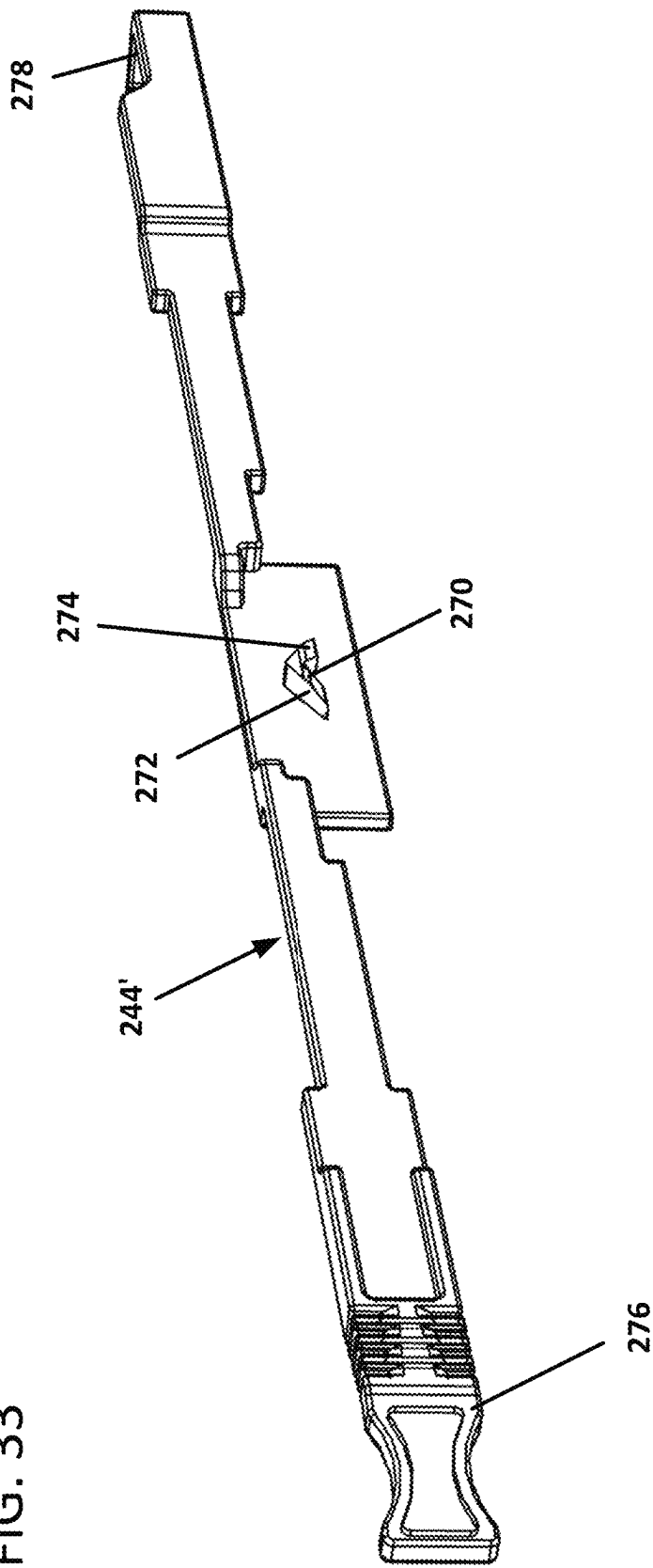


FIG. 34

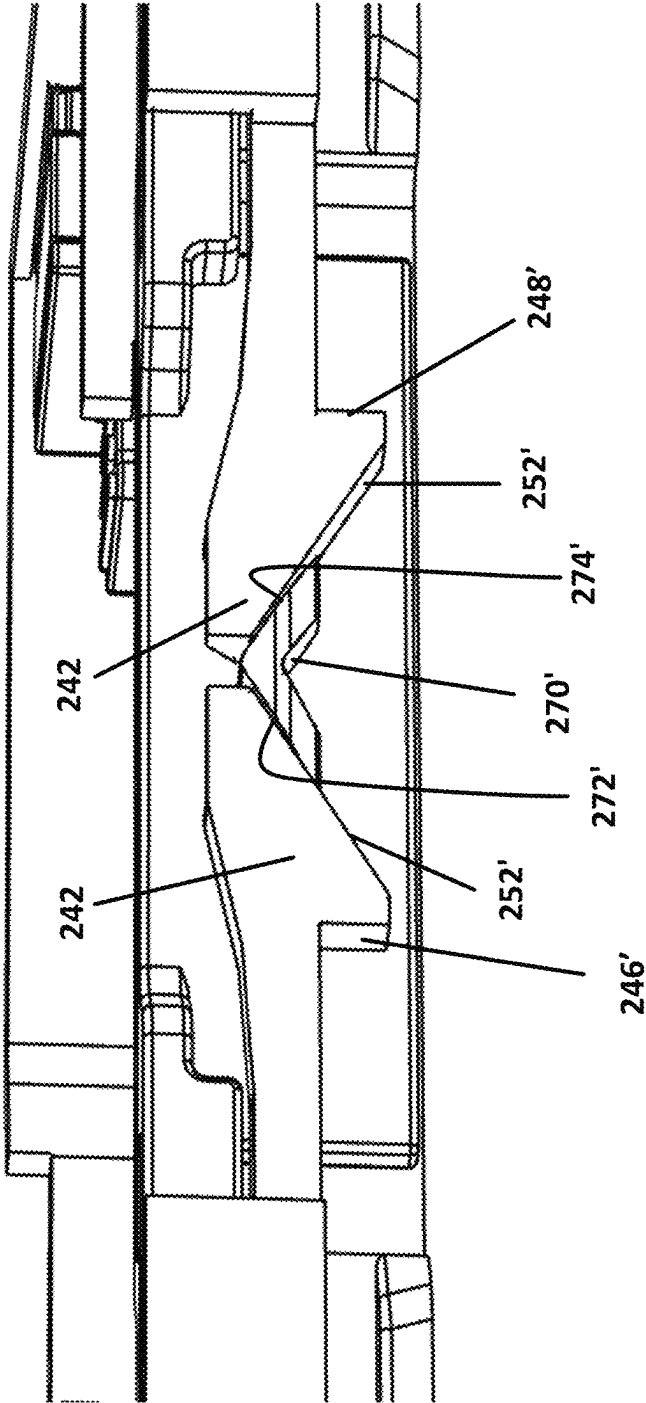
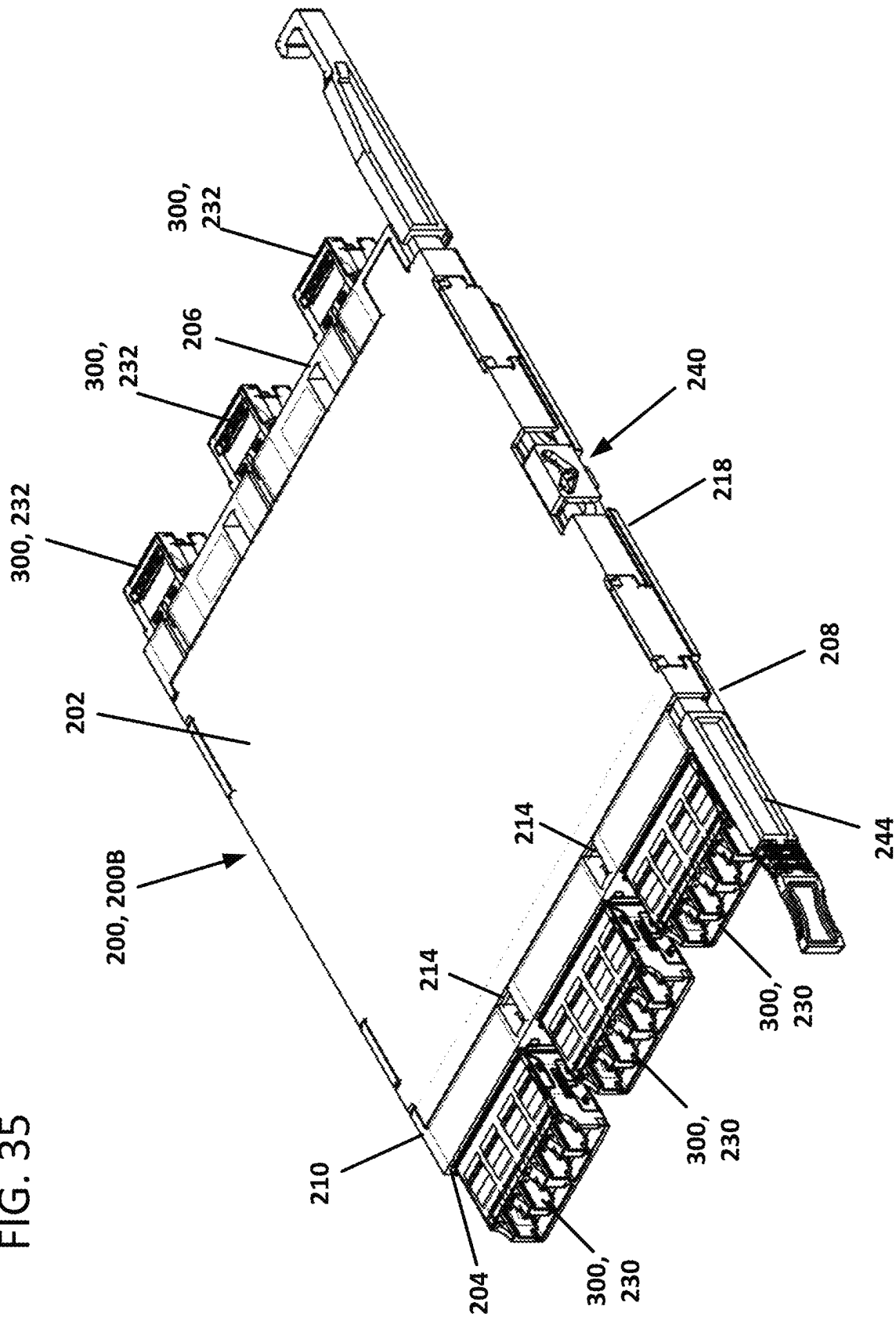
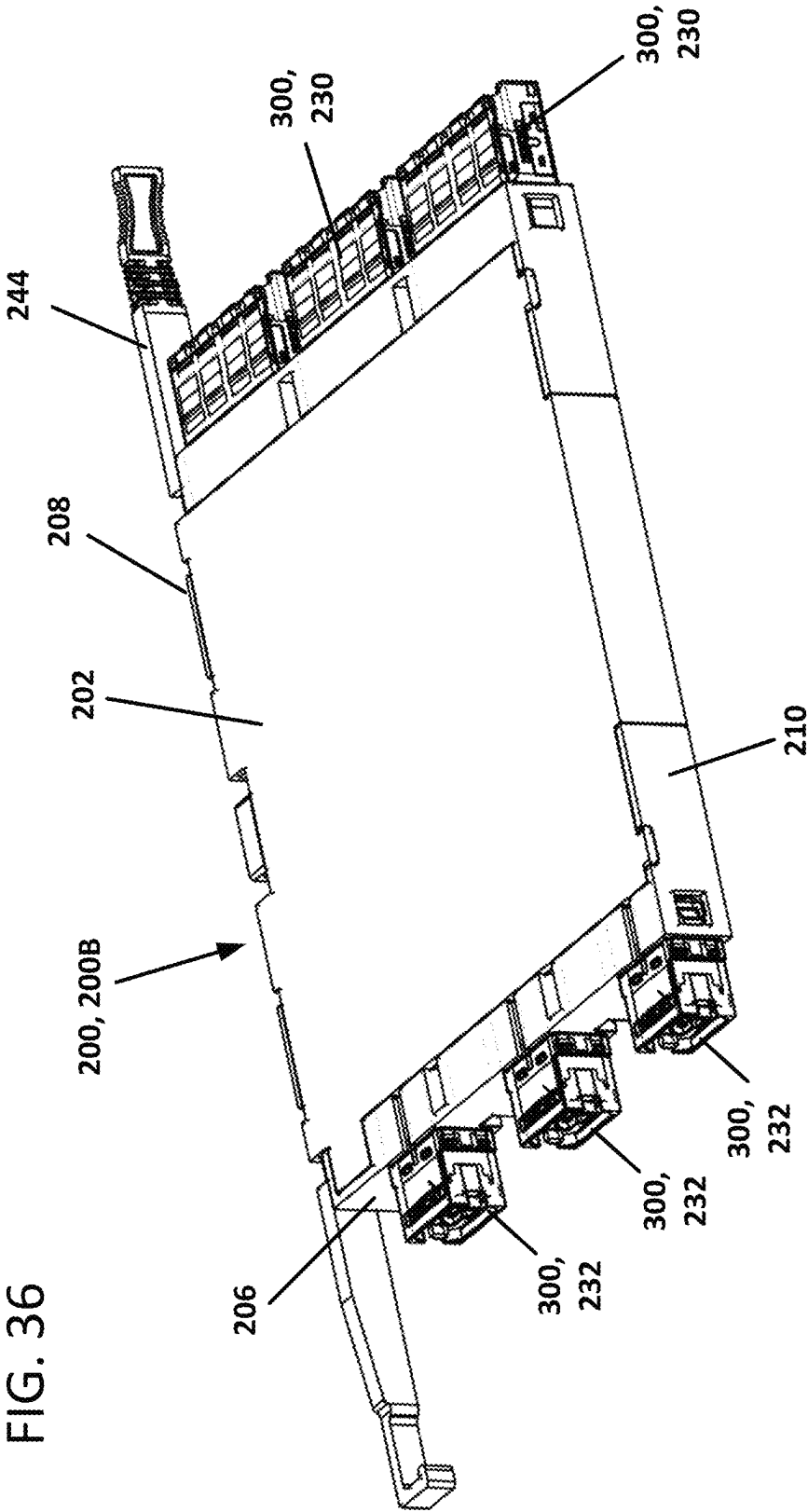
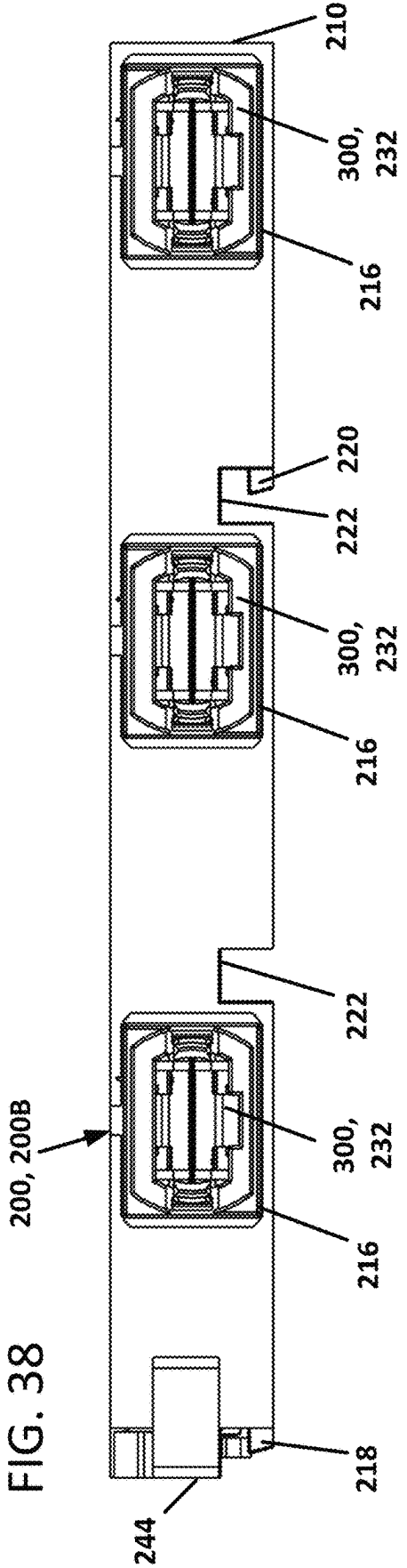
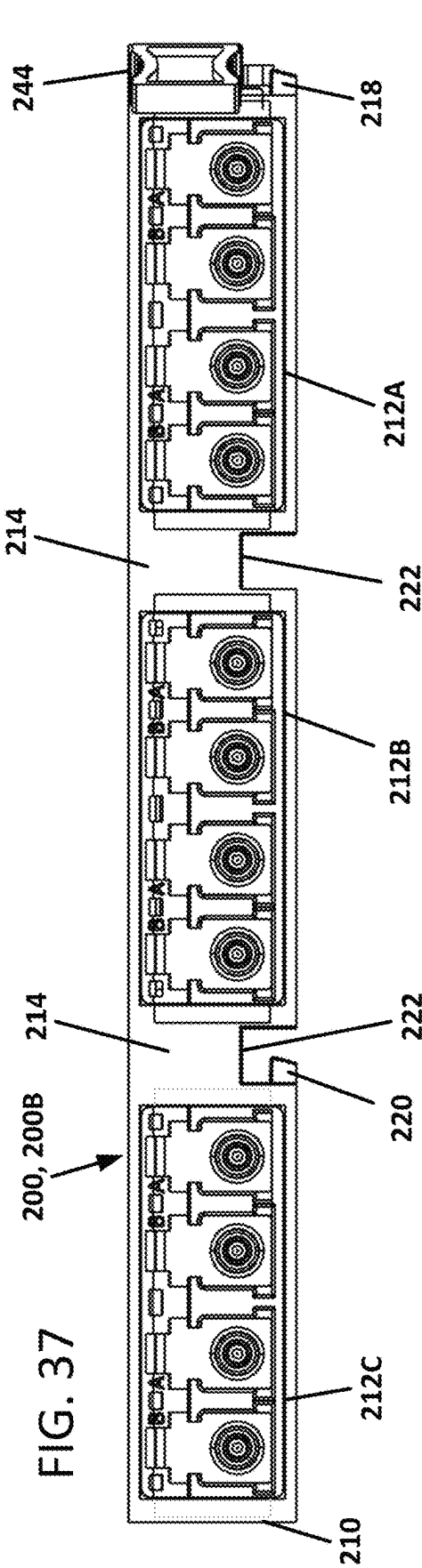
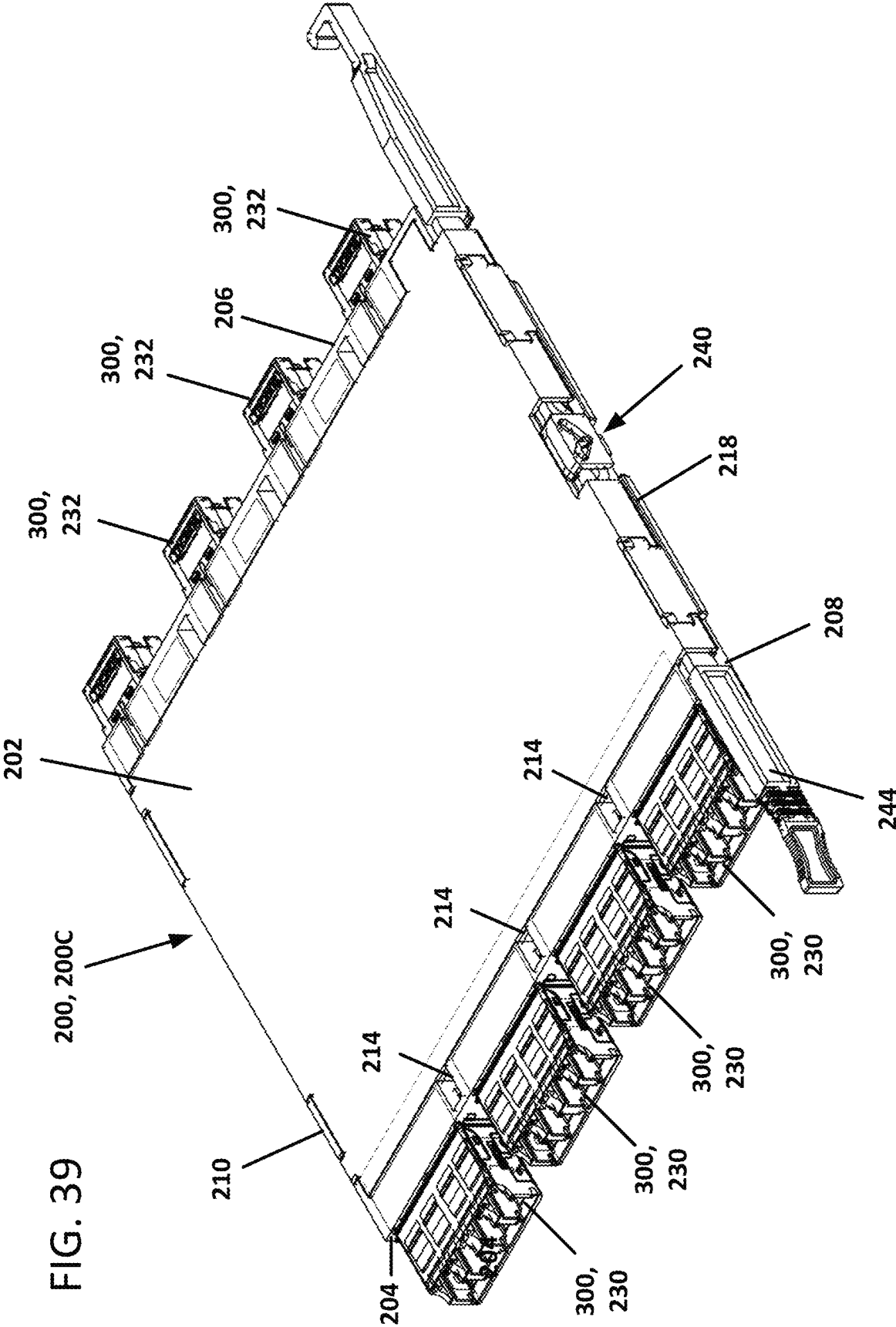


FIG. 35









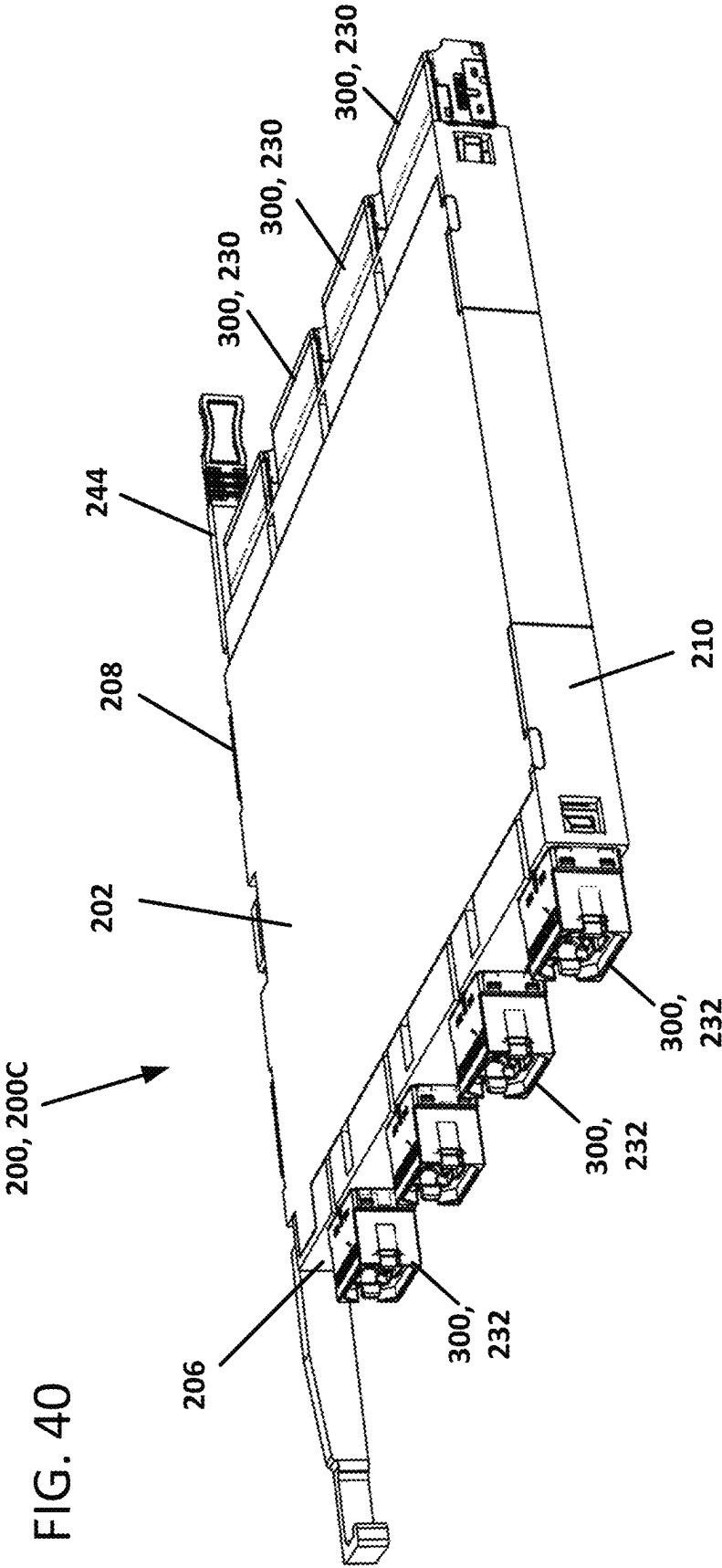


FIG. 41

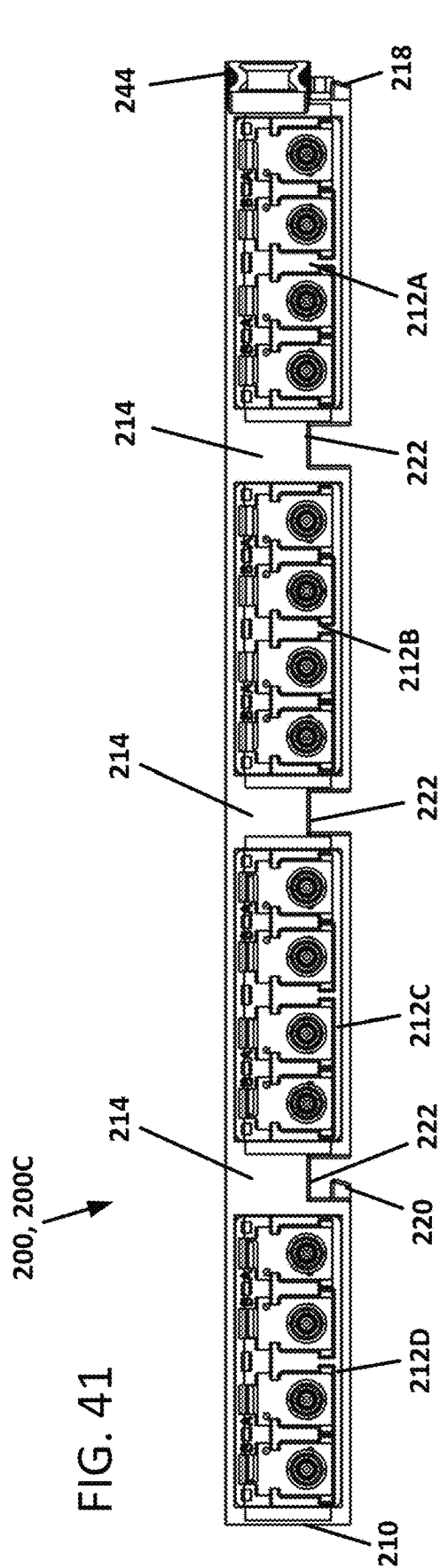


FIG. 42

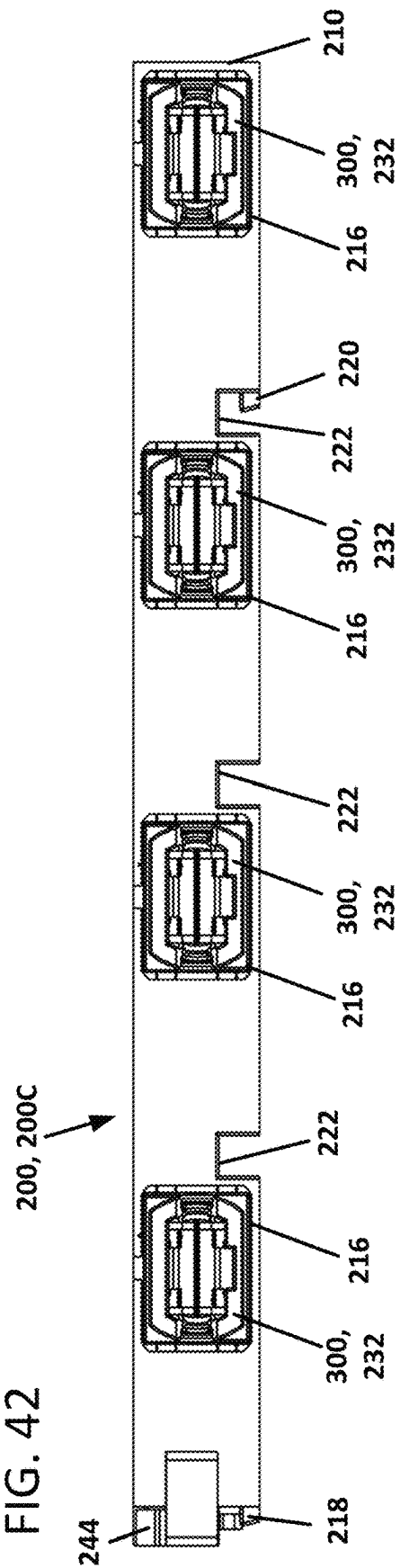
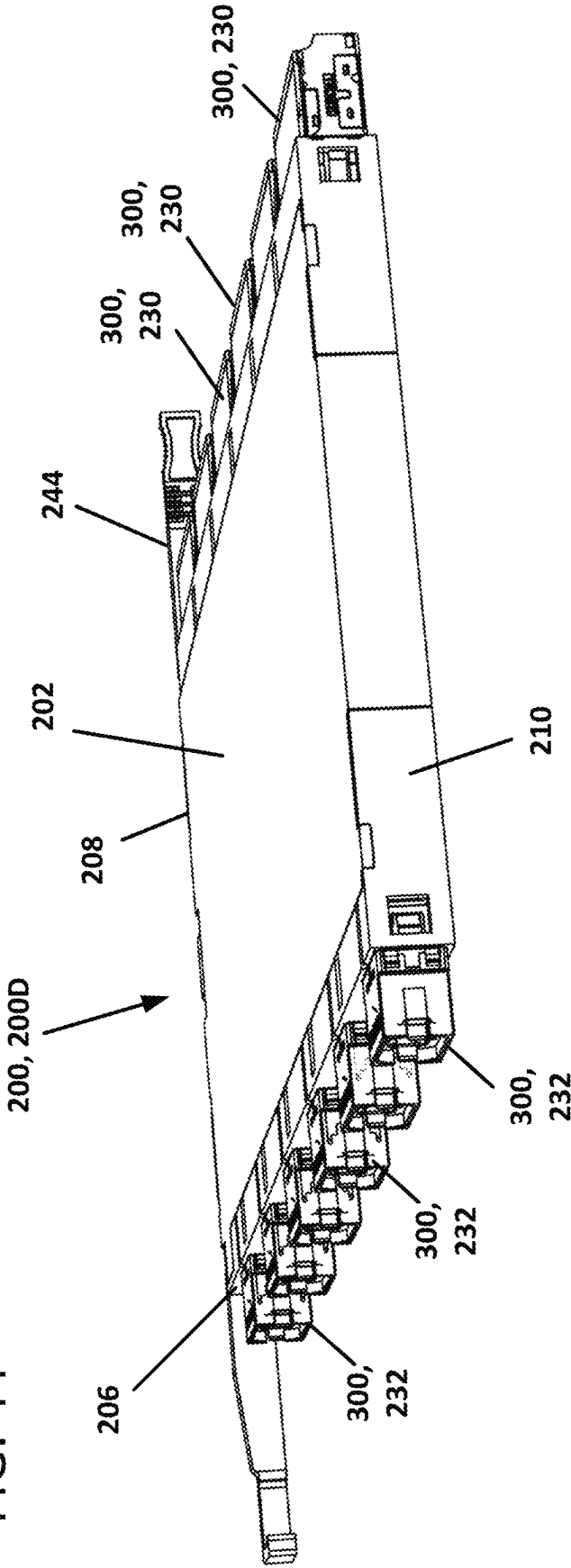
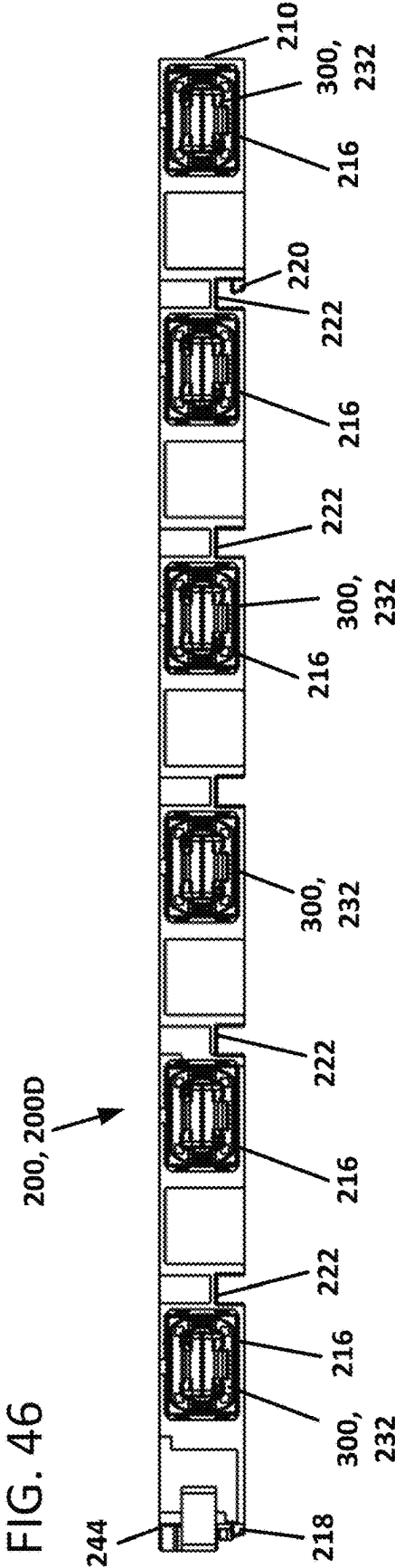
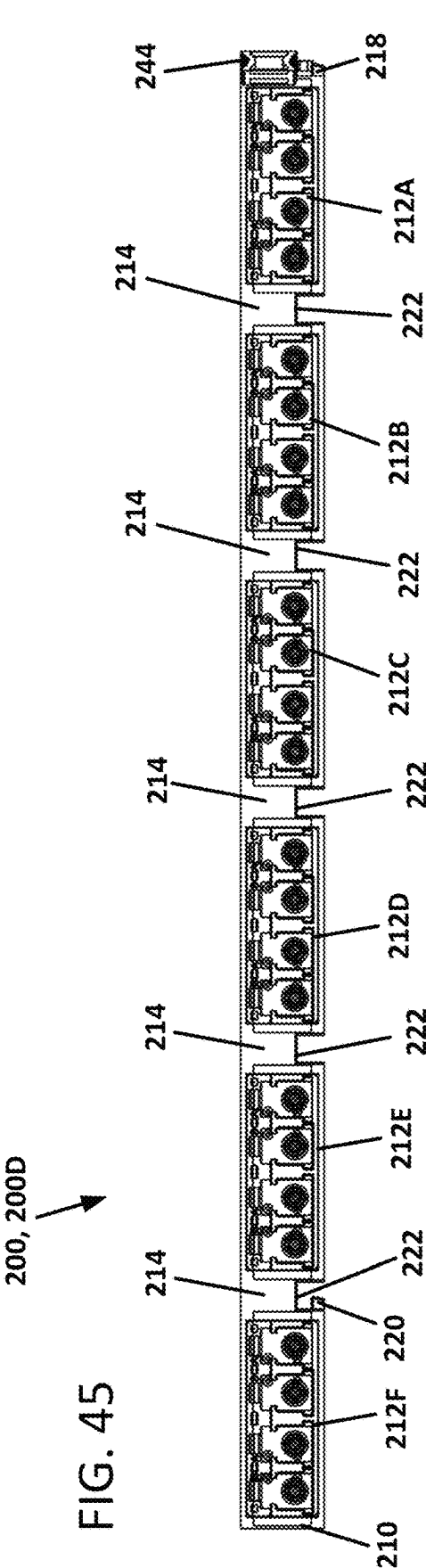


FIG. 44





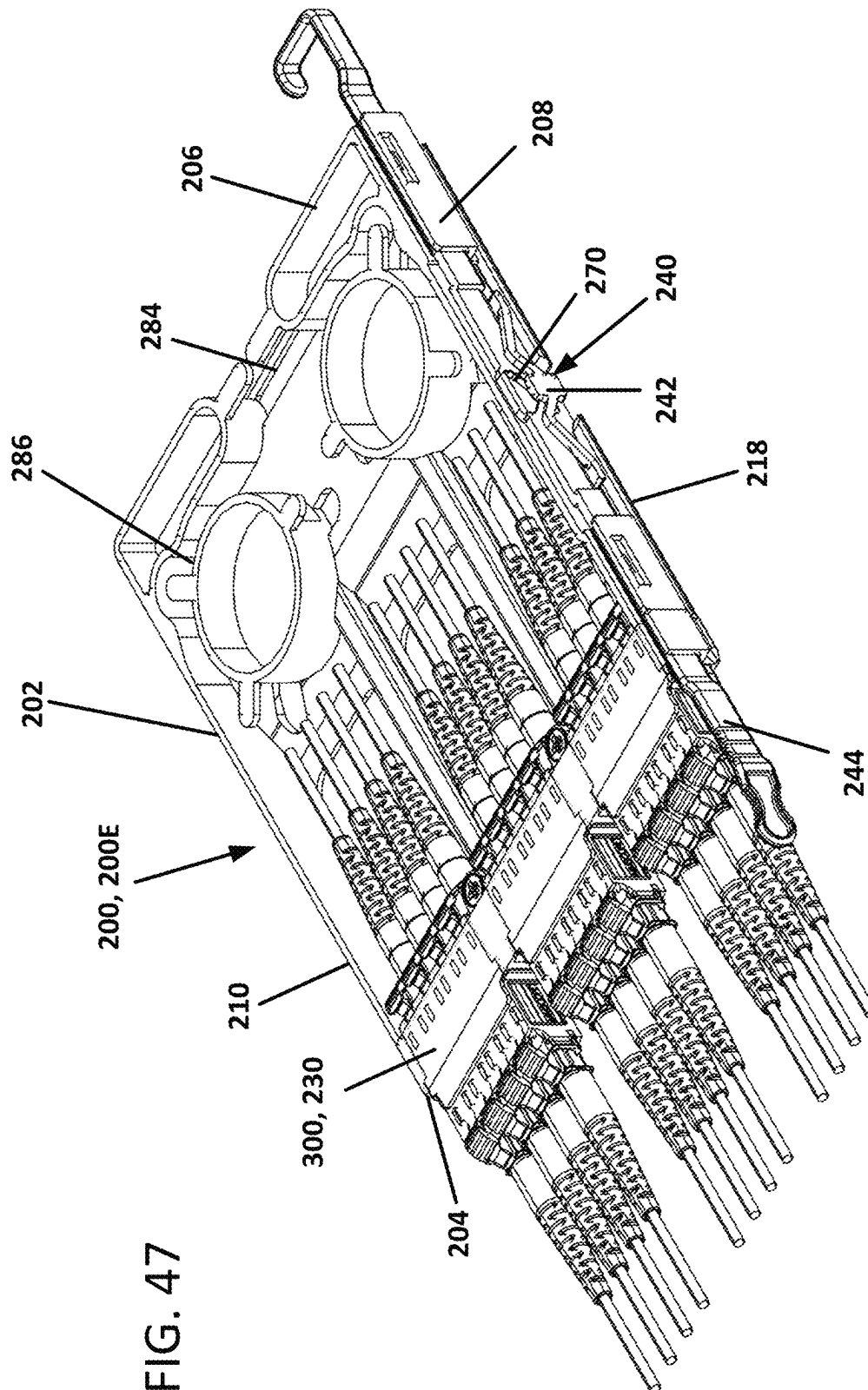
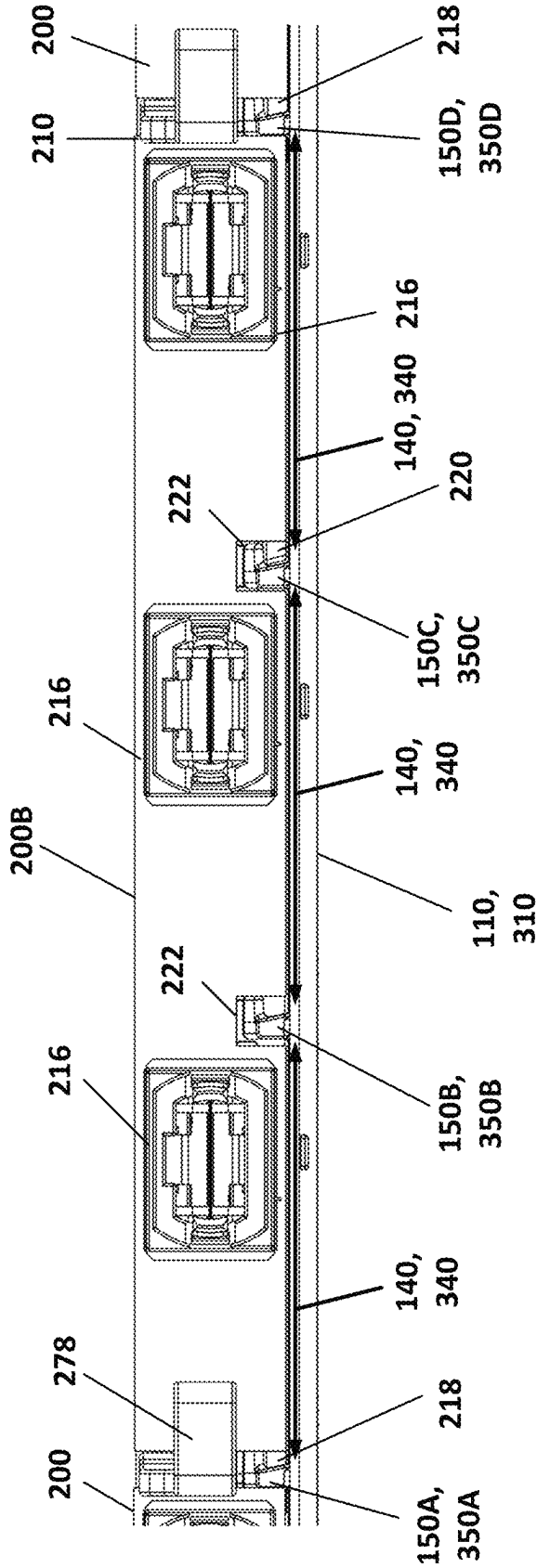


FIG. 48



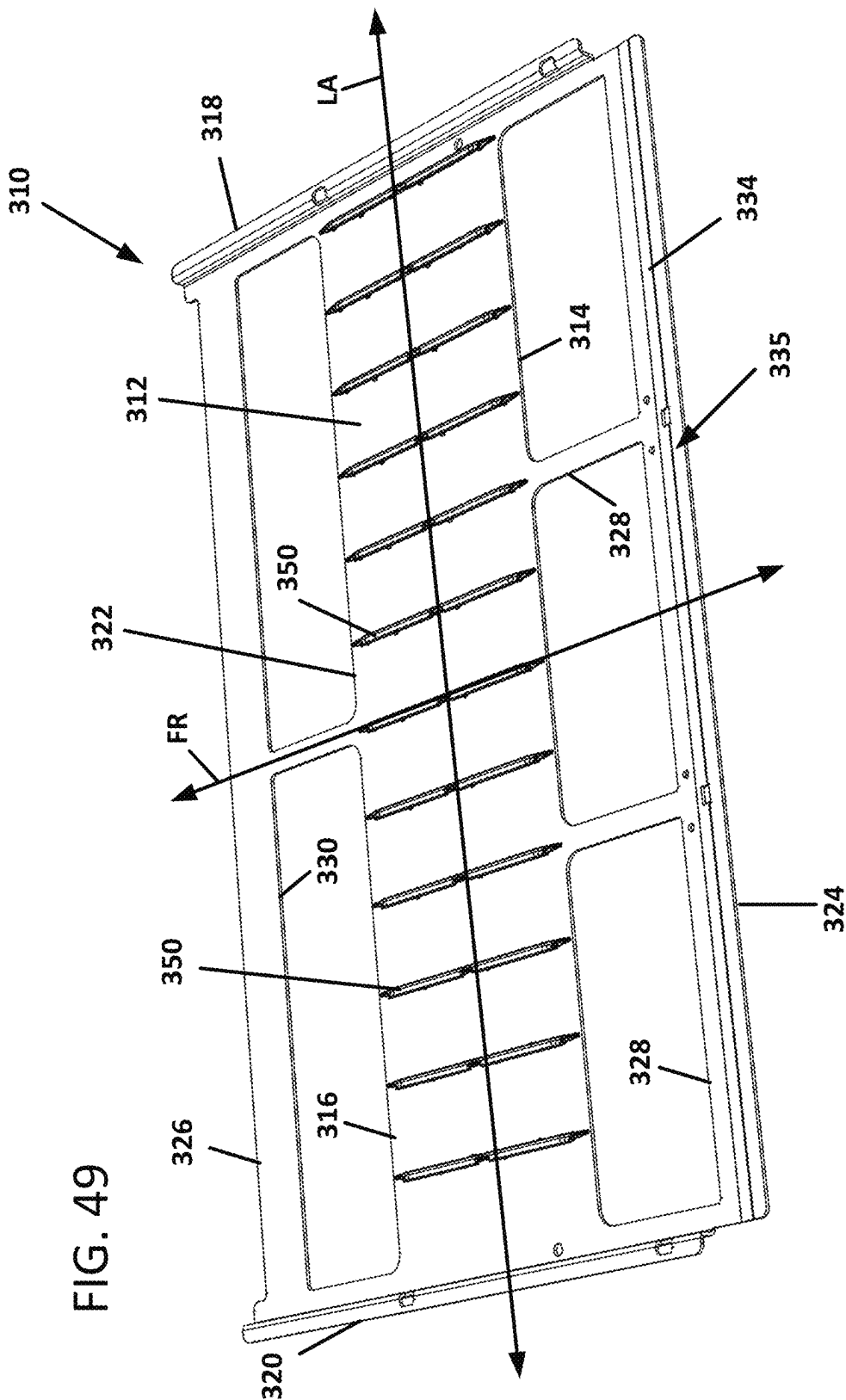


FIG. 50

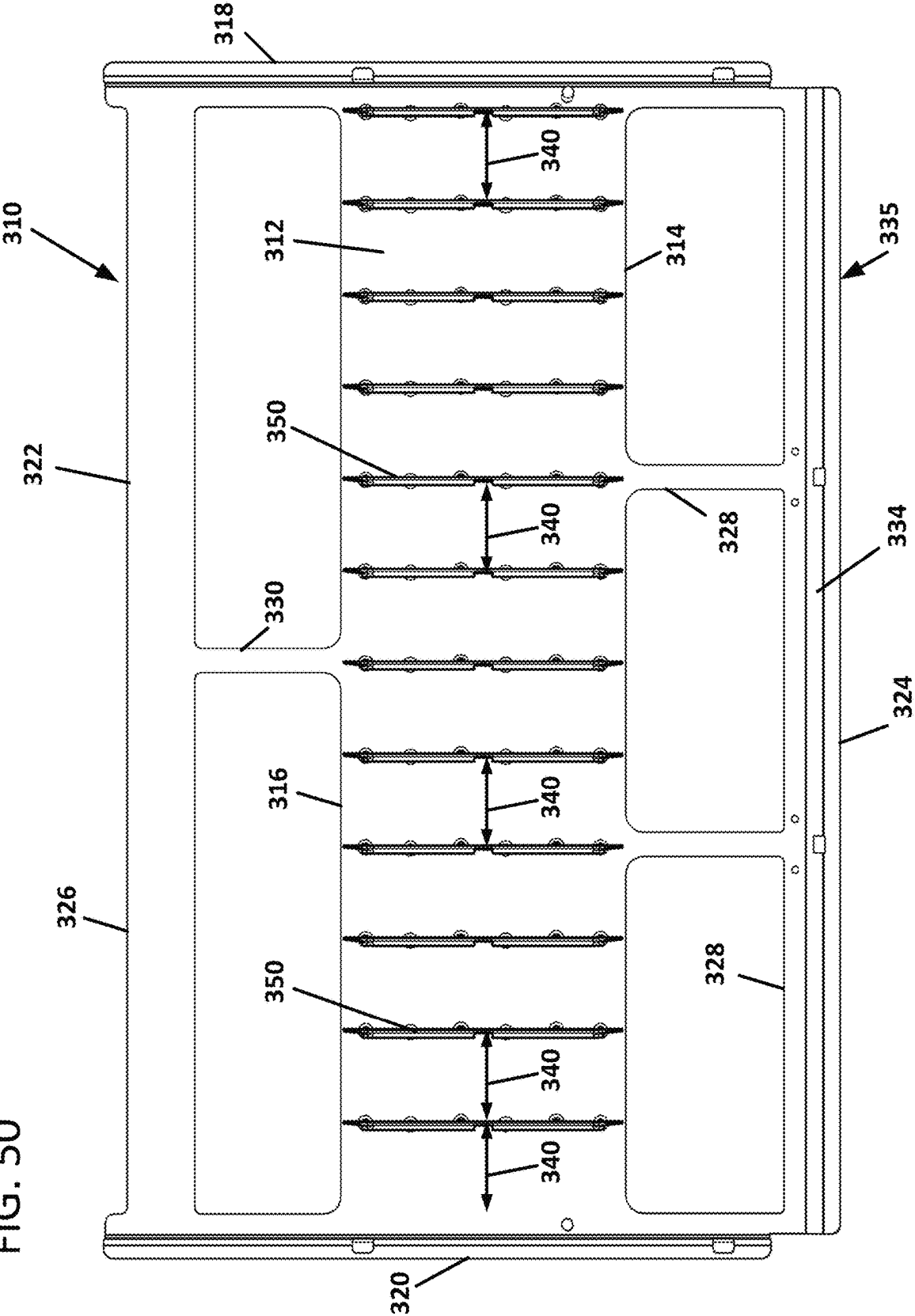


FIG. 51

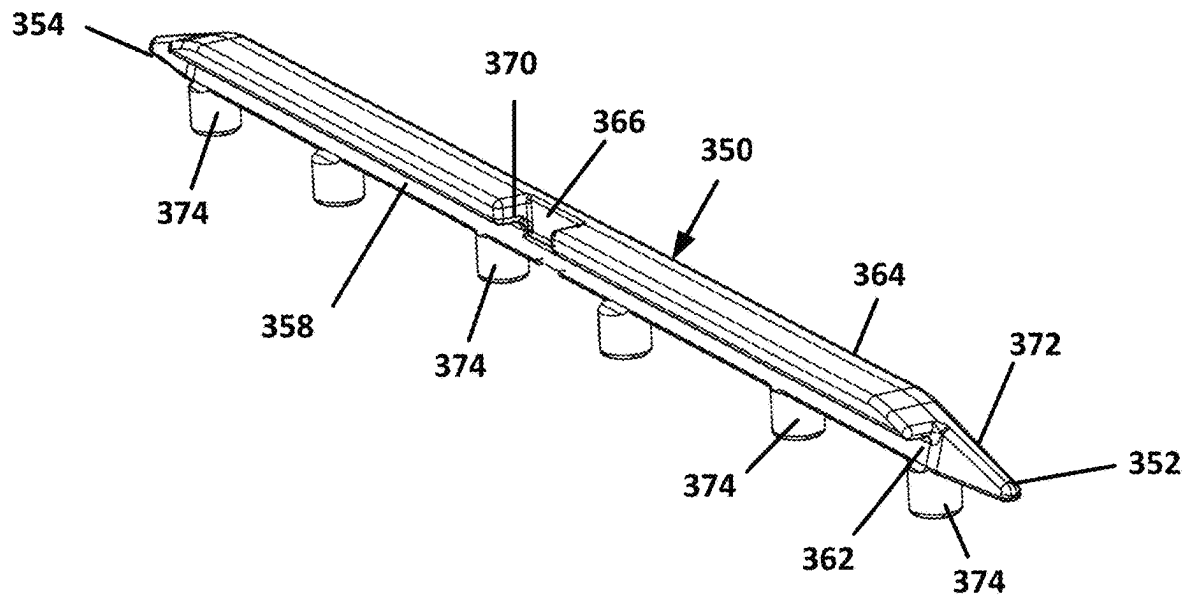


FIG. 52

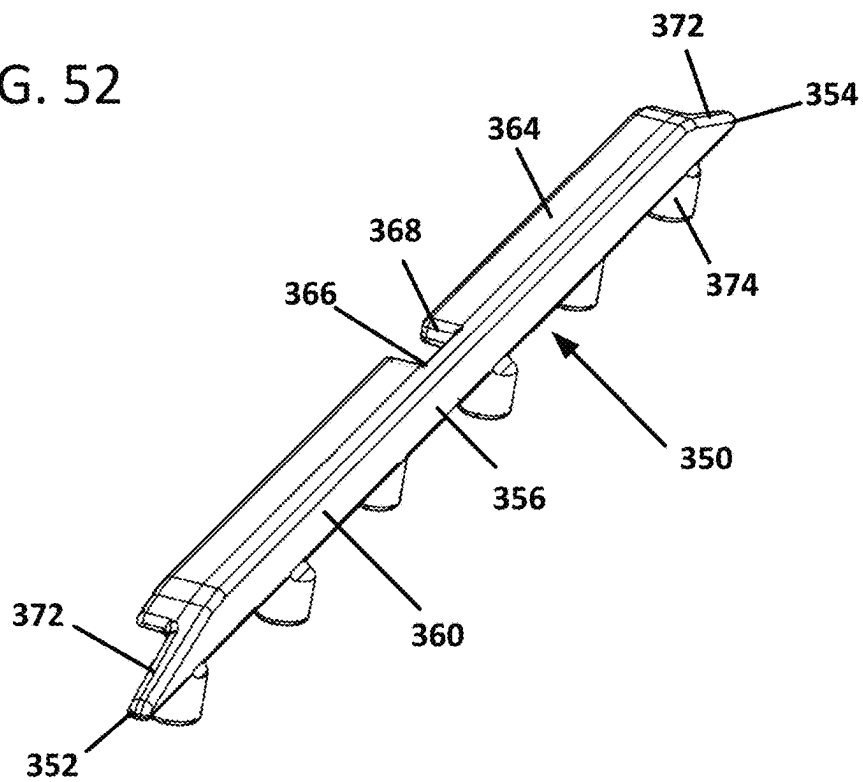


FIG. 53

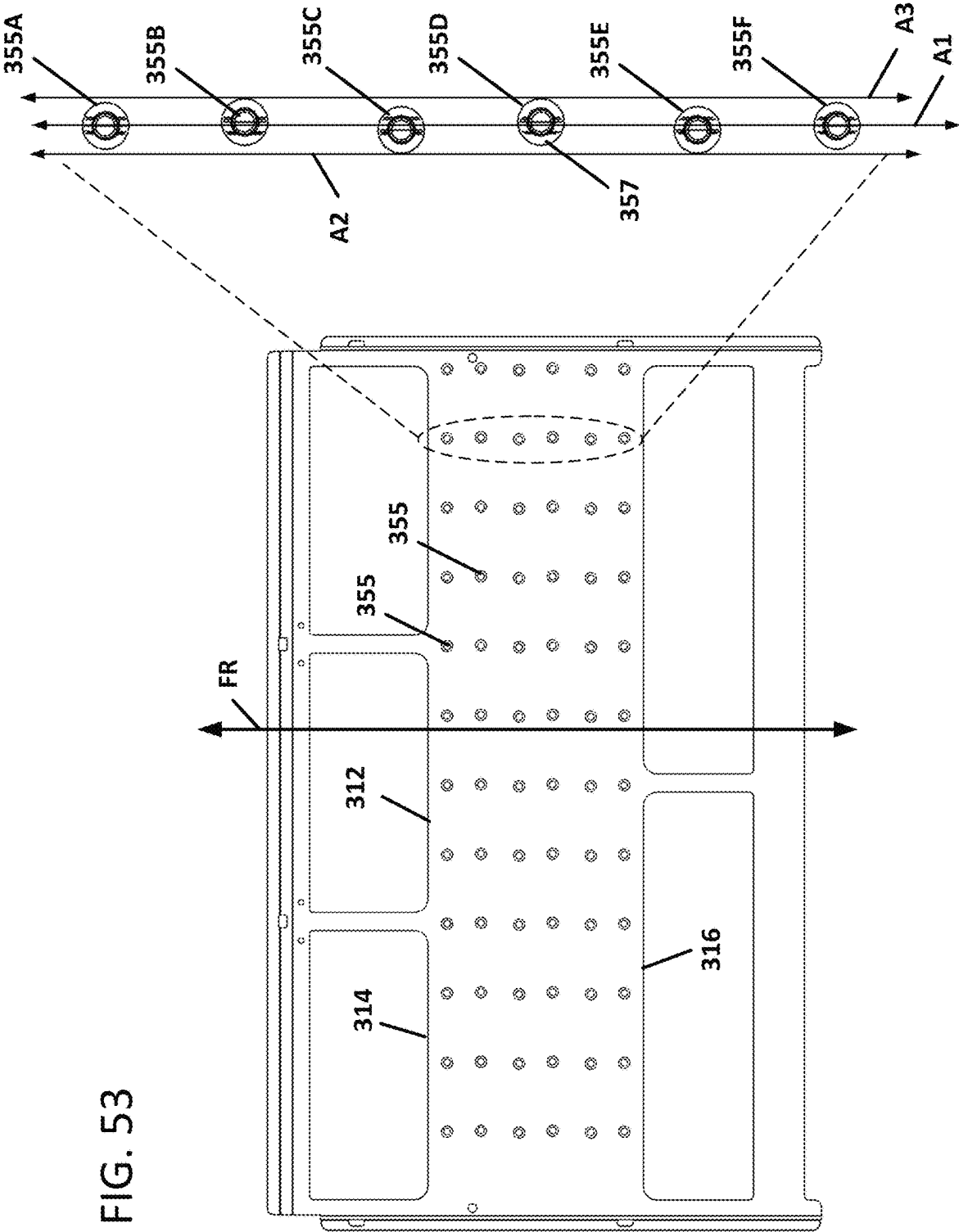


FIG. 54

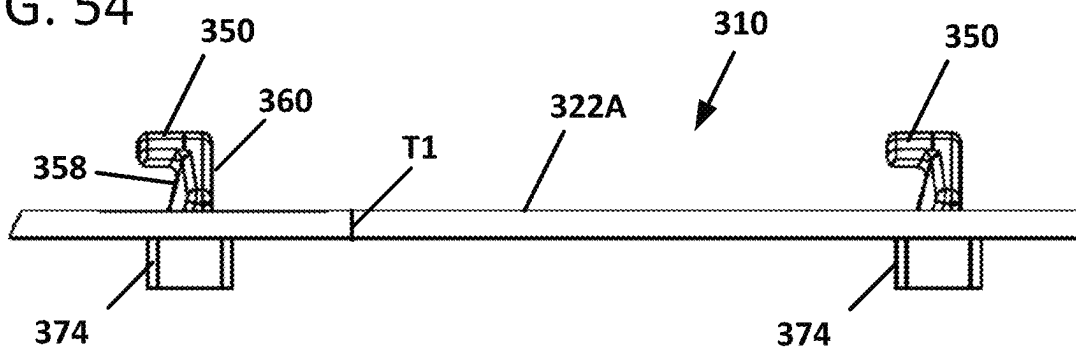


FIG. 55

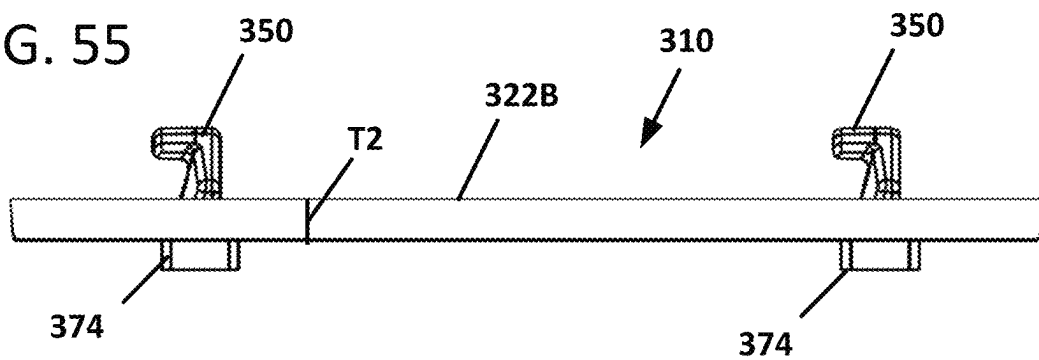


FIG. 56

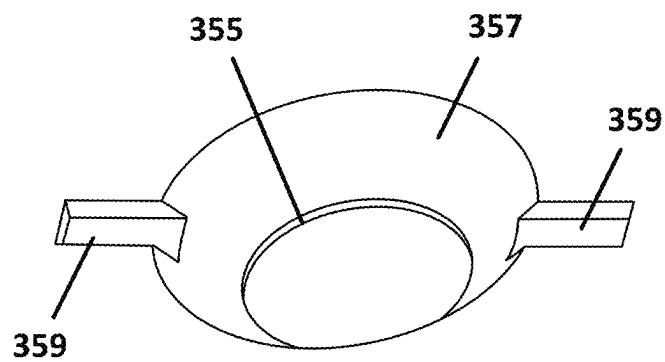


FIG. 57

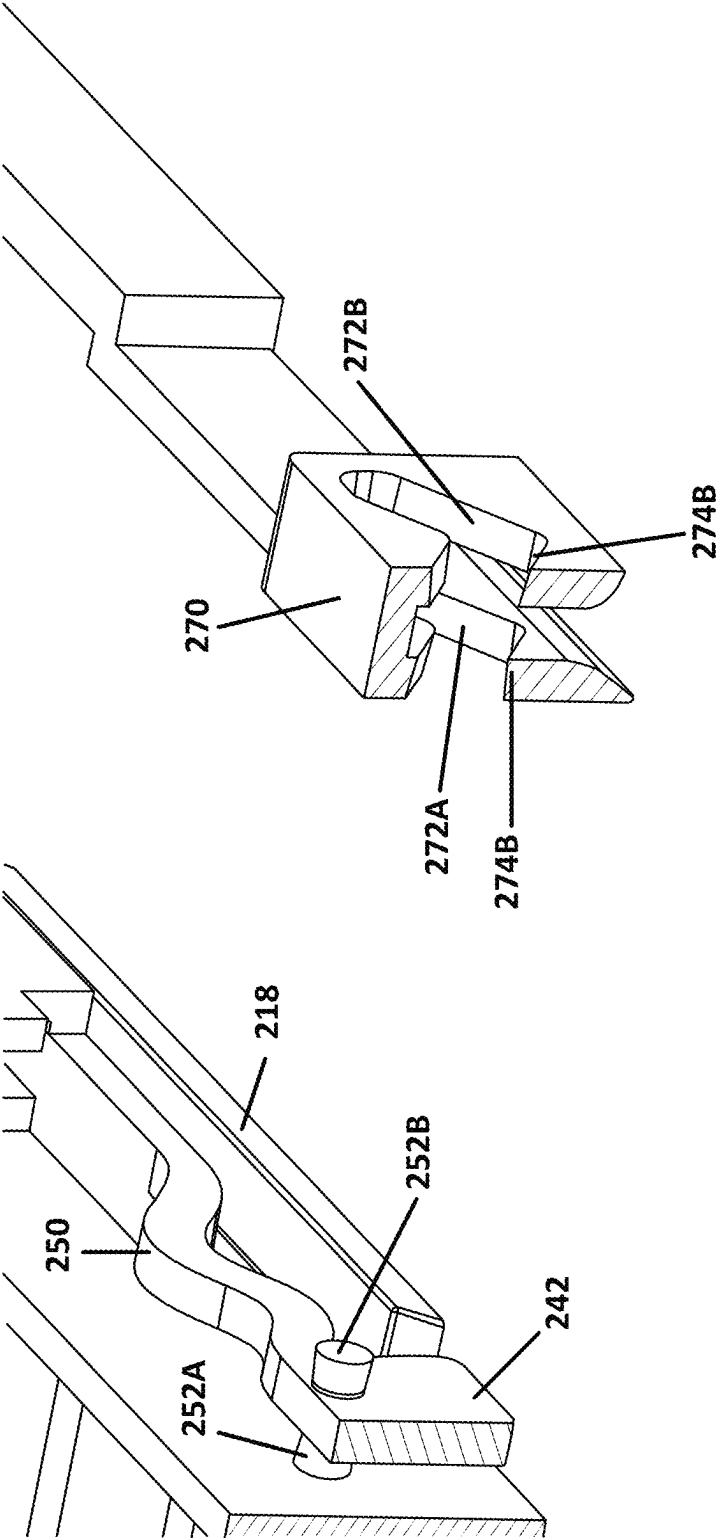
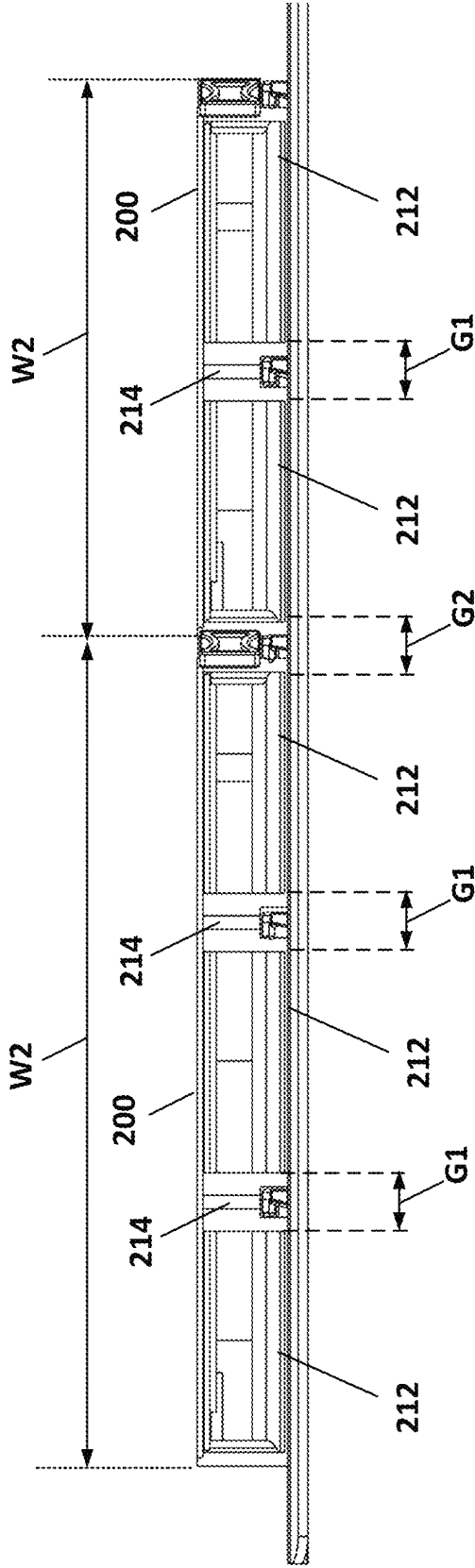


FIG. 58



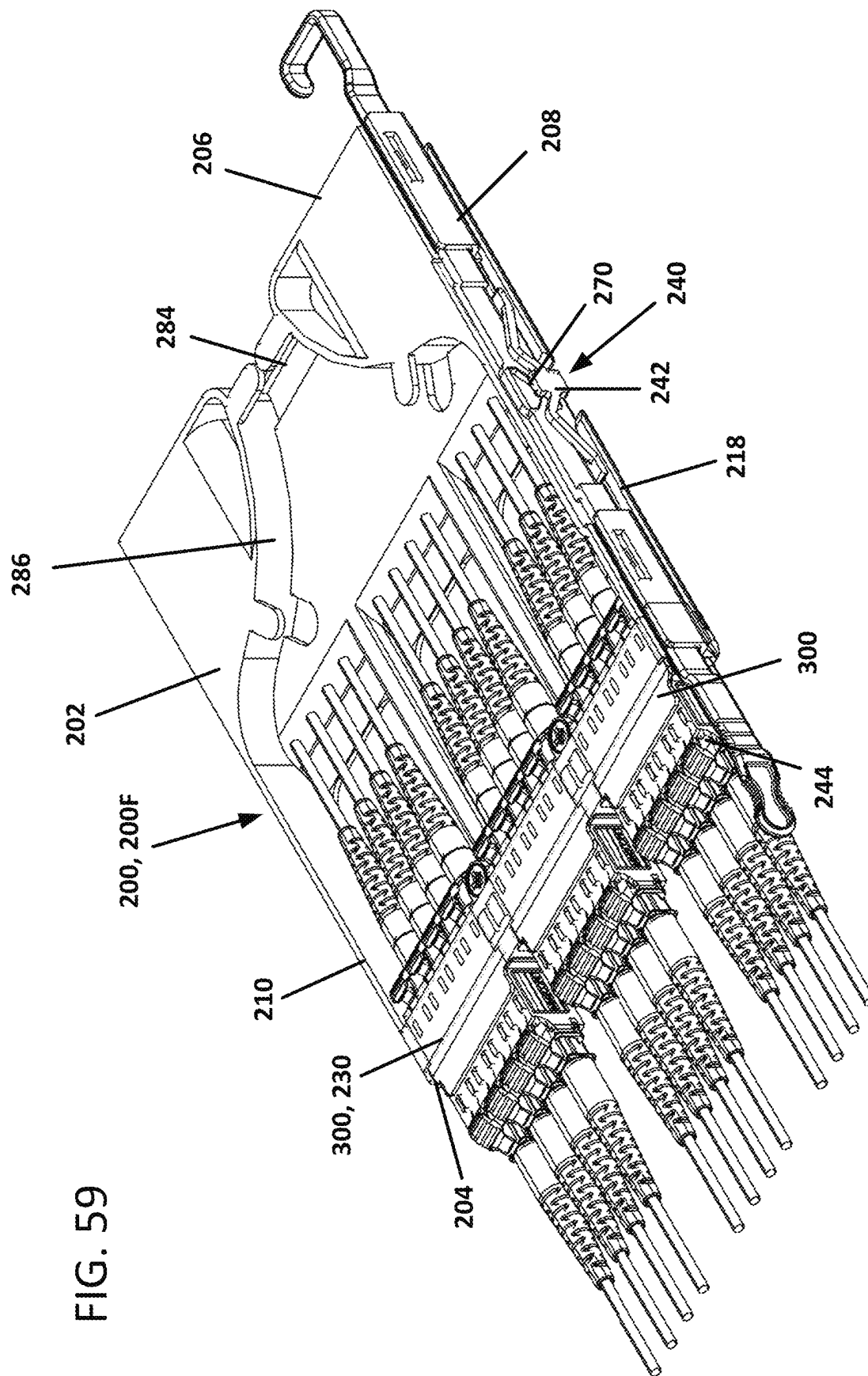


FIG. 59

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TRAY ARRANGEMENTS FOR CASSETTES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of Ser. No. 17/675,866, filed Feb. 18, 2022, which claims priority from U.S. Provisional Application Ser. No. 63/150,851, filed Feb. 18, 2021, and titled "Tray Arrangements for Cassettes," the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

In the telecommunications industry, the demand for added capacity is growing rapidly. This demand is being met in part by the increasing use and density of fiber optic transmission equipment. Even though fiber optic equipment permits higher levels of transmission in the same or smaller footprint than traditional copper transmission equipment, the demand requires even higher levels of fiber density. This has led to the development of high-density fiber handling equipment.

In communications panel systems, port members (e.g., optical adapters, electrical jacks, hybrid port members, etc.) defining front ports are mounted to one or more trays that are disposable within a chassis. The front ports are configured to receive plug connectors at the fronts of the trays. The trays can either be stationary within a chassis or can slide forwardly of the chassis to enhance access to the port members. Improvements are desired.

SUMMARY

A communications panel includes a chassis holding one or more trays. Each tray spans a width and depth of the chassis. In some implementations, the trays are stationary within the chassis. In other implementations, the trays are slidable within the chassis between two or more positions. Each tray is configured to support one or more cassettes carrying ports (e.g., optical ports, electrical ports, or hybrid ports) to receive plug connectors. A tray populated with one or more cassettes is referred to herein as a communications arrangement. The cassettes are slidably mounted to the tray along guides.

Some aspects of the disclosure are directed to a communications arrangement including a tray carrying guides configured to slidably receive a plurality of cassettes along the guides from either the front end of the tray or from the rear end of the tray at the discretion of the user. The guides and cassettes are configured so the cassette sizes need not be in integer increments of each other. For example, the tray may receive a first cassette having a first width and a second cassette having a second width that is 1.5 times the first width. Accordingly, the same tray can be fully populated by a plurality of cassettes having the first width or by a plurality of cassettes having the second width at the discretion of the user without modifying the guides or the tray. Alternatively, trays of different widths can be mixed on the tray.

Other aspects of the disclosure are directed to a communications arrangement including a tray having parallel guides spaced apart along a lateral axis of the tray, the guides extending along a forward-rearward axis of the tray that is perpendicular to the lateral axis. One or more cassettes can be mounted to the tray to slide along the guides. Each cassette has a latching arrangement that releasably locks to a respective one of the guides. The latching arrangement includes a release handle that slides relative to the cassette along an actuation axis that is parallel to the forward-

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rearward axis of the tray. Pulling or pushing the release handle forwardly releases the lock between the cassette and the respective guide. Continuing to pull or push the release handle forwardly slides the cassette forwardly along the guides. Similarly, pulling or pushing the release handle rearwardly releases the lock between the cassette and the respective guide. Continuing to pull or push the release handle rearwardly slides the cassette rearwardly along the guides.

Other aspects of the disclosure are directed to a communications arrangement including a tray carrying a plurality of parallel guides spaced a common distance from each other along a lateral axis of the tray. The space disposed between adjacent guides is referred to herein as a bay. One or more cassettes are mounted to the tray to slide along the guides. Each cassette spans at least two bays. Each guide is configured to inter-connect with no more than one cassette.

Other aspects of the disclosure are directed to a tray carrying generally parallel guides that each define cassette engagement structure facing in a first direction. None of the guides define cassette engagement structure facing in a second direction opposite the first direction. For example, each guide may define a groove facing in the first direction and a wall facing in the second direction.

Other aspects of the disclosure are directed to a cassette extending along a width between opposite first and second ends and along a length between front and rear ends. The length is orthogonal to the width. The cassette defines a bottom channel extending along the length of the cassette. In certain implementations, the cassette includes a bottom rail disposed within the bottom channel. The bottom rail extends from a base end attached to the cassette to a free end. In certain examples, the bottom rail extends parallel with the width of the cassette.

Other aspects of the disclosure are directed to a cassette extending across a width between opposite first and second ends. The cassette includes an end rail at the first end. The cassette is devoid of rails at the second end. In certain implementations, the cassette defines a bottom channel extending along a depth of the cassette orthogonal to the width. In certain examples, the cassette includes a bottom rail disposed within the bottom channel. In an example, the end rail and the bottom rail of each cassette extend in a common direction from a base end to a free end.

Other aspects of the disclosure are directed to a communications arrangement including a tray having parallel guides spaced apart along a lateral axis of the tray, the guides extending along a forward-rearward axis of the tray. The guides each define a notch through a top of the guide. One or more cassettes are mounted to the tray to slide along the guides. Each cassette has a latching arrangement at an end of the cassette that rides over the top of a respective one of the guides. The latching arrangement includes a stop member that drops into the notch of the respective guide to lock the cassette to the guide. The latching arrangement also includes a release handle that is actuated by moving the release handle in a different direction from the stop member. In certain examples, the release handle is moved orthogonally to the movement of the stop member. In an example, the release handle is moved along the forward-rearward axis of the tray. The release handle may extend forwardly and/or rearwardly relative to the front and rear ends of the cassette.

Other aspects of the disclosure are directed to a tray arrangement extending along a lateral axis between opposite first and second ends, along a forward-rearward axis between opposite front and rear ends, and along a height between a bottom and a top. The tray arrangement includes

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a tray and guides. The tray has a forward end portion towards the front end of the tray arrangement, a rearward end portion toward the rearward end of the tray arrangement, and an intermediate portion disposed between the forward end portion and the rearward end portion along the forward-rearward axis. The intermediate portion is raised along the height of the tray arrangement relative to the forward end portion. The guides are disposed at the intermediate portion and extend along the forward-rearward axis. Each of the guides is mounted to the forward end portion of the tray and to the rearward end portion of the tray. Each of the guides extends partially over the forward end portion and partially over the rearward end portion.

Other aspects of the disclosure are directed to a tray arrangement extending along a lateral axis between opposite first and second ends, along a forward-rearward axis between opposite front and rear ends, and along a height between a bottom and a top. The tray arrangement includes a tray and guides. The tray has opposite top and bottom surfaces each extending along the lateral axis and the forward-rearward axis. The tray defines a main region at which a plurality of apertures are defined. Each of the apertures extends between the top and bottom surfaces. The guides are attached to the tray at the main region. Each of the guides includes a plurality of pegs extending into respective ones of the apertures defined in the tray. Each of the guides defines a grooved side facing towards the first end of the tray arrangement and a closed side facing towards the second end of the tray arrangement.

Other aspects of the disclosure are directed to a tray arrangement extending along a lateral axis between opposite first and second ends, along a forward-rearward axis between opposite front and rear ends, and along a height between a bottom and a top. The tray arrangement includes a tray; and a plurality of guides mounted to the tray. Each of the guides extends along the forward-rearward axis. The guides are spaced from each other along the lateral axis. The guides include a first guide and a second guide connected by a webbing so that the first and second guides mount to the tray as a unit.

Other aspects of the disclosure are directed to a method of manufacturing a plurality of tray arrangements. The method includes selecting one of a first tray arrangement and a second tray; and mounting a plurality of guides to the selected one of the first and second trays. The guides each have a plurality of pegs. The method also includes liquefying at least portions of the pegs into the countersinks. In certain implementations, each of the first and second trays defines a plurality of apertures. Each of the first and second trays define countersinks. The first tray have a different thickness than the second tray so that the apertures of the first tray have a different height than the apertures of the second tray. Mounting the guides includes inserting the pegs into the apertures of the selected one of the first and second trays. The pegs of the guides have a common size regardless of which of the first and second trays is selected.

A variety of additional inventive aspects will be set forth in the description that follows. The inventive aspects can relate to individual features and to combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

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BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of the description, illustrate several aspects of the present disclosure. A brief description of the drawings is as follows:

FIG. 1 is a perspective view of an example communications panel system including a chassis mounted to a rack, the chassis being configured to hold a plurality of tray arrangements;

FIG. 2 is a perspective view of an example tray arrangement configured to be mounted within the chassis of FIG. 1, the tray arrangement including a tray carrying a plurality of guides;

FIG. 3 is a perspective view of the tray arrangement of FIG. 2 shown populated with a plurality of cassettes to form a communications arrangement, the cassettes each having a first width;

FIG. 4 is a perspective view of the tray arrangement of FIG. 2 shown populated with a plurality of cassettes each having a second width, the second width being about 1.5 times the first width;

FIG. 5 is a perspective view of the tray arrangement of FIG. 2 shown populated with a mixture of the cassettes of FIGS. 3 and 4;

FIG. 6 is a perspective view of the tray arrangement of FIG. 2 shown populated with a plurality of cassettes each having a third width, the third width being about double the first width;

FIG. 7 is a perspective view of the tray arrangement of FIG. 2 shown populated with a plurality of cassettes each having a fourth width, the fourth width being about triple the first width;

FIG. 8 is a perspective view of the tray arrangement of FIG. 2 showing the guides, a front extension arrangement, and a rear extension arrangement exploded outwardly from the tray;

FIG. 9 is a perspective view of the tray arrangement of FIG. 2 with the front and rear extension arrangements removed from the tray;

FIG. 10 is a side elevational view of the tray arrangement of FIG. 9;

FIG. 11 is a perspective view showing a first side of an example guide suitable for use with the tray arrangement of FIG. 2;

FIG. 12 is a perspective view showing a second side of the guide of FIG. 11;

FIG. 13 is a perspective view of an alternative guide unit suitable for use with the tray arrangement of FIG. 2, the guide unit including two guides coupled together as a unit;

FIG. 14 is a front perspective of a portion of the tray arrangement of FIG. 9;

FIG. 15 is a front elevational view of the tray arrangement of FIG. 9;

FIG. 16 is an enlarged view of a portion of FIG. 15;

FIG. 17 is a front, first side perspective view of a first example cassette suitable for use with the tray arrangement of FIG. 2, the first cassette carrying two groups of LC ports separated by an intermediate region at a front end of the first cassette;

FIG. 18 is a front, second side perspective view of the first cassette of FIG. 17, the second side being opposite the first side;

FIG. 19 is a front elevational view of the first cassette of FIG. 17 with the front and rear port members removed;

FIG. 20 is a rear elevational view of the first cassette of FIG. 19;

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FIG. 21 is a front perspective view of the first cassette of FIG. 17 carrying two groups of MPO ports separated by the intermediate region at the front end of the first cassette;

FIG. 22 is a front perspective view of the first cassette of FIG. 17 carrying two groups of SN ports separated by the intermediate region at the front end of the first cassette;

FIG. 23 is a perspective view of the first cassette of FIG. 17 with the front and rear port members exploded outwardly from the cassette body to show the front and rear apertures and a release handle exploded outwardly from a cassette body to show a stop member carried by the cassette body;

FIG. 24 is a side elevational view of the stop member and a portion of the cassette body of FIG. 23;

FIG. 25 is a front elevational view of the cassette body of the first cassette of FIG. 17;

FIG. 26 is a perspective view of the release handle of FIG. 23 showing an externally-facing side of the release handle, the release handle defining a camming groove having first and second ramped surfaces;

FIG. 27 is an enlarged view of the camming groove of the release handle and stop surfaces of the cassette body of FIG. 26;

FIG. 28 is an enlarged view of the forward grip portion of the release handle;

FIG. 29 is a side elevational view of the cassette of FIG. 17 shown with the release handle in a neutral position relative to the cassette body;

FIG. 29A is an enlarged view of a portion of FIG. 29 showing a camming member disposed along the camming groove of the release handle;

FIG. 30 is a side elevational view of the cassette of FIG. 17 shown with the release handle in a forwardly-extended position relative to the cassette body;

FIG. 30A is an enlarged view of a portion of FIG. 30 showing the camming member disposed along the camming groove of the release handle;

FIG. 31 is a side elevational view of the cassette of FIG. 17 shown with the release handle in a rearwardly-extended position relative to the cassette body;

FIG. 31A is an enlarged view of a portion of FIG. 31 showing the camming member disposed along the camming groove of the release handle;

FIG. 32 is front perspective view of the first cassette of FIG. 17 having an alternative latching arrangement and alternative cover, the cover being raised relative to the cassette body sufficient to expose the alternative stop members;

FIG. 33 is a perspective view of an alternative release handle for use with the alternative latching arrangement of FIG. 32;

FIG. 34 is an enlarged view of the alternative latching arrangement of FIG. 32 showing the stop members resting on the camming member of the release handle when the release handle is disposed in the neutral position;

FIG. 35 is a front, first side perspective view of a second example cassette suitable for use with the tray arrangement of FIG. 2, the second cassette carrying three groups of front ports separated by intermediate regions at a front end of the second cassette;

FIG. 36 is a front, second side perspective view of the second cassette of FIG. 35, the second side being opposite the first side;

FIG. 37 is a front elevational view of the second cassette of FIG. 35;

FIG. 38 is a rear elevational view of the second cassette of FIG. 36;

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FIG. 39 is a front, first side perspective view of a third example cassette suitable for use with the tray arrangement of FIG. 2, the third cassette carrying four groups of front ports separated by intermediate regions at a front end of the third cassette;

FIG. 40 is a front, second side perspective view of the third cassette of FIG. 39, the second side being opposite the first side;

FIG. 41 is a front elevational view of the third cassette of FIG. 39;

FIG. 42 is a rear elevational view of the third cassette of FIG. 39;

FIG. 43 is a front, first side perspective view of a fourth example cassette suitable for use with the tray arrangement of FIG. 2, the fourth cassette carrying six groups of front ports separated by intermediate regions at a front end of the fourth cassette;

FIG. 44 is a front, second side perspective view of the fourth cassette of FIG. 43, the second side being opposite the first side;

FIG. 45 is a front elevational view of the fourth cassette of FIG. 43;

FIG. 46 is a rear elevational view of the fourth cassette of FIG. 43; and

FIG. 47 is a perspective view of a fifth example cassette configured to connect a like number of front and rear plug connectors, the fifth cassette being shown with an open top;

FIG. 48 is a rear elevational view of a portion of a tray arrangement including a cassette mounted to guides carried by the tray arrangement;

FIG. 49 is a perspective view of a second example tray arrangement configured to be mounted within the chassis of FIG. 1, the tray arrangement including a tray carrying a plurality of guides;

FIG. 50 is a top plan view of the tray arrangement of FIG. 49;

FIG. 51 is a first perspective view of an example guide suitable for use with the tray arrangement of FIG. 49;

FIG. 52 is a second perspective view of the guide of FIG. 51;

FIG. 53 is a bottom plan view of the tray of FIG. 49 with a portion of the tray enlarged for ease in viewing;

FIG. 54 is a front elevational view of a portion of the main region of the tray arrangement of FIG. 49;

FIG. 55 shows the guides of FIG. 54 mounted to a thicker tray;

FIG. 56 is a perspective view of an example countersink disposed at the main region of the tray arrangement of FIG. 53;

FIG. 57 shows a portion of an example transition arrangement of an example release handle shown exploded from an example stop member of an example latching arrangement of any of the cassettes shown herein;

FIG. 58 is a front elevational view of a portion of a communications arrangement including first and second cassettes mounted to the tray arrangement of FIG. 2, the first and second cassettes having the port members removed for ease in viewing the front apertures to show the even spacing between the front apertures; and

FIG. 59 is a perspective view of a sixth example cassette configured to connect a like number of front and rear plug connectors, the sixth cassette being shown with an open top.

DETAILED DESCRIPTION

Reference will now be made in detail to exemplary aspects of the present disclosure that are illustrated in the

accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

The present disclosure is directed to a communications panel **100** configured to mount to a rack **R**. The communications panel **100** includes a chassis **105** holding one or more tray arrangements **110**, **310** (e.g., see FIGS. 3-7). Each tray arrangement **110**, **310** spans a width **W** of the chassis **105**. In some implementations, the tray arrangements **110**, **310** are stationary within the chassis **105**. In other implementations, the tray arrangements **110**, **310** are slidable along a depth **D** of the chassis **105** between two or more positions. Each tray arrangement **110**, **310** is configured to support one or more cassettes **200** carrying ports (e.g., optical ports, electrical ports, hybrid ports, etc.) configured to receive plug connectors. A tray arrangement **110**, **310** populated with one or more cassettes **200** is referred to herein as a communications arrangement **108**.

As will be discussed in more detail herein, a cassette **200** carries one or more front ports at a front of the cassette **200**. The front ports are configured to receive plug connectors. In some implementations, the front ports are optical ports configured to receive optical plug connectors. In other implementations, the front ports may be electrical ports configured to receive electrical plug connectors, hybrid ports configured to receive hybrid plug connectors, or a mixture thereof. In some implementations, a cassette **200** carries single-fiber front ports (e.g., LC front ports). In other implementations, a cassette **200** carries multi-fiber front ports (e.g., MPO front ports, SN front ports offered by Senko, etc.). In still other implementations, the same cassette **200** may carry a mix of two or more types of front ports (e.g., a mixture of MPO and LC ports).

In some implementations, the cassette **200** carries one or more rear ports (e.g., optical port, electrical port, hybrid port, etc.) at the rear of the cassette **200**. Each rear port is configured to receive a plug connector. Intermediate optical fibers within the cassette connect the front and rear ports. In other implementations, the cassette **200** carries ports only at the front of the cassette **200**. In such implementations, a rear of the cassette **200** is configured to enable optical fibers or one or more optical fiber cables to extend into/onto the cassette **200** and be routed towards the front ports.

The cassettes **200** are slidably mounted to the tray arrangement **110**, **310** along guides **150**, **350**. As shown in FIG. 2, the tray arrangement **110**, **310** includes a main region **112**, **312** that extends along a forward-rearward axis **FR** between a front end **114**, **314** and a rear end **116**, **316** of the main region **112**, **312**. The main region **112**, **312** also extends along a lateral axis **LA** between opposite first and second ends **118**, **120**, **318**, **320** of the main region **112**, **312**. The main region **112**, **312** of the tray arrangement **110**, **310** carries the guides **150**, **350**. Each guide **150**, **350** extends parallel to the forward-rearward axis **FR**. The guides **150**, **350** are spaced from each other along the lateral axis **LA**.

The guides **150**, **350** are configured to slidably receive a plurality of cassettes **200** along the guides **150**, **350** from either the front end **114**, **314** of the main region **112**, **312** or from the rear end **116**, **316** of the main region **112**, **312** at a discretion of a user. Interaction between the guides **150**, **350** and the cassettes **200** will be discussed in more detail herein.

The guides **150**, **350** and the cassettes **200** are configured so the sizes of the cassettes **200** need not be in integer increments of each other to each fit on the same tray arrangement **110**, **310** with the same configuration of guides **150**, **350**. For example, the tray arrangement **110**, **310** may receive a first cassette **200** having a first width and a second

cassette having a second width that is 1.5 times the first width. Accordingly, the same tray can be fully populated by a plurality of cassettes having the first width or by a plurality of cassettes having the second width at the discretion of the user without modifying the guides or the tray. In other cases, trays of different widths can be mixed on the tray.

In FIG. 3, the tray arrangement **110** of FIG. 2 is shown fully populated by first cassettes **200A** having the first width to form a first example communications arrangement **108**. FIG. 4 shows the tray arrangement **110** of FIG. 2 fully populated by second cassettes **200B** having a second width that is larger than the first width to form a second example communications arrangement **108**. The configuration (e.g., positioning or spacing) of the guides **150** has not been modified between the two images. In the example shown, the second width is about 1.5 times the first width. FIG. 5 shows the tray arrangement **110** of FIG. 2 populated by a combination of the first and second cassettes **200A**, **200B**. FIG. 6 shows the tray arrangement **110** of FIG. 2 fully populated by third cassettes **200C** having a third width that is larger than the second width (e.g., approximately twice the width of the first width). FIG. 7 shows the tray arrangement **110** of FIG. 2 fully populated by fourth cassettes **200D** having a fourth width that is larger than the third width (e.g., approximately three times the width of the first width). The positioning of the guides **150** remains constant in all five of these iterations. These same cassette configurations can be mounted to the tray arrangement **350** of FIGS. 49-51 as well.

In various implementations, a cassette **200** (e.g., any of cassettes **200A**, **200B**, **200C**, **200D**) can be a base-8 cassette, a base-12 cassette, or another type of cassette (e.g., a base-2 cassette, a base-16 cassette, a base-24 cassette, etc.). A base-8 cassette receives fibers at the rear (e.g., through a rear adapter or through fibers extending into the cassette) in increments of eight whereas a base-12 cassette receives fiber at the rear in increments of twelve. In certain examples, a base-16 cassette receives fibers at the rear in increments of sixteen and a base-24 cassette receives fibers at the rear in increments of twenty-four. In certain implementations, a base-8 cassette receives plug connectors having eight live fibers at a rear of the cassette **200**, a base-12 cassette receives plug connectors having twelve live fibers at a rear of the cassette **200**, a base-16 cassette receives plug connectors having sixteen live fibers at a rear of the cassette **200**, and a base-24 cassette receives plug connectors having twenty-four live fibers at a rear of the cassette **200**.

In certain implementations, a base-12 cassette has a larger width than a base-8 cassette. For example, a base-12 cassette providing a row of six duplex-LC front ports (i.e., twelve LC front ports) may be wider than a base-8 cassette providing a row of four duplex-LC front ports (i.e., eight LC front ports). In an example, such a base-12 cassette may be 1.5 times as wide as the base-8 cassette. In such implementations, the tray arrangement **110** of FIG. 2 could be populated with the base-8 cassettes, the base-12 cassettes, or a combination thereof without modification to the arrangement of the guides **150**. In other implementations, however, a base-8 cassette and a base-12 cassette can have the same width depending on port type.

FIG. 8 shows the example tray arrangement **110** of FIG. 2 including a tray **122** onto which various components (e.g., guides **150**) are mounted to form the tray arrangement **110**. In certain implementations, the tray **122** extends along the lateral axis **LA** between the first and second ends **118**, **120** of the tray arrangement **110**. In certain implementations, the guides **150** are mounted to the tray **122**.

In the example shown, the tray 122 includes a front portion 124, a rear portion 126, and an intermediate portion 128. The front portion 124 is disposed toward the front end 114 of the tray arrangement 110 and the rear portion 126 is disposed toward the rear end 116 of the tray arrangement 110. The intermediate portion 128 is disposed between the front portion 124 and the rear portion 126 along the forward-rearward axis FR. The intermediate portion 128 is raised along a height H of the tray arrangement 110 relative to the forward portion 124 (e.g., see FIG. 10).

The guides 150 are mounted to the tray arrangement 110 so that each guide 150 extends across the intermediate portion 128 along the forward-rearward axis FR. In certain implementations, the guides 150 are mounted to the forward portion 124 of the tray 122 and to the rearward portion 126 of the tray 122. In certain examples, each guide 150 also extends across the forward and rearward portions 124, 126 along the forward-rearward axis FR.

In certain implementations, the tray arrangement 110 includes a front extension arrangement 130 that mounts to the forward portion 124 of the tray 122 (e.g., see FIGS. 2 and 8). In some examples, the front extension arrangement 130 forms part of the main region 112 of the tray arrangement 110 and defines the front end 114 of the main region 112. In other examples, the forward portion 124 of the tray 122 defines the forward end 114 of the main region 112. In certain implementations, the front extension arrangement 130 extends between the first and second ends 118, 120 of the tray arrangement 110. In certain implementations, the front extension arrangement 130 is disposed no higher than the intermediate portion 128 of the tray 122. In certain examples, the front extension arrangement 130 is disposed lower than the intermediate portion 128 of the tray 122.

In certain implementations, the tray arrangement 110 includes a rear extension arrangement 132 that mounts to the rearward portion 126 of the tray 122. In some examples, the rear extension arrangement 133 forms part of the main region 121 of the tray arrangement 110 and defines the rear end 116 of the main region 112. In other examples, the rearward portion 126 of the tray 122 defines the rearward end 116 of the main region 112. In certain implementations, the rear extension arrangement 132 extends between the first and second ends 118, 120 of the tray arrangement 110. In certain implementations, the rear extension arrangement 132 is disposed no higher than the intermediate portion 128 of the tray 122. In certain examples, the rear extension arrangement 132 is disposed lower than the intermediate portion 128 of the tray 122.

In certain implementations, the front extension arrangement 130 and/or the rear extension arrangement 132 are formed of a different material than the tray 122. For example, the tray 122 may be formed of metal while the front and/or rear extension arrangements 130, 132 are formed of plastic or other materials. In certain implementations, the front extension arrangement 130 and/or the rear extension arrangement 132 defines notches in which portions of the guides 150 extend.

In certain implementations, the tray arrangement 110 includes a cable support arrangement 135 extending forwardly of the main region 112 of the tray arrangement 110. The cable support arrangement 135 includes a cable support bar 134 that extends parallel to the lateral axis LA of the tray arrangement 110 to support cables routed to the front ports of the cassettes 200 mounted to the tray arrangement 110. The cable support bar 134 is coupled to the main region 112 of the tray arrangement 110 by arms 136, 138. In certain examples, the arms 136, 138 space the support bar 134 away

from the front end 114 of the tray main region 112 sufficient to provide finger access to plug connectors received at the front ports of the cassettes 200. In some examples, the support bar 134, first arm 136, and second arm 138 are monolithically formed with the tray 122. In other examples, the cable support arrangement 135 is a separate piece coupled to the main region 112 of the tray arrangement 110 using fasteners, latches, or other attachment mechanisms.

FIGS. 11 and 12 illustrate an example guide 150 suitable for use with the tray arrangement 110 of FIG. 2. The guide 150 extends along a length GL (FIG. 9) between a front end 152 and a rear end 154 and along a height H (FIG. 10) between a bottom and a top surface 164. The guide 150 also has opposite first and second sides 156, 158 facing orthogonally to length GL and the height H.

The guide 150 is configured to engage (e.g., interconnect) with a cassette from only the second side 158. For example, the second side 158 defines cassette engagement structure and the first side 156 does not. In the example shown, the first side 156 defines a closed surface 160 and the second side 158 defines a groove 162 that extends along the length GL between opposite open ends. In some examples, the guide 150 has an L-shaped cross-sectional profile (e.g., see FIG. 15). In other examples, the guide 150 has a C-shaped cross-sectional profile along at least part of the length GL of the guide 150 (e.g., see FIG. 13).

The guides 150 each define a top surface 164 extending along the length GL of the guide 150. In certain examples, the top surface 164 of a guide 150 defines a ramped surface 172 at the first and second open ends of the groove 162. The top surface 164 defines a notch 166 at an intermediate position between the first and second open ends of the groove 162. In certain examples, the notch provides access to the groove 162 through the top surface 164. In certain implementations, the notch 166 provides access to a forward-facing catch surface 168 and a rearward-facing catch surface 170 at opposite ends of the notch 166.

In certain implementations, the guides 150 are configured to be mounted to the main region 112 of the tray arrangement 110 (e.g., to the tray 122). In some examples, the guides 150 include pegs 174 that can be inserted into apertures defined through the main region 112 to attach the guides 150 to the tray arrangement 110. In various examples, the pegs 174 can be attached to the main region 112 by heating, ultrasonically welding, adhesive, friction, or another mechanism. In other examples, the guides 150 can be fastened, soldered, latched, or otherwise attached to the tray.

In certain implementations, a guide 150 includes side tabs 176 extending outwardly from the first and second sides 156, 158 of the guide 150. The side tabs 176 may carry pegs 174 or other attachment structure. In certain examples, the side tabs 176 provide stability to the guides 150. In certain examples, an example guide 150 includes a first pair of side tabs 176 at the front end 152 and a second pair of side tabs 176 at the rear end 154. In the example shown in FIG. 9, the first pair of side tabs 176 are mounted to the front end portion 124 of the tray 122 and the second pair of side tabs 176 are mounted to the rear end portion 126 of the tray 122. In certain examples, an example guide 150 includes a third pair of side tabs 176 extending outwardly from an intermediate position along the length GL of the guide 150. In an example, the third pair of side tabs 176 align with the notch 166.

FIG. 13 shows an alternative guide unit 178 including a plurality of guides 150 connected together with webbing 180. In the example shown, the webbing 180 connects two

guides. In other examples, the webbing 180 can connect three, four, six, twelve, or any desired number of guides 150. The guide unit 178 enables multiple guides 150 to be mounted to the main region 112 of the tray arrangement 110 simultaneously. In certain implementations, the guide unit 178 also may include pegs 174 for heat staking, ultrasonically welding, or otherwise securing the guide unit 178 to the tray arrangement 110. In certain implementations, the guide unit 178 also may include side tabs 176.

The guides 150, 350 extend across the main region 112, 312 of the tray arrangement 110, 310 along the lateral axis LA. In certain implementations, the guides 150, 350 are spaced a common distance CD apart from each other along the lateral axis LA (e.g., see FIGS. 14-16). A bay 140, 340 is a space extending outwardly from the second side 158, 358 of a guide 150, 350 by the common distance CD. In most cases, the bay 140, 340 extends to the first side 156, 356 of an adjacent guide 150, 350. In certain cases, a bay 140, 340 is bordered by only one guide 150, 350 (e.g., see the left-most bay 140, 340 of FIGS. 15 and 50). In such a case, the bay 140, 340 extends towards a sidewall of the chassis 105.

FIGS. 49-51 illustrate another example implementation of a tray arrangement 310 suitable for use in receiving any of the cassettes 200 disclosed herein. FIG. 49 shows the example tray arrangement 310 including a planar tray 322 onto which various components (e.g., guides 350) are mounted to form the tray arrangement 310. In certain implementations, the tray 322 extends along the lateral axis LA between the first and second ends 318, 320 of the tray arrangement 310. In the example shown, the tray 322 defines the main region 312 of the tray arrangement 310 at which the guides 350 mount.

In certain implementations, the tray 322 defines a front cable management region 324 and a rear cable management region 326. In certain examples, the front cable management region 324 and rear cable management region 326 are monolithically formed with the tray 322. In certain implementations, the front cable management region 324 includes a cable support arrangement 335. The cable support arrangement 335 includes a cable support bar 334 that extends parallel to the lateral axis LA of the tray arrangement 310 to support cables routed to the front ports of the cassettes 200 mounted to the tray arrangement 310. In certain implementations, the rear cable management region 326 defines a cable support surface for cables routed to the rear ends 206 of the cassettes 200.

One or more front finger access apertures 328 are defined between the main region 312 and the front cable management region 324. One or more rear finger access apertures 330 are defined between the main region 312 and the rear cable management region 326. In some examples, the tray 322 defines a different number of front finger access apertures 328 than rear finger access apertures 330. The finger access apertures 328, 330 provide finger access to the front and/or rear ports carried by any cassettes 200 mounted to the tray arrangement 310.

FIGS. 51 and 52 illustrate another example guide 350 suitable for use with the tray arrangement 310. The guide 350 extends along a length between a front end 352 and a rear end 354 and along a height between a bottom and a top surface 364. The guide 350 also has opposite first and second sides 356, 358 facing orthogonally to length and the height. In certain implementations, the guide 350 is substantially similar to the guide 150 of FIGS. 11 and 12, except that the guide 350 does not include side tabs 176.

The guide 350 is configured to engage (e.g., interconnect) with a cassette from only the second side 358. For example, the second side 358 defines cassette engagement structure and the first side 356 does not. In the example shown, the first side 356 defines a closed surface 360 and the second side 358 defines a groove 362 that extends along the length between opposite open ends. In some examples, the guide 350 has an L-shaped cross-sectional profile (e.g., see FIG. 54).

The guides 350 each define a top surface 364 extending along the length of the guide 350. In certain examples, the top surface 364 of a guide 350 defines a ramped surface 372 at the first and second open ends of the groove 362. The top surface 364 defines a notch 366 at an intermediate position between the first and second open ends of the groove 362. In certain examples, the notch 366 provides access to the groove 362 through the top surface 364. In certain implementations, the notch 366 provides access to a forward-facing catch surface 368 and a rearward-facing catch surface 370 at opposite ends of the notch 366.

In certain implementations, the guides 350 are configured to be mounted to the main region 312 of the tray arrangement 310 (e.g., to the tray 322). In some examples, the guides 350 are individually mounted to the tray 322. In other examples, the guides 350 are mounted to the tray 322 in one or more units with each unit including two or more guides 350.

In some examples, the guides 350 include pegs 374 that can be inserted into apertures 355 defined through the main region 312 and heated to attach the guides 350 to the tray arrangement 310. In other examples, the guides 350 can be attached to the main region 312 by fasteners, latches, friction-fit pegs, welding, soldering, or other attachment mechanisms.

As shown in FIG. 50, the guides 350 are mounted to the main region 312 of the tray 322 so that each guide 350 terminates before reaching the finger access apertures 328, 330. As shown in FIG. 53, the apertures 355 may lead to countersinks or pockets 357 at the bottom of the tray 322. For example, the countersinks 357 may be sized to receive material melted or otherwise liquefied from the pegs of the guides 350. In an example, the countersinks 357 are sized so that the flowable material is flush with the bottom of the tray 322 when the guides 350 are attached to the tray 322.

In certain implementations, the main region 312 defines rows of apertures 355 through which the guides 350 can be staked, fastened, or otherwise coupled to the tray 322. In some implementations, the apertures 355 of each row are aligned along respective axes A1. In other implementations, one or more of the apertures 355 in a row may be laterally offset from each other. For example, as shown in FIG. 53, front and rear apertures 355A, 355F of each row may be aligned along the axis A1 while intermediate apertures 355B-355E are offset along the lateral axis A relative to the axis A1. In the example shown, apertures 355C and 355E are offset to be tangent to a second axis A2 spaced in a first direction from the first axis A1 and apertures 355B and 355D are offset to be tangent to a third axis A3 spaced in an opposite second direction from the first axis A1. Offsetting the apertures 355 in a row may increase the stability of the guide 350. In certain implementations, the apertures 355 may be positioned in a configuration to add a poke-yoke to inhibit installing the guides 350 backwards. For example, the guide 358 shown in FIG. 53 may be modified to have an odd number of pegs.

As shown in FIGS. 54 and 55, a thickness of the tray arrangement 310 may vary. For example, FIG. 54 illustrates

two example guides **350** mounted to a first tray **322A** having a first thickness **T1** and the same two guides **350** mounted to a second tray **322B** having a second thickness **T2**. In other examples, additional components can be mounted to the tray **322A** to create a thickness of **T2**.

In some implementations, different guides **350** with different sizes of pegs **374** are used with each tray **322A**, **322B**. For example, the guide **350** used with the thicker tray **322B** may have a longer peg **374**. In other implementations, the same guides **350** can be utilized with trays **322A**, **322B** of different thicknesses **T1**, **T2**. For example, each of the trays **322A**, **322B** may have a respective countersink **357** sized based on the length of the peg **374** extend through the aperture **355**. In still other examples, the same countersink **357** may be used with both trays **322A**, **322B**.

In some such examples, the, each tray **322A**, **322B** may have a common countersink **357** that is configured to accommodate different amounts of peg material. For example, FIG. **56** illustrates a countersink **357** sized to receive the flowable material from a peg **374** extending through the thicker tray **322B** so that the flowable material is about flush with the bottom of the tray **322B**. The countersink **357** of FIG. **56** defines channels or wells **359** to accommodate additional flowable material when the peg **374** is utilized with the thinner tray **322A**. It will be understood that a greater or lesser number of channels or wells **359** can be provided at the countersink **357** and in other configurations.

In other such examples, the countersink **357** may be sized so that the flowable material is flush with a bottom of the tray **322A** for the thinner tray **322A**, but fills only a portion of the countersink **357** for the thicker tray **322B**. In certain implementations, a tool that melts or otherwise liquefies the pegs **374** may include a protrusion that extends partially into the countersink **357** to press or displace the flowable material towards the periphery of the countersink **357** for better adhesion.

Referring now to FIGS. **17-20**, the cassettes **200** (e.g., cassettes **200A**, **200B**, **200C**, and **200D**) are configured to slidably mount to the tray arrangement **110** using the guides **150**, **350**. A cassette **200** includes a cassette body **202** that extends along a width **W2** between opposite first and second ends **208**, **210**, along a length **L2** between front and rear ends **204**, **206**, and along a height **H2** (FIG. **19**) between a bottom and a top. The length **L2**, width **W2**, and height **H2** are orthogonal to each other. The cassette **200** includes a bottom rail **220** (FIG. **19**) that is configured to fit (i.e., inter-connect) within the groove **162**, **362** of a respective one of the guides **150**, **350**. The bottom rail **220** is configured to slide along the groove **162**, **362** from either the open front end or the open rear end of the groove **162**, **362**.

In certain implementations, the cassette body **202** includes an end rail **218** (FIGS. **19** and **23**) at the first end **208** of the cassette body **202**. The end rail **218** is configured to fit within the groove **162**, **362** of a respective one of the guides **150**, **350**. The end rail **218** is configured to slide along the groove **162**, **362** from either the open front end or the open rear end of the groove **162**, **362**. In the example shown in FIGS. **19** and **20**, the bottom rail **220** and the end rail **218** would slide along adjacent guides **150**, **350**. In other examples, however, the bottom rail **220** and the end rail **218** are configured to slide along non-adjacent guides **150**, **350** (e.g., see FIG. **37**). In certain examples, each guide **150**, **350** is configured to inter-connect with no more than one cassette **200**.

In certain examples, the end rail **218** and the bottom rail **220** of each cassette body **202** extend in a common direction

from a base end **218a**, **220a** attached to the cassette body **202** to a free end **218b**, **220b**. In an example, the common direction extends along the width **W2** of the cassette body **202**. In some examples, the free ends **218b**, **220b** of the rails **218**, **220** extend parallel with the height **H2** of the cassette body **202**. In other examples, the free ends **218b**, **220b** of the rails **218**, **220** are angled relative to the height **H2** of the cassette body **202** (e.g., see FIGS. **19** and **20**).

The second end **210** of the cassette body **202** is not configured to engage a guide **150**, **350**. In certain examples, the cassette body **202** is devoid of rails (i.e., does not include any rails) at the second end **210** (e.g., see FIG. **18**). Rather, the second end **210** of the cassette body **202** directly opposes the first side **156**, **356** (e.g., the closed surface **160**, **360**) of the guide **150**, **350**. In certain examples, the second end **210** of the cassette body **202** defines a generally flat surface. In certain examples, the second end **210** of the cassette body **202** is spaced from the first side **156**, **356** of the guide **150**, **350** sufficient to avoid contacting the guide **150**, **350**.

The cassette body **202** defines a two or more front apertures (e.g., three front apertures, four front apertures, six front apertures, etc.) **212** at the front end **204**. Each of the front apertures **212** is configured to hold one or more optical adapters **230**, **232**, **234**, **236**. In certain implementations, the front apertures **212** are separated from each other by respective intermediate regions **214**. For example, in FIG. **19**, the cassette body **202** defines a first front aperture **212A** separated from a second front aperture **212B** by an intermediate region **214**.

As shown in FIG. **58**, the intermediate regions **214** of a cassette **200** have a consistent size so that adjacent front apertures **212** of a cassette **200** are spaced from each other by a gap **G1** that extends along the width **W2** of the cassette **200**. In certain examples, the front apertures **212** of a cassette **200** also have a consistent size extending along the width **W2** of the cassette **200**. In certain examples, all cassettes **200** that fit on the tray arrangement **110** have equally sized front apertures **212**. In certain examples, the cassettes **200** are configured so that adjacent front apertures **212** of adjacent cassettes **200** (e.g., the right-most front aperture **212** of a first cassette **200** and the left-most front aperture **212** of a second cassette **200**) are spaced from each other by a spacing **G2** that is the same (within manufacturing tolerances) as the spacing **G1** of the adjacent front apertures **212** of the same cassette **200** (e.g., see FIG. **58**). Making the spacing between adjacent front apertures **212** of adjacent cassettes **200** consistent with the spacing between adjacent front apertures **212** of the same cassette **200** facilitates mixing different sizes of cassettes **200** on the tray arrangement **110** at a discretion of the user.

As shown in FIGS. **19** and **20**, the cassette body **202** defines a bottom channel **222** extending along the length **L2** of the cassette body **202**. The bottom channels **222** are sized to receive guides **150**, **350** to enable a cassette **200** to extend over one or more guides **150**, **350**. In some implementations, the channels **222** are sized so that a top of the channel **222** is spaced from the top surface **164**, **364** of the guide **150**, **350** received therein. In other implementations, the channels **222** are sized so that a top of the channel **222** rests on and slides along the top **164**, **364** of the guide **150**, **350**. Each cassette **200** spans at least two bays **140**, **340** of the tray arrangement **110**, **310**. In certain examples, the bottom channel **222** is disposed at an intermediate region **214** of the cassette **200** between two front apertures **212**.

In certain implementations, each intermediate region **214** defines a bottom channel **222**. Accordingly, a cassette **200** having two front apertures **212A**, **212B** defines a single

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bottom channel 222. A cassette 200 having three front apertures 212 defines two bottom channels 222—a first bottom channel 222 disposed between the first front aperture 212A and the second front aperture 212B and a second bottom channel 222 disposed between the second front aperture 212B and a third front aperture 212 (e.g., see FIG. 37). A cassette 200 having four front apertures 212 defines three bottom channels 222 (e.g., see FIG. 41). A cassette 200 having six front apertures 212 defines five bottom channels 222 (e.g., see FIG. 45).

In certain implementations, a punch-out cover may be initially provided at one or more of the front apertures 212 so a respective optical adapter is mounted at fewer than all of the front apertures 212. In such cases, a user has the flexibility to select a position for a front optical adapter to facilitate cable routing.

In certain implementations, the bottom rail 220 of the cassette body 202 is disposed within the bottom channel 222. The bottom rail 220 extends less than a width of the bottom channel 222 to accommodate the guide 150, 350 within the bottom channel 222 (e.g., see FIG. 48). In certain examples, the bottom rail 220 is flush with a bottom of the cassette body 202. In certain examples, the bottom rail 220 and the end rail 218 are coplanar with each other.

In the example shown in FIG. 48, a cassette 200B is mounted to guides 150, 350 of a tray arrangement 110. The cassette 200B spans three bays 140, 340—a first bay 140, 340 disposed between the first guide 150A, 350A and the second guide 150B, 350B, a second bay 140, 340 disposed between the second and third guides 150B, 350B, 150C, 350C and a third bay 140, 340 disposed between the third guide 150C, 350C and a fourth guide 150D, 350D. The cassette 200B defines bottom channels 222 that receive the second and third guides 150B, 350B, 150C, 350C. The first guide 150A, 350A inter-connects with the end rail 218 of the cassette 200B. The second guide 150B, 350B does not inter-connect with any structure of the cassette 200B, but rather slides within one of the bottom channels 222. In an example, the cassette 200B may slide along a top 164, 364 of the second guide 150B, 350B. The third guide 150C, 350C inter-connects with the bottom rail 220 of the cassette 200B within the other bottom channel 222. In the example shown, the cassette 200B extends towards, but does not engage a fourth guide 150D, 350D at an opposite side of the third bay 140, 340. Rather, the fourth guide 150D, 350D receives the end rail 218 of an adjacent cassette 200. In other implementations, the tray arrangement 110, 310 may not include a fourth guide 150D, 350D. Similarly, the first guide 150, 350 does not engage the cassette 200 disposed at an opposite side of the first guide 150A, 350A from the cassette 200B.

In certain implementations, each front aperture 212 of a cassette 200 has a common size. In certain examples, each of the front apertures 212 has a footprint sized to two standardized ports. As the term is used herein, a standardized port refers to a sufficient spacing to accommodate an MPO port or a duplex-LC port at a discretion of the user. In FIG. 17, the cassette body 202 is shown carrying a quadruplex LC adapter 230 at each of the front apertures 212. Each quadruplex LC adapter 230 defines two duplex-LC ports (i.e., four LC ports). In other examples, the cassette body 202 may carry a duplex MPO adapter 234 (e.g., see FIG. 21), two quadruplex SN adapters 236 (e.g., see FIG. 22), two SC adapters, or any desired adapter type that fits in a quadruplex LC adapter footprint. In still other implementations, the front apertures 212 may be sized to handle other port footprints.

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In certain examples, cassette 200 is configured to fit one front aperture 212 per bay 140. In certain examples, each tray arrangement 110 defines at least twelve bays 140. In certain examples, each front aperture 212 is configured to fit four LC ports. In such examples, each tray arrangement 110 is configured to hold forty-eight LC ports. In certain examples, each front aperture 212 is configured to hold four SN ports. In such examples, each tray arrangement 110 is configured to hold forty-eight SN ports (or ninety-six fibers). In certain examples, each front aperture 212 is configured to hold a duplex MPO ports—each receiving eight, twelve, sixteen, or twenty-four live fibers. In such examples, each tray arrangement 110 is configured to hold twenty-four MPO ports.

In certain implementations, three tray arrangements 110 fit within a 1 RU chassis. In certain examples, each 1 RU chassis may hold at least 144 LC ports. In certain examples, each 1 RU chassis may hold at least 144 SN ports (or 288 fiber connections). In certain examples, each 1 RU chassis may hold at least 72 MPO ports (for 576 fiber connections, 864 fiber connections, 1,152 fiber connections, and 1,728 fiber connections depending on the number of live fibers received).

In certain implementations, the cassettes 200 are configured to lock to the guides 150, 350 in at least one discrete position. Each cassette 200 includes a latching arrangement 240 that rides over the top 164, 364 of a respective one of the guides 150, 350. The latching arrangement 240 includes a stop member 242 that drops into the notch 166, 366 of the respective guide 150, 350 to lock the cassette 200 to the guide 150, 350. The latching arrangement 240 also includes a release handle 244 to raise the stop member 242 out of the notch 166, 366.

The stop member 242 is carried by the cassette body 202 at the first end 208. The stop member 242 is movable relative to the cassette body 202 along a movement axis SM that extends parallel to the height H2 of the cassette 200. In certain implementations, the stop member 242 is movable along the movement axis SM between a lock position and a release position. In certain examples, the stop member 242 is biased to the lock position.

In certain implementations, the stop member 242 rides over the ramped surface 172, 372 at the front end 152, 352 or rear end 154, 354 of the guide 150, 350 when the cassette 200 is first mounted to the guides 150, 350. The ramped surfaces 172, 372 transition the stop member 242 from the lock position to the release position. The top 164, 364 of the guide 150, 350 maintains the stop member 242 in the release position until the stop member 242 vertically aligns with the notch 166, 366. The notch 166, 366 allows the stop member 242 to transition to the lock position.

The release handle 244 is configured to move relative to the cassette body 202 along an actuation axis A between a neutral position (FIG. 29) and a forward extended position (FIG. 30). When in the neutral position, the release handle 244 enables the stop member 242 to be in the lock position. When disposed in the forward extended position, the release handle 244 transitions the stop member 242 to the release position. In certain implementations, the release handle 244 also can be moved relative to the cassette body 202 along the actuation axis A to a rearward extended position (FIG. 31). When disposed in the rearward extended position, the release handle 244 transitions the stop member 242 to the release position.

When in the lock position, engagement surfaces 246, 248 of the stop member 242 extend at least partially through a recess or gap 218e in the end rail 218. The first engagement

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surface 246 is configured to oppose the first catch surface 168, 368 within the notch 166, 366 of the guide 150, 350 and the second engagement surface 248 is configured to oppose the second catch surface 170, 370 when the stop member 242 is engaged with the notch 166, 366. Abutment between the first engagement surface 246 and the first catch surface 168, 368 inhibits forward movement of the cassette 200 along the guides 150, 350. Abutment between the second engagement surface 248 and the second catch surface 170, 370 inhibits rearward movement of the cassette 200 along the guides 150, 350. When in the release position, the engagement surfaces 246, 248 of the stop member 242 are raised out of the notch 166, 366 sufficient so that the engagement surfaces 246, 248 do not abut the catch surfaces 168, 368, 170, 370 upon movement of the cassette 200 relative to the guides 150, 350.

In certain examples, the end rail 218 includes a front end rail 218c and a rear end rail 218d aligned along the length L2 of the cassette 200 (e.g., FIG. 24). The rear end rail 218d is spaced from the front end rail 218c along the length L2 of the cassette 200 to provide the gap 218e. In other implementations, a recess can be provided in a continuous end rail 218. When in the lock position, the stop member 242 extends at least partially into the recess or gap 218e (e.g., see FIG. 24). When in the release position, the engagement surfaces 246, 248 of the stop member 242 are raised out of the recess or gap 218e.

In certain implementations, the stop member 242 is coupled to the cassette body 202 using one or more spring members 250. In the example shown, first and second spring members 250 extend from opposite sides of the stop member 242. In certain examples, the spring members 250 limit the movement of the stop member 242 to the movement axis SM. The spring members 250 bias the stop member 242 to the lock position.

The release handle 244 is actuated by moving the release handle 244 in a different direction than the stop member 242 moves. In certain examples, the release handle 244 is moved orthogonally to the movement of the stop member 242. In an example, the release handle 244 is moved along the forward-rearward axis FR of the tray arrangement 110 while the stop member 242 moves along the height H2 of the cassette 200.

In certain implementations, the stop member 242 is carried outside a main interior of the cassette body 202. Accordingly, movement of the stop member 242 does not affect the optical fibers or other connections disposed within the main interior of the cassette body 202. In certain examples, the cassette body 202 includes a base 260 and a cover 262 that cooperate to define the main interior. In certain examples, the stop member 242 is carried at an exterior of the base 260, but is at least partially covered by the cover 262. For example, the cover 262 may include an overhang portion 264 that extends over the stop member 242 (e.g., see FIGS. 20 and 25). In some implementations, the cover 262 may include one or more latching members 266 to secure the cover 262 to the base 260. In other implementations, the cover 262 may be secured to the base 260 by fasteners, welding, or other securement mechanisms.

In certain implementations, the stop member 242 includes a camming member 252 that engages the release handle 244. Interaction between the camming member 252 and the release handle 244 transitions the stop member 242 between the lock and release positions as will be described in more detail herein. In certain examples, the camming member 252 is disposed at an opposite end of the stop member 242 from the engagement surfaces 246, 248. For example, the camming member 252 may be disposed at a top of the stop

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member 242 while the engagement surfaces 246, 248 are disposed at a bottom of the stop member 242. In certain examples, the camming member 252 extends outwardly from the stop member 242 along the width W2 of the cassette 200. In an example, the camming member 252 extends from the stop member 242 away from the cassette body 202 (e.g., see FIG. 23). In another example, the camming member 252 extends from the stop member 242 towards the cassette body 202.

In certain implementations, the base 260 defines a channel 268 along the exterior of the cassette 200 at the first end 208. The channel 268 is sized to hold the release handle 244. The channel 268 extends along the length L2 of the cassette 200 past the stop member 242. At least a portion of the stop member 242 is disposed within the channel 268. In certain implementations, the overhang portion 264 of the cover 262 extends over the release handle 244.

As shown in FIGS. 26, 27, and 57, the release handle 244 includes a transition arrangement 270 and a forward grip portion 276. As best seen in FIG. 28A, the transition arrangement 270 includes a ramped surface 272 along which the camming member 252 rides as the release handle 244 is moved along the actuation axis A between the neutral position and the forwardly extended position (e.g., see FIGS. 29 and 30). As the camming member 252 rides along the ramped surface 272, the stop member 242 moves with the camming member 252 along the movement axis SM (compare FIGS. 29 and 30).

In certain implementations, the ramped surface 272 is a first ramped surface and the transition arrangement 270 also includes a second ramped surface 274 extending in an opposite direction from the first ramped surface 272. As the camming member 252 rides along the second ramped surface 274, the stop member 242 moves with the camming member 252 along the movement axis SM. The camming member 252 rides along the second ramped surface 274 when the release handle 244 moves along the actuation axis A in a rearward direction relative to the cassette body 202 to the rearward extended position (FIG. 31).

In certain implementations, the transition arrangement 270 straddles the stop member 242. For example, the transition arrangement 270 may define a first pair of inner and outer ramped surfaces 272A, 272B disposed on opposite sides of the stop member 242 from each other and a second pair of ramped surfaces 274A, 274B disposed on opposite sides of the stop member 242 from each other. The stop member 242 may include a first camming member 252A extending towards the cassette body 202 and a second camming member 252B extending outwardly away from the cassette body 202. The first camming member 252A is configured to ride along the inner first and second ramped surfaces 272A, 274A and the second camming member 252B is configured to ride along the outer first and second ramped surfaces 272B, 274B. By straddling the stop member 242, the transition member 270 provides more stability to the movement of the stop member 242.

In other implementations, the stop member 242 may define a single camming member 252 and the transition arrangement 270 may define a single first ramped surface and a single second ramped surface along which the camming member 252 rides. In the example shown in FIG. 47, the stop member 242 includes a single camming member extending towards the cassette body 202. In the example shown in FIG. 47, the first and second ramped surfaces define part of a triangular shaped aperture in the transition arrangement 270.

In certain implementations, the cassette body 202 and the release handle 244 are configured to limit movement of the release handle 244 relative to the cassette body 202 along the actuation axis A to between the forwardly extended position and the rearwardly extended position. For example, the cassette body 202 may define forward and rearward stop surfaces 254 that oppose forward and rearward abutment surfaces 256 on the release handle 244 (e.g., see FIG. 27). When the release handle 244 is moved to the forwardly extended position, the forward abutment surface 256 engages or is close to engaging the forward stop surface 254 of the cassette body 202. When the release handle 244 is moved to the rearwardly extended position, the rearward abutment surface 256 engages or is close to engaging the rearward stop surface 254 of the cassette body 202.

Accordingly, pulling the release handle 244 forwardly relative to the cassette body 202 releases the cassette 200 from the guide 150, 350. Continuing to pull forwardly on the release handle 244 causes the release handle 244 to entrain the cassette body 202 in the forward direction, thereby sliding the cassette body 202 along the guides 150, 350 in the forward direction. Pushing the release handle 244 rearwardly relative to the cassette body 202 also releases the cassette 200 from the guide 150, 350. Continuing to push rearwardly on the release handle 244 causes the release handle 244 to entrain the cassette body 202 in the rearward direction, thereby sliding the cassette body 202 along the guides 150, 350 in the rearward direction.

In certain implementations, the first and second ramped surfaces 272, 274 are defined by a cutout or aperture in the release handle 244. In the example shown in FIG. 28A, the first and second ramped surfaces 272, 274 are defined by a V-shaped groove. In other examples, the first and second ramped surface 272, 274 may be defined by a triangular aperture in the transition arrangement 270. In still other examples, the cutout or aperture may have any desired shape including the ramped surface 272, 274.

The release handle 244 extends forwardly of the cassette body 202 so that the forward grip portion 276 is disposed forwardly of the front end 204 of the cassette 200 (e.g., see FIGS. 17 and 18). In certain implementations, the forward grip portion 276 is angled relative to a majority of the release handle 244 to facilitate grasping of the forward grip portion 276. In certain examples, the forward grip portion 276 may be angled along the lateral axis LA. In an example, the forward grip portion 276 is angled away from the first end 118 of the tray arrangement 110 to facilitate grasping the grip portion 276 without interference from the chassis 105 of any components mounted at the first end 118 of the tray arrangement 110.

In certain implementations, the forward grip portion 276 is configured to flex or deflect relative to a remainder of the release handle 244. In certain examples, the forward grip portion 276 may include a grip member 280 attached to a remainder of the release handle 244 by a flexible section 282. In certain examples, the flexible section 282 enables the grip member 280 to deflect along the width W2 of the cassette 200. This deflection ability facilitates grasping of the grip member 280 when the cassettes 200 are densely packed on the tray arrangement 110, especially when cables are routed to the front ports. In some examples, the flexible section 282 enables deflection of the grip member 280 along the height H2 of the cassette 200. In certain examples, the flexible section 282 enables a reduced amount of deflection along the height H2 compared to along the width W2. In

other examples, the flexible section 282 inhibits deflection of the grip member 280 along the height H2 of the cassette 200.

In certain implementations, the handle 244 also includes a rearward grip portion 278. The release handle 244 extends rearwardly of the cassette body 202 so that the rearward grip portion 276 is disposed rearwardly of the rear end 206 of the cassette 200 (e.g., see FIGS. 17 and 18). In certain examples, the rearward grip portion 278 has a different shape than the forward grip portion 276. In certain examples, the rearward grip portion 278 is hook shaped. In certain examples, the rearward grip portion 278 is not deflectable relative to the transition arrangement 270.

FIGS. 32-34 illustrate an alternative latching arrangement 240' suitable for use in locking the cassette 200 to a guide 150, 350. The latching arrangement 240' includes first stop member 242' and second stop members 242' extending along the channel 268 at the first end 208 of the cassette body 202. The latching arrangement 240' also includes a release handle 244' that defines first and second ramped surfaces 272', 274' that transition the stop members 242' from lock positions to release positions.

Each stop member 242' includes a latching hook deflectable along the height H2 of the cassette body 202. The first stop member 242' extends rearwardly and defines a first engagement surface 246' facing in the forwardly direction and the second stop member 242' extends forwardly and defines a second engagement surface 248' facing in the rearward direction. When the stop members 242' align with the notch 166, 366 through the top 164, 364 of the guide 150, 350, the first engagement surface 246' abuts against the rearward-facing catch surface 170, 370 to inhibit movement of the cassette 200 relative to the guide 150, 350 in the forward direction. The second engagement surface 248' abuts against the forward-facing catch surface 168, 368 to inhibit movement of the cassette 200 relative to the guide 150, 350 in the rearward direction.

The release handle 244' includes a transition arrangement 270' that defines the first and second ramped surfaces 272', 274'. In the example shown, the transition arrangement 270' includes a protrusion extending outwardly from the release handle 244 away from the cassette body 202. In other implementations, the protrusion may extend from the release handle 244 towards the cassette body 202.

Each latching hook defines a camming surface 252' that seats on a respective one of the first and second ramped surfaces 272', 274'. In the example shown, the tips of the latching hooks seat on the ramped surfaces 272', 274'. When the release handle 244 is moved along the actuation axis A in the forward direction, the transition arrangement 270' moves towards the first latching hook so that the camming surface 252' of the first latching hook rides over the first ramped surface 272', thereby deflecting the first engagement surface 246' to the release position. As the cassette body 202 begins to move forwardly relative to the guide 150, 350, the camming surface 252' of the second latching hook rides over the rearward-facing catch surface 170, 370 of the guide 150, 350. Both latching hooks ride over the top 164, 364 of the guide 150, 350 in the release position as the cassette 200 is slid along the guide 150, 350.

Similarly, when the release handle 244 is moved along the actuation axis A in the rearward direction, the transition arrangement 270' moves towards the second latching hook so that the camming surface 252' of the second latching hook rides over the second ramped surface 274', thereby deflecting the second engagement surface 248' to the release position. As the cassette body 202 begins to move rear-

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wardly relative to the guide **150**, **350**, the camming surface **252'** of the first latching hook rides over the forward-facing catch surface **168** of the guide **150**, **350**. Both latching hooks ride over the top **164**, **364** of the guide **150** in the release position as the cassette **200** is slid along the guide **150**, **350**.

Referring to the figures in general, some implementations of a cassette **200** are configured to receive a like number of plug connectors from the front and from the rear. Such cassettes **200** carry port members (e.g., optical adapters) only at the front end **204** of the cassette body **202**. Examples of such cassettes **200** are shown in FIG. **47** as cassette **200E** and FIG. **59** as cassette **200F**. The rear end **206** of the cassette body **202** defines a fiber port or cable port **284** through which fibers or one or more cables may extend into the cassette **200E**, **200F**. The cassette **200E**, **200F** may carry fiber management members **286** (e.g., a spool, a bend radius limiter, a retention finger, etc.) to facilitate routing fibers or cables between the front port members and the rear port **284**. The cassette **200E** of FIG. **47** includes a spool about which excess length of the fibers can be stored. The cassette **200F** of FIG. **59** is not configured to store excess length of the fibers.

Connectorized ends of the fibers or cables routed into the cassette **200** can be plugged into rear ports defined by the forward port members. For example, a plurality of connectorized optical fibers can be plugged into rear ports of LC optical adapters **230** disposed at the front end **204** of the cassette **200** to mate with connectorized optical fibers plugged into front ports of the LC optical adapters **230**. In another example, a connectorized end of a multi-fiber cable can be plugged into a rear port of an MPO optical adapter disposed at the front end **204** of the cassette to mate with an MPO plug connector received at the front port.

In some implementations, cassettes **200** connecting like numbers of plug connectors include a cover **262**. For example, a stub length of fibers or cable may be pre-terminated at the rear ports of the front port members and then routed out of the cassette **200** through the fiber or cable port **284**. In such examples, access to the rear ports of the front port members is not needed. The opposite end of the stub can be unterminated to facilitate splicing to one or more network cables. In other implementations, cassettes **200** connecting like numbers of plug connectors may have an open top that does not receive a cover (e.g., FIG. **47**). The open top may facilitate access to the rear ports of the front port members when stub fibers are not included with the cassette **200**. Rather, terminated ends of the network cables are routed directly to the front port members.

In other implementations, the cassettes **200** are configured to connect unlike numbers of plug connectors. For example, a cassette **200** may carry one or more port members (e.g., single MPO adapters **232**) at the rear end **206** of the cassette body **202**. Media segments (e.g., optical fibers) extend between the rear port members and the front port members to form connections therebetween. In some implementations, the rear port members include multi-fiber optical adapters (e.g., MPO adapters) and the front port members include single-fiber optical adapters (e.g., LC adapters). In other implementations, the cassette **200** is configured to receive higher fiber count cables at the rear than at the front. For example, a twenty-four fiber MPO plug connector received at the rear end **206** of the cassette **200** may connect through the cassette **200** to three eight-fiber MPO plug connectors at the front, to twenty-four single-fiber LC plug connectors at the front, or to twelve dual-fiber SN plug connectors at the front.

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In certain implementations, the cassette body **202** defines one or more rear apertures **216** at the rear end **206** at which to receive the rear port members. In certain examples, each rear aperture **216** is configured to hold one or more multi-fiber optical adapters. In certain examples, each of the rear apertures **216** has a footprint sized to one standardized port. For example, each rear aperture **216** may receive an MPO optical adapter. In certain examples, the rear apertures **216** are sized smaller than the front apertures **212**. In an example, each cassette **200** has a single rear aperture **216**. In another example, each cassette **200** has two rear apertures **216**. In another example, each cassette **200** has three rear apertures **216**. In certain examples, each rear aperture **216** is sized to hold one standardized port. In certain implementations, a punch-out cover may be initially provided at each of the rear apertures **216** so a respective optical adapter is mounted at fewer than all of the rear apertures **216**. In such cases, a user has the flexibility to select a position for a rear optical adapter to facilitate cable routing.

Aspects of the Disclosure

Aspect 1. A communications arrangement comprising:

- a tray arrangement carrying a plurality of spaced apart guides extending parallel to each other along a forward-rearward axis of the tray arrangement; and
- a cassette configured to mount to the tray arrangement, the cassette carrying a plurality of port members, the cassette being sized to extend from a first guide, over a second guide, to a third guide, the cassette being configured to inter-connect with the first guide and the second guide, but not the third guide.

Aspect 2. A communications arrangement comprising:

- a tray extending along a forward-rearward axis between opposite front and rear ends and along a lateral axis between opposite first and second sides carrying a plurality of guides spaced from each other along the lateral axis and extending generally parallel to each other, each of the guides extending along the forward-rearward axis, each of the guides defining a notch at a top of the guide; and
- a cassette mounted to the tray and selectively slidable along the forward-rearward axis, the cassette extending along the forward-rearward axis between opposite front and rear ends of the cassette and along the lateral axis between opposite first and second ends of the cassette, the cassette carrying a latching arrangement at the first end, the latching arrangement including a stop member aligned over the top of one of the guides when the cassette is mounted to the tray, the stop member riding over the top of the guide as the cassette is slid along the forward-rearward axis until the stop member aligns with the notch defined in the top of the guide, the stop member entering the notch when aligned with the notch along the forward-rearward axis to lock the cassette relative to the tray.

Aspect 3. A communications arrangement comprising:

- a tray arrangement carrying a plurality of guides; the guides being configured to receive cassettes having widths of X and 1.5X without modifying the guides.

Aspect 4. A communications arrangement comprising:

- a tray carrying a plurality of guides extending along a forward-rearward axis of the tray, the guides being parallel and spaced apart from each other along a lateral axis, each of the guides defining a groove facing in a first direction along the lateral axis, and each of the

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- guides being devoid of grooves facing in an opposite second direction along the lateral axis;
- a first cassette mountable to the tray by engaging and sliding along first and second ones of the guides, the first and second guides being adjacent each other; and
 - a second cassette mountable to the tray by engaging and sliding along third and fourth ones of the guides, the third and fourth guides being non-adjacent so that at least a fifth guide is disposed between the third and fourth guides along the lateral axis.
- Aspect 5. A communications arrangement comprising:
- a tray carrying a guide, the guide extending along a length between opposite first and second ends of the guide, the guide defining a groove extending along the length of the guide, the guide also extending between a bottom and a top, the bottom of the guide being configured to mount to the tray, the top of the guide defining a notch at an intermediate position between the first and second ends;
 - a cassette extending along a length between opposite front and rear ends, along a width between opposite first and second ends, and along a height between a top and a bottom, the cassette including an end rail at the first side that is configured to slide along the groove defined by the guide, the cassette including a latching arrangement disposed at the first end above the end rail, the latching arrangement including:
 - a stop member configured to move along the height of the cassette between a lock position and a release position; and
 - a release handle that extends along the length of the cassette between opposite front and rear ends of the release handle so that the front end of the handle extends outwardly beyond the front end of the cassette and the rear end of the handle extends outwardly beyond the rear end of the cassette, the release handle being slidable along a forward-rearward axis relative to the cassette, the release handle being configured to transition the stop member from the lock position to the release position.
- Aspect 6. A communications arrangement comprising:
- a tray extending along a forward-rearward axis between a front end and a rear end, the tray also extending along a lateral axis between opposite first and second sides;
 - a plurality of guides mounted to the tray, the guides extending along lengths between opposite front and rear ends of the guides, the lengths of the guides being parallel to the forward-rearward axis of the tray, the guides being spaced apart along the lateral axis of the tray;
 - a cassette configured to mount to the tray by sliding along first and second ones of the guides from either the front ends of the first and second guides or from the rear ends of the first and second guides at a discretion of a user, the cassette including a locking arrangement configured to lock to the first guide to maintain the cassette in position relative to the first guide, the cassette carrying a release handle that moves relative to the cassette along an actuation axis that extends parallel to the forward-rearward axis of the tray, the release handle releasing the locking arrangement from the first guide when moved along the actuation axis in a forward direction or a rearward direction at the discretion of the user.
- Aspect 7. A communications arrangement comprising:
- a tray extending along a forward-rearward axis and along a lateral axis that is orthogonal to the forward-rearward

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- axis, the tray defining a row of bays extending along the lateral axis, each of the bays extending along a width between an engagement end of a first guide and a non-engagement end of a second guide that is adjacent the first guide, the width extending along the lateral axis;
- a plurality of cassettes each configured to mount to the tray across a respective plurality of the bays, each cassette being configured to slide over the respective plurality of bays along the forward-rearward axis of the tray, each cassette including first and second engagement members that inter-connect with the engagement ends of first and second guides, wherein the cassette does not inter-connect with the non-engagement ends of the guides.
- Aspect 8. A communications cassette comprising:
- a body extending along a length between opposite front and rear ends, along a width between opposite first and second ends, and along a height between a top and a bottom, the body including a latching arrangement disposed at the first end of the body, the latching arrangement including:
 - a stop member configured to move along the height of the body between a lock position and a release position; and
 - a release handle that extends along the length of the body between opposite front and rear ends of the release handle so that the front end of the release handle extends outwardly beyond the front end of the body and the rear end of the release handle extends outwardly beyond the rear end of the body, the release handle being slidable along a forward-rearward axis relative to the body, the release handle being configured to transition the stop member from the lock position to the release position.
- Aspect 9. A communications cassette comprising:
- a body having a width extending between opposite first and second ends of the body and a depth extending between opposite front and rear ends of the body;
 - a row of optical adapters carried at the front end of the body so that front ports of the optical adapters are accessible from the front end of the body;
 - an end rail disposed at the first end of the body; and
 - a bottom rail disposed at an intermediate position between the first and second ends of the body, the bottom rail having a common orientation with the end rail;
- the body being devoid of rails at the second end of the body.
- Aspect 10. A tray arrangement extending along a lateral axis between opposite first and second ends, along a forward-rearward axis between opposite front and rear ends, and along a height between a bottom and a top, the tray arrangement comprising:
- a tray having a forward end portion towards the front end of the tray arrangement, a rearward end portion toward the rearward end of the tray arrangement, and an intermediate portion disposed between the forward end portion and the rearward end portion along the forward-rearward axis, the intermediate portion being raised along the height of the tray arrangement relative to the forward end portion; and
 - a plurality of guides extending across the intermediate portion along the forward-rearward axis, each of the guides being mounted to the forward end portion of the tray and to the rearward end portion of the tray, each of the guides extending partially over the forward end portion and partially over the rearward end portion.

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Aspect 11. A tray arrangement comprising:
 a tray extending along a forward-rearward axis and along
 a lateral axis that is orthogonal to the forward-rearward
 axis; and
 a plurality of guides mounted to the tray, the guides being
 configured to slidably receive a plurality of cassettes
 along the forward-rearward axis, the guides being par-
 allel to each other and spaced apart along the lateral
 axis so that the plurality of cassettes includes a plurality
 of base-8 cassettes or a plurality of base-12 cassettes at
 a discretion of the user without modifying a spacing
 between the guides, the base-12 cassette being wider
 than the base-8 cassette.

Aspect 12. A tray arrangement comprising:
 a tray extending along a forward-rearward axis and along
 a lateral axis that is orthogonal to the forward-rearward
 axis; and
 a plurality of guides mounted to the tray, each of the
 guides having an L-shaped transverse cross-sectional
 profile.

Aspect 13. A communications panel comprising:
 a chassis defining an interior;
 a tray arrangement configured to mount within the interior
 of the chassis, the tray arrangement extending along a
 forward-rearward axis between opposite front and rear
 ends and along a lateral axis between opposite first and
 second sides carrying a plurality of guides spaced from
 each other along the lateral axis and extending gener-
 ally parallel to each other, each of the guides extending
 along the forward-rearward axis, each of the guides
 defining a notch at a top of the guide; and
 a cassette mounted to the tray arrangement and selectively
 slidable along the forward-rearward axis, the cassette
 extending along the forward-rearward axis between
 opposite front and rear ends of the cassette and along
 the lateral axis between opposite first and second ends
 of the cassette, the cassette carrying a latching arrange-
 ment at the first end, the latching arrangement includ-
 ing a stop member aligned over the top of one of the
 guides when the cassette is mounted to the tray, the stop
 member riding over the top of the guide as the cassette
 is slid along the forward-rearward axis until the stop
 member aligns with the notch defined in the top of the
 guide, the stop member entering the notch when
 aligned with the notch along the forward-rearward axis
 to lock the cassette relative to the tray.

Aspect 14. A communications panel system comprising:
 a chassis defining an interior extending along a forward-
 rearward axis between opposite front and rear ends of
 the chassis and along a lateral axis between opposite
 first and second ends of the chassis;
 a tray mountable within the interior of the chassis through
 the front end, the tray carrying a plurality of guides
 spaced apart from each other along the lateral axis, each
 of the guides extending along the forward-rearward
 axis;
 a plurality of cassettes configured to slide along the guides
 to mount to the tray, each of the cassettes extending
 along a width between opposite first and second ends,
 the first end of each cassette carrying an end rail
 configured to inter-connect with a respective first one
 of the guides, an intermediate portion of the cassette
 carrying a bottom rail configured to inter-connect with
 a respective second one of the guides, the end rail and
 the bottom rail extending outwardly in a common
 direction from a base end contacting the cassette to a
 free end.

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Aspect 15. A method of removing a cassette from a tray
 having a plurality of parallel guides each extending parallel
 to a forward-rearward axis of the tray, the cassette being
 slidable along the tray relative to the guides, each cassette
 carrying a release handle, the method comprising:
 choosing one of a forward direction and a rearward
 direction along which to remove the cassette from the
 tray;
 pulling the release handle relative to the cassette in the
 chosen direction to unlatch the cassette from a first of
 the guides; and
 continuing to pull the release handle in the chosen direc-
 tion to slide the cassette in the chosen direction relative
 to the first guide.

Aspect 16. A tray arrangement extending along a lateral
 axis between opposite first and second ends, along a for-
 ward-rearward axis between opposite front and rear ends,
 and along a height between a bottom and a top, the tray
 arrangement comprising:
 a tray having opposite top and bottom surfaces each
 extending along the lateral axis and the forward-rear-
 ward axis, the tray defining a main region at which a
 plurality of apertures are defined, each of the apertures
 extending between the top and bottom surfaces; and
 a plurality of guides heat attached the tray at the main
 region, each of the guides including a plurality of pegs
 extending into respective ones of the apertures defined
 in the tray, each of the guides defining a grooved side
 facing towards the first end of the tray arrangement and
 a closed side facing towards the second end of the tray
 arrangement.

Aspect 17. A tray arrangement extending along a lateral
 axis between opposite first and second ends, along a for-
 ward-rearward axis between opposite front and rear ends,
 and along a height between a bottom and a top, the tray
 arrangement comprising:
 a tray; and
 a plurality of guides mounted to the tray, each of the
 guides extending along the forward-rearward axis, the
 guides being spaced from each other along the lateral
 axis, the guides including a first guide and a second
 guide connected by a webbing so that the first and
 second guides mount to the tray as a unit.

Aspect 18. A method of manufacturing a plurality of tray
 arrangements, the method comprising:
 selecting one of a first tray arrangement and a second tray,
 each of the first and second trays defining a plurality of
 apertures, each of the first and second trays defining
 countersinks, the first tray having a different thickness
 than the second tray so that the apertures of the first tray
 have a different height than the apertures of the second
 tray;
 mounting a plurality of guides to the selected one of the
 first and second trays, the guides each having a plurality
 of pegs, wherein mounting the guides includes insert-
 ing the pegs into the apertures of the selected one of the
 first and second trays, wherein the pegs of the guides
 have a common size regardless of which of the first and
 second trays is selected; and
 liquefying at least portions of the pegs into the counter-
 sinks.

Aspect 19. A communications arrangement comprising:
 a tray arrangement carrying a plurality of guides spaced
 apart from each other along a lateral axis of the tray
 arrangement, each of the guides extending along a
 forward-rearward axis of the tray arrangement;

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a plurality of cassettes mounted to the tray arrangement in a row along the lateral axis of the tray arrangement, each of the cassettes defining a plurality of front apertures each configured to receive at least one port member, the front apertures of each cassette being spaced from each other along the lateral axis of the tray arrangement by a first gap, wherein the cassettes are configured so that adjacent front apertures of adjacent cassettes on the tray arrangement are separated from each other by a second gap that is about equal to the first gap.

Having described the preferred aspects and implementations of the present disclosure, modifications and equivalents of the disclosed concepts may readily occur to one skilled in the art. However, it is intended that such modifications and equivalents be included within the scope of the claims which are appended hereto.

What is claimed is:

1. A tray arrangement extending along a lateral axis between opposite first and second ends, along a forward-rearward axis between opposite front and rear ends, and along a height between a bottom and a top, the tray arrangement comprising:

a tray having opposite top and bottom surfaces each extending along the lateral axis and the forward-rearward axis, the tray defining a main region at which a plurality of apertures are defined, each of the apertures extending between the top and bottom surfaces; and
a plurality of guides attached to the tray at the main region, each of the guides including a plurality of pegs extending into respective ones of the apertures defined in the tray, each of the guides defining a grooved side facing towards the first end of the tray arrangement and a closed side facing towards the second end of the tray arrangement.

2. The tray arrangement of claim 1, wherein each guide extends along a length from a respective first end to a respective second end.

3. The tray arrangement of claim 2, wherein the grooved side defines a groove that extends along the length between the first and second ends.

4. The tray arrangement of claim 2, wherein the grooved side is partially defined by a top surface extending along the length of the guide, the top surface defining a ramped surface at the first and second ends.

5. The tray arrangement of claim 4, wherein the top surface defines a notch at an intermediate position between

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the first and second ends, the notch providing access to the grooved side through the top surface.

6. The tray arrangement of claim 5, wherein the notch provides access to a forward-facing catch surface and a rearward-facing catch surface at opposite ends of the notch.

7. The tray arrangement of claim 1, wherein the grooved side defines a notch at an intermediate position between the first and second ends.

8. The tray arrangement of claim 7, wherein the notch provides access to a forward-facing catch surface and a rearward-facing catch surface at opposite ends of the notch.

9. The tray arrangement of claim 1, wherein each of the guides has an L-shaped transverse cross-section.

10. The tray arrangement of claim 1, wherein each of the guides has a C-shaped transverse cross-section.

11. The tray arrangement of claim 1, wherein the tray arrangement is configured to mount to a chassis and to move relative to the chassis between a retracted position and an extended position.

12. The tray arrangement of claim 1, wherein the pegs are heat-staked to the tray.

13. The tray arrangement of claim 1, wherein the tray defines a plurality of finger access openings disposed forward of the main region of the tray.

14. The tray arrangement of claim 1, wherein the apertures lead to countersinks at the bottom surface of the tray.

15. The tray arrangement of claim 1, wherein the apertures are aligned in rows.

16. The tray arrangement of claim 15, wherein one or more of the apertures in a first of the rows are laterally offset from each other.

17. The tray arrangement of claim 16, wherein front and rear ones of the apertures of the first row are aligned along a reference axis while intermediate ones of the apertures of the first row are offset along the lateral axis relative to the reference axis.

18. The tray arrangement of claim 1, wherein a first of the guides includes a side tab extending outwardly from the grooved side.

19. The tray arrangement of claim 7, wherein a first of the guides includes a side tab extending outwardly from the grooved side, the side tab aligning with the notch.

20. The tray arrangement of claim 1, wherein the guide is configured to engage with a cassette at only the grooved side.

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